SRI (k-inflation): N04LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 4$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

For k-inflation, $c_t = 1$. The scalar amplitude includes transport from the acoustic reference to the common tensor reference. The sound speed is $c_{s*} > 0$, with

$$\alpha_n^{c_s} = \frac{d^{n+1}\ln c_s}{d(\ln k)^{n+1}}.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*} c_{s*}} \left[\sum_{d=0}^{4} F_{s,d} + O(\mathrm{SR}^5)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$\begin{aligned} F_{s,1} = {} & 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1 \\ & - 1.2703628454615\alpha_0^{c_s} - 1.0\alpha_0^{c_s}\log(c_s) \\ & - 1.0\epsilon_2\log(c_s) - 2.0\epsilon_1\log(c_s) \end{aligned}$$

Relative order 2.

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & - 0.39567736285244\alpha_1^{c_s} - 0.22975141576082\left(\alpha_0^{c_s}\right)^2 \\ & + 0.5\left(\alpha_0^{c_s}\right)^2\log(c_s)^2 + 0.5\epsilon_2^2\log(c_s)^2 \\ & + 2.0\epsilon_1^2\log(c_s)^2 + 0.27036284546148\left(\alpha_0^{c_s}\right)^2\log(c_s) \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 1.0809943369567\alpha_0^{c_s}\epsilon_1 \\ & + 0.14504832807052\epsilon_2\epsilon_3 - 0.5\alpha_1^{c_s}\log(c_s)^2 \\ & - 0.91854861815409\epsilon_1^2\log(c_s) - 0.72986567698311\alpha_0^{c_s}\epsilon_2 \\ & - 1.2703628454615\alpha_1^{c_s}\log(c_s) - 0.72963715453852\epsilon_2^2\log(c_s) \\ & + 1.0\alpha_0^{c_s}\epsilon_2\log(c_s)^2 + 1.0\epsilon_1\epsilon_2\log(c_s)^2 \\ & + 2.0\alpha_0^{c_s}\epsilon_1\log(c_s)^2 + 1.0814513818459\alpha_0^{c_s}\epsilon_1\log(c_s) \\ & + 0.72963715453852\epsilon_2\epsilon_3\log(c_s) - 0.5\epsilon_2\epsilon_3\log(c_s)^2 \\ & - 0.45927430907704\alpha_0^{c_s}\epsilon_2\log(c_s) \\ & - 2.459274309077\epsilon_1\epsilon_2\log(c_s) \end{aligned}$$

Relative order 3.

$$\begin{aligned} F_{s,3} = {} & 0.20072294544592\epsilon_2^3 + 1.6057835635673\epsilon_1^3 \\ & - 0.2199597209462\alpha_2^{c_s} - 0.028914209092607\left(\alpha_0^{c_s}\right)^3 \\ & + 4.2977813647689\epsilon_1^2\epsilon_2 + 1.1508618640566\alpha_1^{c_s}\epsilon_1 \\ & + 0.50011426122229\left(\alpha_0^{c_s}\right)^3\log(c_s) + 0.17975356917588\alpha_1^{c_s}\epsilon_2 \end{aligned}$$

$$
\begin{aligned}
&+ 0.36481857726926 \left(\alpha_0^{c_s}\right)^3 \log\left(c_s\right)^2 \\
&+ 0.36481857726926\epsilon_2^3 \log\left(c_s\right)^2 + 2.9185486181541\epsilon_1^3 \log\left(c_s\right)^2 \\
&+ 1.959600275274\epsilon_1\epsilon_2^2 + 0.030669313835717\epsilon_2\epsilon_3^2 \\
&- 0.39567736285244\alpha_2^{c_s} \log\left(c_s\right) - 1.0906092762856\alpha_0^{c_s}\alpha_1^{c_s} \\
&- 0.12746831820078\alpha_0^{c_s}\epsilon_2^2 - 0.25470811395696 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \\
&- 0.63518142273074\alpha_2^{c_s} \log\left(c_s\right)^2 - 0.32050476429836\alpha_0^{c_s}\epsilon_1^2 \\
&- 0.16666666666667\alpha_2^{c_s} \log\left(c_s\right)^3 \\
&- 0.16666666666667 \left(\alpha_0^{c_s}\right)^3 \log\left(c_s\right)^3 \\
&- 0.16666666666667\epsilon_2^3 \log\left(c_s\right)^3 - 1.3333333333333\epsilon_1^3 \log\left(c_s\right)^3 \\
&- 0.3571054727393 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 - 0.23520055946694\epsilon_2^2\epsilon_3 \\
&- 0.49988573877771\epsilon_2^3 \log\left(c_s\right) - 3.9990859102216\epsilon_1^3 \log\left(c_s\right) \\
&+ 0.5\alpha_0^{c_s}\alpha_1^{c_s} \log\left(c_s\right)^3 + 0.5\alpha_1^{c_s}\epsilon_2 \log\left(c_s\right)^3 \\
&+ 0.5\epsilon_2^2\epsilon_3 \log\left(c_s\right)^3 + 1.0\alpha_1^{c_s}\epsilon_1 \log\left(c_s\right)^3 \\
&+ 3.0\epsilon_1^2\epsilon_2 \log\left(c_s\right)^2 \\
&+ 0.50034278366688 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 \log\left(c_s\right) \\
&+ 2.2898599312799\alpha_0^{c_s}\epsilon_1\epsilon_2 \\
&+ 1.1889114636156 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \log\left(c_s\right)^2 \\
&+ 2.3778229272311\alpha_0^{c_s}\epsilon_1^2 \log\left(c_s\right)^2 \\
&+ 1.8110885363844\alpha_1^{c_s}\epsilon_1 \log\left(c_s\right)^2 + 0.49443659507158\alpha_0^{c_s}\epsilon_2\epsilon_3 \\
&+ 1.0944557318078\alpha_0^{c_s}\epsilon_2^2 \log\left(c_s\right)^2 \\
&+ 1.0944557318078 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 \log\left(c_s\right)^2 \\
&+ 0.14504832807052\epsilon_2\epsilon_3^2 \log\left(c_s\right) \\
&+ 1.2296371545385\epsilon_1\epsilon_2^2 \log\left(c_s\right)^2 \\
&+ 0.36481857726926\epsilon_2\epsilon_3^2 \log\left(c_s\right)^2 \\
&+ 0.40554426819222\alpha_0^{c_s}\alpha_1^{c_s} \log\left(c_s\right)^2 \\
&+ 0.40554426819222\alpha_1^{c_s}\epsilon_2 \log\left(c_s\right)^2 \\
&+ 0.85472314948489\epsilon_2^2\epsilon_3 \log\left(c_s\right) \\
&+ 0.45995987641081 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \log\left(c_s\right) \\
&+ 1.8723490626616\alpha_1^{c_s}\epsilon_1 \log\left(c_s\right) + 0.030669313835717\epsilon_2\epsilon_3\epsilon_4 \\
&- 0.5\alpha_0^{c_s}\epsilon_2^2 \log\left(c_s\right)^3 \\
&- 0.5 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 \log\left(c_s\right)^3 \\
&- 1.0 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \log\left(c_s\right)^3 \\
&- 2.0\alpha_0^{c_s}\epsilon_1^2 \log\left(c_s\right)^3 \\
&- 0.33418831413068\alpha_1^{c_s}\epsilon_2 \log\left(c_s\right) \\
&- 4.957559194624\epsilon_1^2\epsilon_2 \log\left(c_s\right) \\
&- 2.0800802471784\alpha_0^{c_s}\epsilon_1^2 \log\left(c_s\right) \\
&- 1.0944557318078\epsilon_2^2\epsilon_3 \log\left(c_s\right)^2 - 0.65969152604436\epsilon_1\epsilon_2\epsilon_3 \\
&- 0.16666666666667\epsilon_2\epsilon_3^2 \log\left(c_s\right)^3 \\
&- 0.33333333333333\epsilon_1\epsilon_2^2 \log\left(c_s\right)^3 \\
&- 1.3341883141307\alpha_0^{c_s}\alpha_1^{c_s} \log\left(c_s\right) \\
&- 2.7294086320939\epsilon_1\epsilon_2^2 \log\left(c_s\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 0.49965721633312\alpha_0^{c_s}\epsilon_2^2\log(c_s)\\
&+ 0.5\alpha_0^{c_s}\epsilon_2\epsilon_3\log(c_s)^3\\
&+ 1.4588172641879\epsilon_1\epsilon_2\epsilon_3\log(c_s)\\
&+ 0.14504832807052\epsilon_2\epsilon_3\epsilon_4\log(c_s)\\
&+ 0.66666666666667\epsilon_1\epsilon_2\epsilon_3\log(c_s)^3\\
&+ 0.36481857726926\epsilon_2\epsilon_3\epsilon_4\log(c_s)^2\\
&+ 3.1889114636156\alpha_0^{c_s}\epsilon_1\epsilon_2\log(c_s)^2\\
&- 1.0\alpha_0^{c_s}\epsilon_1\epsilon_2\log(c_s)^3\\
&- 0.14527685051511\alpha_0^{c_s}\epsilon_2\epsilon_3\log(c_s)\\
&- 1.0944557318078\alpha_0^{c_s}\epsilon_2\epsilon_3\log(c_s)^2\\
&- 2.459274309077\epsilon_1\epsilon_2\epsilon_3\log(c_s)^2\\
&- 0.16666666666667\epsilon_2\epsilon_3\epsilon_4\log(c_s)^3\\
&- 0.74845687543973\alpha_0^{c_s}\epsilon_1\epsilon_2\log(c_s)
\end{aligned}
$$

Relative order 4.

$$
\begin{aligned}
F_{s,4} = {} & 0.086152301850743\left(\alpha_0^{c_s}\right)^4 + 0.083467005785127\epsilon_2^4\\
&+ 2.9412556561294\epsilon_1^4 - 0.13350316714425\alpha_3^{c_s}\\
&- 0.65234365420886\left(\alpha_1^{c_s}\right)^2 + 0.125\left(\alpha_1^{c_s}\right)^2\log(c_s)^4\\
&+ 0.055787239966231\epsilon_2\epsilon_3^3 + 8.1576664436908\epsilon_1^2\epsilon_2^2\\
&+ 0.13518142273074\left(\alpha_1^{c_s}\right)^2\log(c_s)^3 + 9.1655563981061\epsilon_1^3\epsilon_2\\
&+ 0.35823966089031\alpha_2^{c_s}\epsilon_1 + 6.9176345283757\epsilon_1^4\log(c_s)^2\\
&+ 0.11476144665811\left(\alpha_0^{c_s}\right)^4\log(c_s)^2\\
&+ 0.5290284703149\left(\alpha_0^{c_s}\right)^4\log(c_s) + 0.034705229808718\alpha_0^{c_s}\epsilon_2^3\\
&+ 0.27764183846974\alpha_0^{c_s}\epsilon_1^3 + 2.2830486720555\epsilon_1\epsilon_2^3\\
&+ 0.041666666666667\left(\alpha_0^{c_s}\right)^4\log(c_s)^4\\
&+ 0.041666666666667\epsilon_2^4\log(c_s)^4 + 0.66666666666667\epsilon_1^4\log(c_s)^4\\
&+ 0.029535258092405\alpha_1^{c_s}\epsilon_2^2 + 0.24994286938885\epsilon_2^4\log(c_s)^2\\
&- 0.63704346883458\alpha_1^{c_s}\epsilon_1^2 - 0.19535327126822\left(\alpha_0^{c_s}\right)^3\epsilon_1\\
&- 0.20996470057763\epsilon_2^3\epsilon_3 - 1.2600809095931\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\\
&- 0.19783868142622\alpha_3^{c_s}\log(c_s)^2\\
&- 1.0906092762856\left(\alpha_1^{c_s}\right)^2\log(c_s)\\
&- 0.21172714091025\alpha_3^{c_s}\log(c_s)^3 - 0.2199597209462\alpha_3^{c_s}\log(c_s)\\
&- 0.12659084472672\left(\alpha_0^{c_s}\right)^3\epsilon_2 - 0.26150492255387\left(\alpha_0^{c_s}\right)^2\epsilon_2^2\\
&- 0.5615601604977\left(\alpha_0^{c_s}\right)^2\epsilon_1^2\\
&- 0.28827285908975\left(\alpha_0^{c_s}\right)^4\log(c_s)^3 - 0.04083989050104\alpha_2^{c_s}\epsilon_2\\
&- 0.65647697429967\alpha_0^{c_s}\alpha_2^{c_s} - 0.041666666666667\alpha_3^{c_s}\log(c_s)^4\\
&- 0.66709415706534\left(\alpha_1^{c_s}\right)^2\log(c_s)^2\\
&- 0.20072294544592\epsilon_2^4\log(c_s) - 3.2790324121027\epsilon_1^4\log(c_s)^3\\
&- 7.2106530373563\epsilon_1^4\log(c_s) - 0.058660421836305\epsilon_2^2\epsilon_3^2\\
&- 0.12160619242309\epsilon_2^4\log(c_s)^3\\
&+ 0.25\left(\alpha_0^{c_s}\right)^2\epsilon_2^2\log(c_s)^4\\
&+ 1.0\left(\alpha_0^{c_s}\right)^2\epsilon_1^2\log(c_s)^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.055787239966231\epsilon_2\epsilon_3\epsilon_4^2 \\
&+ 0.45881726418786\left(\alpha_0^{c_s}\right)^3\epsilon_1\log\left(c_s\right)^2 \\
&+ 0.59422720936319\left(\alpha_0^{c_s}\right)^3\epsilon_2\log\left(c_s\right)^2 \\
&+ 0.16736171989869\epsilon_2\epsilon_3^2\epsilon_4 \\
&+ 0.34690856364099\alpha_2^{c_s}\epsilon_2\log\left(c_s\right)^3 \\
&+ 0.38664546255166\epsilon_1^2\epsilon_2\epsilon_3 \\
&+ 0.78219898544963\epsilon_2^2\epsilon_3^2\log\left(c_s\right)^2 \\
&+ 0.1995931888627\alpha_0^{c_s}\epsilon_2^2\epsilon_3 \\
&+ 0.22963715453852\alpha_1^{c_s}\epsilon_2^2\log\left(c_s\right)^3 \\
&+ 0.77249640855299\left(\alpha_0^{c_s}\right)^3\epsilon_1\log\left(c_s\right) \\
&+ 1.590781305949\alpha_2^{c_s}\epsilon_1\log\left(c_s\right) \\
&+ 1.0271504606153\alpha_2^{c_s}\epsilon_1\log\left(c_s\right)^3 + 0.26604756153058\alpha_1^{c_s}\epsilon_2\epsilon_3 \\
&+ 1.0941129481409\left(\alpha_0^{c_s}\right)^2\epsilon_2^2\log\left(c_s\right)^2 \\
&+ 0.27437646670959\left(\alpha_0^{c_s}\right)^2\epsilon_2\epsilon_3 \\
&+ 0.072524164035261\epsilon_2\epsilon_3^3\log\left(c_s\right)^2 \\
&+ 0.29166666666667\epsilon_2^2\epsilon_3^2\log\left(c_s\right)^4 \\
&+ 2.4802661893204\epsilon_1^2\epsilon_2^2\log\left(c_s\right)^2 \\
&+ 0.15965542786054\alpha_0^{c_s}\epsilon_2\epsilon_3^2 + 2.6436935116586\alpha_0^{c_s}\epsilon_1^2\epsilon_2 \\
&+ 1.3318518941832\alpha_2^{c_s}\epsilon_1\log\left(c_s\right)^2 \\
&+ 0.083333333333333\epsilon_1\epsilon_2^3\log\left(c_s\right)^4 \\
&+ 0.16666666666667\alpha_0^{c_s}\alpha_2^{c_s}\log\left(c_s\right)^4 \\
&+ 0.16666666666667\alpha_0^{c_s}\epsilon_2^3\log\left(c_s\right)^4 \\
&+ 0.16666666666667\alpha_2^{c_s}\epsilon_2\log\left(c_s\right)^4 \\
&+ 0.16666666666667\left(\alpha_0^{c_s}\right)^3\epsilon_2\log\left(c_s\right)^4 \\
&+ 0.16666666666667\epsilon_1^2\epsilon_2^2\log\left(c_s\right)^4 \\
&+ 0.33333333333333\alpha_2^{c_s}\epsilon_1\log\left(c_s\right)^4 \\
&+ 0.33333333333333\left(\alpha_0^{c_s}\right)^3\epsilon_1\log\left(c_s\right)^4 \\
&+ 1.3333333333333\alpha_0^{c_s}\epsilon_1^3\log\left(c_s\right)^4 \\
&+ 2.687540328948\left(\alpha_0^{c_s}\right)^2\epsilon_1^2\log\left(c_s\right)^2 \\
&+ 1.364704316047\epsilon_1\epsilon_2^3\log\left(c_s\right)^2 \\
&+ 0.18024189697432\alpha_0^{c_s}\alpha_2^{c_s}\log\left(c_s\right)^3 \\
&+ 0.72963715453852\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\log\left(c_s\right)^3 \\
&+ 0.72963715453852\epsilon_2^3\epsilon_3\log\left(c_s\right)^3 \\
&+ 5.9584732844024\epsilon_1^3\epsilon_2\log\left(c_s\right)^2 \\
&+ 0.39971329012208\alpha_2^{c_s}\epsilon_2\log\left(c_s\right) + 1.6213444503115\alpha_1^{c_s}\epsilon_1\epsilon_2 \\
&+ 0.83736939580468\epsilon_2^3\epsilon_3\log\left(c_s\right) \\
&+ 0.86459005482467\alpha_0^{c_s}\epsilon_2^3\log\left(c_s\right)^2 \\
&+ 6.9167204385974\alpha_0^{c_s}\epsilon_1^3\log\left(c_s\right)^2 \\
&+ 0.88636246549879\left(\alpha_0^{c_s}\right)^3\epsilon_2\log\left(c_s\right) \\
&+ 1.822809817091\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\log\left(c_s\right)^2 \\
&+ 3.7110251386503\alpha_0^{c_s}\epsilon_1\epsilon_2^2
\end{aligned}
$$

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\begin{aligned}
&+ 0.12160619242309\epsilon_2\epsilon_3^3 \log(c_s)^3 \\
&+ 0.030669313835717\epsilon_2\epsilon_3^3 \log(c_s) \\
&+ 0.030744524360879\alpha_2^{c_s}\epsilon_2 \log(c_s)^2 \\
&+ 1.9904900950312\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_1 \\
&- 0.25\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\log(c_s)^4 \\
&- 0.25\alpha_1^{c_s}\epsilon_2^2\log(c_s)^4 - 0.25\epsilon_2^3\epsilon_3\log(c_s)^4 \\
&- 1.0\alpha_1^{c_s}\epsilon_1^2\log(c_s)^4 \\
&- 13.260103415181\epsilon_1^3\epsilon_2\log(c_s) \\
&- 0.15309143635901\epsilon_1^2\epsilon_2^2\log(c_s)^3 \\
&- 0.3303216651229\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\log(c_s) \\
&- 0.57314036602284\alpha_0^{c_s}\epsilon_2^3\log(c_s) \\
&- 4.5851229281828\alpha_0^{c_s}\epsilon_1^3\log(c_s)
\end{aligned}
$$

$$
\begin{aligned}
&- 0.054300921230608\epsilon_1^3\epsilon_2\log(c_s)^3 \\
&- 0.016062214045699\epsilon_2^2\epsilon_3\epsilon_4 \\
&- 0.65309143635901\alpha_0^{c_s}\epsilon_2^3\log(c_s)^3 \\
&- 5.2247314908721\alpha_0^{c_s}\epsilon_1^3\log(c_s)^3 \\
&- 2.6309687842081\epsilon_1\epsilon_2^3\log(c_s) \\
&- 0.39486939801397\left(\alpha_0^{c_s}\right)^2\epsilon_1\epsilon_2 \\
&- 0.79065642344446\left(\alpha_0^{c_s}\right)^2\epsilon_1^2\log(c_s) \\
&- 0.015083425393042\left(\alpha_0^{c_s}\right)^2\epsilon_2^2\log(c_s) \\
&- 7.81010212695\epsilon_1^2\epsilon_2^2\log(c_s) \\
&- 0.5010704327696\epsilon_2^2\epsilon_3^2\log(c_s) \\
&- 1.1044368983699\alpha_0^{c_s}\alpha_2^{c_s}\log(c_s)^2 \\
&- 1.177190182909\epsilon_2^3\epsilon_3\log(c_s)^2 \\
&- 0.30722188737666\alpha_1^{c_s}\epsilon_2^2\log(c_s) \\
&- 1.2296371545385\left(\alpha_0^{c_s}\right)^2\epsilon_2^2\log(c_s)^3 \\
&- 1.2663269181918\alpha_0^{c_s}\alpha_2^{c_s}\log(c_s) \\
&- 0.079171041433698\epsilon_1\epsilon_2\epsilon_3^2 \\
&- 0.081451381845913\alpha_1^{c_s}\epsilon_1^2\log(c_s)^3 \\
&- 0.08273445110122\alpha_1^{c_s}\epsilon_2^2\log(c_s)^2 \\
&- 0.041666666666667\epsilon_2\epsilon_3^3\log(c_s)^4 \\
&- 0.66666666666667\epsilon_1^3\epsilon_2\log(c_s)^4 \\
&- 0.095364228769985\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_2 \\
&- 0.40987905151284\epsilon_1\epsilon_2^3\log(c_s)^3 \\
&- 1.6395162060514\left(\alpha_0^{c_s}\right)^3\epsilon_1\log(c_s)^3 \\
&- 1.6420263407893\alpha_1^{c_s}\epsilon_1^2\log(c_s)^2 \\
&- 0.85124334696161\epsilon_2^2\epsilon_3^2\log(c_s)^3 \\
&- 0.91389525644738\epsilon_1\epsilon_2^2\epsilon_3 - 1.8308737667068\alpha_1^{c_s}\epsilon_1^2\log(c_s) \\
&- 3.9185486181541\left(\alpha_0^{c_s}\right)^2\epsilon_1^2\log(c_s)^3 \\
&- 0.98642476969235\left(\alpha_0^{c_s}\right)^3\epsilon_2\log(c_s)^3
\end{aligned}
$$

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\begin{aligned}
&+ 0.25\alpha_1^{c_s}\epsilon_2\epsilon_3 \log(c_s)^4 \\
&+ 0.25\epsilon_1\epsilon_2\epsilon_3^2 \log(c_s)^4 \\
&+ 0.5(\alpha_0^{c_s})^2\epsilon_1\epsilon_2 \log(c_s)^4 \\
&+ 2.3932895232622\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_1 \log(c_s) \\
&+ 4.9739619735052\epsilon_1\epsilon_2^2\epsilon_3 \log(c_s) \\
&+ 0.055787239966231\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\
&+ 5.0759816810454\epsilon_1^2\epsilon_2\epsilon_3 \log(c_s) \\
&+ 0.60881560930639\alpha_0^{c_s}\epsilon_2\epsilon_3^2 \log(c_s) \\
&+ 0.91854861815409\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_1 \log(c_s)^3 \\
&+ 2.3260301557806\alpha_1^{c_s}\epsilon_1\epsilon_2 \log(c_s) \\
&+ 2.9276156949378(\alpha_0^{c_s})^2\epsilon_1\epsilon_2 \log(c_s)^2 \\
&+ 0.092007941507151\epsilon_2\epsilon_3^2\epsilon_4 \log(c_s) \\
&+ 1.7400753659898\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_2 \log(c_s)^2 \\
&+ 3.8236358414571\alpha_0^{c_s}\epsilon_1\epsilon_2^2 \log(c_s)^2 \\
&+ 0.95927430907704\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_2 \log(c_s)^3 \\
&+ 0.95927430907704\alpha_1^{c_s}\epsilon_1\epsilon_2 \log(c_s)^3 \\
&+ 2.0493952575642\epsilon_1\epsilon_2^2\epsilon_3 \log(c_s)^3 \\
&+ 0.072524164035261\epsilon_2\epsilon_3\epsilon_4^2 \log(c_s)^2 \\
&+ 0.59514129914155\alpha_1^{c_s}\epsilon_1\epsilon_2 \log(c_s)^2 \\
&+ 1.2296371545385(\alpha_0^{c_s})^2\epsilon_2\epsilon_3 \log(c_s)^3 \\
&+ 0.15965542786054\alpha_0^{c_s}\epsilon_2\epsilon_3\epsilon_4 \\
&+ 0.16666666666667\alpha_0^{c_s}\epsilon_2\epsilon_3^2 \log(c_s)^4 \\
&+ 0.16666666666667\epsilon_2^2\epsilon_3\epsilon_4 \log(c_s)^4 \\
&+ 0.33333333333333\alpha_0^{c_s}\epsilon_1\epsilon_2^2 \log(c_s)^4 \\
&+ 2.6690621955951\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_1 \log(c_s)^2 \\
&+ 0.67419016424746\alpha_1^{c_s}\epsilon_2\epsilon_3 \log(c_s) \\
&+ 0.34055047747869\alpha_0^{c_s}\epsilon_2^2\epsilon_3 \log(c_s) \\
&+ 2.7883647605464\alpha_0^{c_s}\epsilon_1\epsilon_2\epsilon_3 \\
&+ 0.35483741070718\epsilon_2^2\epsilon_3\epsilon_4 \log(c_s)^2 \\
&+ 0.36481857726926\epsilon_2\epsilon_3^2\epsilon_4 \log(c_s)^3 \\
&+ 3.306182872718\epsilon_1^2\epsilon_2\epsilon_3 \log(c_s)^3 \\
&+ 0.21757249210578\epsilon_2\epsilon_3^2\epsilon_4 \log(c_s)^2 \\
&+ 0.21965598797644\alpha_0^{c_s}\epsilon_2\epsilon_3^2 \log(c_s)^2 \\
&+ 7.1464706582396\alpha_0^{c_s}\epsilon_1^2\epsilon_2 \log(c_s)^2 \\
&+ 0.94917888129267\epsilon_1\epsilon_2\epsilon_3^2 \log(c_s)^2 \\
&+ 0.12160619242309\epsilon_2\epsilon_3\epsilon_4^2 \log(c_s)^3 \\
&+ 1.959274309077\alpha_0^{c_s}\epsilon_2^2\epsilon_3 \log(c_s)^3 \\
&+ 0.030669313835717\epsilon_2\epsilon_3\epsilon_4^2 \log(c_s) \\
&- 0.125\epsilon_2\epsilon_3^2\epsilon_4 \log(c_s)^4 \\
&- 0.25(\alpha_0^{c_s})^2\epsilon_2\epsilon_3 \log(c_s)^4 \\
&- 0.5\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_2 \log(c_s)^4
\end{aligned}
$$

$$
\begin{aligned}
&- 0.5\alpha_0^{c_s}\epsilon_2^2\epsilon_3\log\left(c_s\right)^4 \\
&- 0.5\alpha_1^{c_s}\epsilon_1\epsilon_2\log\left(c_s\right)^4 \\
&- 1.0\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_1\log\left(c_s\right)^4 \\
&- 3.0\alpha_0^{c_s}\epsilon_1^2\epsilon_2\log\left(c_s\right)^3 \\
&- 0.77164591471679\left(\alpha_0^{c_s}\right)^2\epsilon_2\epsilon_3\log\left(c_s\right)^2 \\
&- 8.8847452959004\alpha_0^{c_s}\epsilon_1^2\epsilon_2\log\left(c_s\right) \\
&- 2.0943544454802\left(\alpha_0^{c_s}\right)^2\epsilon_1\epsilon_2\log\left(c_s\right) \\
&- 6.3968711705769\epsilon_1^2\epsilon_2\epsilon_3\log\left(c_s\right)^2 \\
&- 0.22963715453852\alpha_1^{c_s}\epsilon_2\epsilon_3\log\left(c_s\right)^3 \\
&- 0.26586987330266\epsilon_2^2\epsilon_3\epsilon_4\log\left(c_s\right) \\
&- 0.13754355249956\alpha_0^{c_s}\alpha_1^{c_s}\epsilon_2\log\left(c_s\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 2.4338437485217\alpha_0^{c_s}\epsilon_1\epsilon_2^2\log\left(c_s\right) \\
&- 5.1785875133866\epsilon_1\epsilon_2^2\epsilon_3\log\left(c_s\right)^2 \\
&- 0.65309143635901\alpha_0^{c_s}\epsilon_2\epsilon_3^2\log\left(c_s\right)^3 \\
&- 1.806182872718\alpha_0^{c_s}\epsilon_1\epsilon_2^2\log\left(c_s\right)^3 \\
&- 0.23973258232289\alpha_1^{c_s}\epsilon_2\epsilon_3\log\left(c_s\right)^2 \\
&- 0.57598182564527\epsilon_1\epsilon_2\epsilon_3^2\log\left(c_s\right) \\
&- 1.2296371545385\epsilon_1\epsilon_2\epsilon_3^2\log\left(c_s\right)^3 \\
&- 0.079171041433698\epsilon_1\epsilon_2\epsilon_3\epsilon_4 \\
&- 0.041666666666667\epsilon_2\epsilon_3\epsilon_4^2\log\left(c_s\right)^4 \\
&- 0.33333333333333\epsilon_1^2\epsilon_2\epsilon_3\log\left(c_s\right)^4 \\
&- 2.959274309077\left(\alpha_0^{c_s}\right)^2\epsilon_1\epsilon_2\log\left(c_s\right)^3 \\
&- 0.41666666666667\epsilon_1\epsilon_2^2\epsilon_3\log\left(c_s\right)^4 \\
&- 0.48642476969235\epsilon_2^2\epsilon_3\epsilon_4\log\left(c_s\right)^3 \\
&- 1.9488360976258\alpha_0^{c_s}\epsilon_2^2\epsilon_3\log\left(c_s\right)^2 \\
&- 0.99681891832599\left(\alpha_0^{c_s}\right)^2\epsilon_2\epsilon_3\log\left(c_s\right) \\
&+ 0.25\epsilon_1\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right)^4 \\
&+ 0.60881560930639\alpha_0^{c_s}\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right) \\
&+ 3.6123657454361\alpha_0^{c_s}\epsilon_1\epsilon_2\epsilon_3\log\left(c_s\right)^3 \\
&+ 0.072524164035261\epsilon_2\epsilon_3\epsilon_4\epsilon_5\log\left(c_s\right)^2 \\
&+ 0.16666666666667\alpha_0^{c_s}\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right)^4 \\
&+ 2.6896298543859\alpha_0^{c_s}\epsilon_1\epsilon_2\epsilon_3\log\left(c_s\right) \\
&+ 0.21965598797644\alpha_0^{c_s}\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right)^2 \\
&+ 0.94917888129267\epsilon_1\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right)^2 \\
&+ 0.12160619242309\epsilon_2\epsilon_3\epsilon_4\epsilon_5\log\left(c_s\right)^3 \\
&+ 0.030669313835717\epsilon_2\epsilon_3\epsilon_4\epsilon_5\log\left(c_s\right) \\
&- 3.6472716829143\alpha_0^{c_s}\epsilon_1\epsilon_2\epsilon_3\log\left(c_s\right)^2 \\
&- 0.65309143635901\alpha_0^{c_s}\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right)^3 \\
&- 0.57598182564527\epsilon_1\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right) \\
&- 1.2296371545385\epsilon_1\epsilon_2\epsilon_3\epsilon_4\log\left(c_s\right)^3
\end{aligned}
$$

$$- 0.041666666666667\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \log(c_s)^4$$
$$- 0.66666666666667\alpha_0^{c_s}\epsilon_1\epsilon_2\epsilon_3 \log(c_s)^4$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2}\left[\sum_{d=0}^{4} F_{t,d} + O(\mathrm{SR}^5)\right].$$

Relative order 0.

$$F_{t,0} = 1.0$$

Relative order 1.

$$F_{t,1} = -0.54072569092296\epsilon_1$$

Relative order 2.

$$F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2$$

Relative order 3.

$$\begin{aligned} F_{t,3} = {} & 1.6057835635673\epsilon_1^3 + 1.8891060194179\epsilon_1^2\epsilon_2 \\ & - 0.18929040711048\epsilon_1\epsilon_2^2 - 0.18929040711048\epsilon_1\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 4.

$$\begin{aligned} F_{t,4} = {} & 2.9412556561294\epsilon_1^4 + 4.0859368676311\epsilon_1^3\epsilon_2 \\ & + 1.8087367742501\epsilon_1^2\epsilon_2^2 - 0.077715927178017\epsilon_1\epsilon_2^3 \\ & + 1.696905383796\epsilon_1^2\epsilon_2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3^2 \\ & - 0.23314778153405\epsilon_1\epsilon_2^2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3\epsilon_4 \end{aligned}$$

2 Consistency relations

2.1 $n = 0$

For $p = 4$, the absolute cutoff is $W = 5$:

$$\alpha_0^t = \sum_{d=1}^{5} R_{0,d} + O(\mathrm{SR}^6).$$

Absolute order 1.

$$R_{0,1} = -\frac{0.125r}{c_s}$$

Absolute order 2.

$$\begin{aligned} R_{0,2} = {} & \frac{0.0078125r^2}{c_s^2} + \frac{0.125\alpha_0^s r}{c_s} \\ & + \frac{0.28379535568268\alpha_0^{c_s} r}{c_s} - \frac{0.125\alpha_0^s r \log(c_s)}{c_s} \end{aligned}$$

Absolute order 3.

$$
\begin{aligned}
R_{0,3} = & -\frac{0.0030712246746565 r^3}{c_s^3} + \frac{0.125 \alpha_1^s r}{c_s} \\
& + \frac{0.33325502603924 \alpha_1^{c_s} r}{c_s} - \frac{0.10280837917801 \left(\alpha_0^s\right)^2 r}{c_s} \\
& - \frac{0.20825502603924 \left(\alpha_0^{c_s}\right)^2 r}{c_s} - \frac{0.041327094794504 \alpha_0^s r^2}{c_s^2} \\
& - \frac{0.061176514254839 \alpha_0^{c_s} r^2}{c_s^2} + \frac{0.015625 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.0625 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.125 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
& - \frac{0.0625 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
& - \frac{0.125 \alpha_1^s r \log\left(c_s\right)}{c_s} - \frac{0.46478233218736 \alpha_0^{c_s} \alpha_0^s r}{c_s} \\
& + \frac{0.28379535568268 \alpha_0^{c_s} \alpha_0^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

Absolute order 4.

$$
\begin{aligned}
R_{0,4} = & \frac{0.0014926374043055 r^4}{c_s^4} + \frac{0.125 \alpha_2^s r}{c_s} \\
& + \frac{0.076954635474829 \left(\alpha_0^{c_s}\right)^3 r}{c_s} + \frac{0.065244100954277 \left(\alpha_0^s\right)^3 r}{c_s} \\
& + \frac{0.36074999115751 \alpha_2^{c_s} r}{c_s} + \frac{0.036259602621199 \alpha_0^{c_s} r^3}{c_s^3} \\
& + \frac{0.024554893470954 \alpha_0^s r^3}{c_s^3} + \frac{0.10320649151137 \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
& + \frac{0.21414669687749 \left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} - \frac{0.11879209843874 \alpha_1^{c_s} r^2}{c_s^2} \\
& - \frac{0.060109233906372 \alpha_1^s r^2}{c_s^2} \\
& + \frac{0.015625 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
& + \frac{0.0625 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.0625 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.34966216656143 \alpha_0^{c_s} \alpha_0^s r^2}{c_s^2} \\
& + \frac{0.60410986672479 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} \\
& + \frac{0.041327094794504 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.46184803450948 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r}{c_s} \\
& - \frac{0.0078125 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.125\alpha_2^s r \log(c_s)}{c_s} \\
&- \frac{0.0092136740239694\alpha_0^s r^3 \log(c_s)}{c_s^3} \\
&- \frac{0.10280837917801 \left(\alpha_0^s\right)^3 r \log(c_s)}{c_s} \\
&- \frac{0.51223106590429\alpha_0^{c_s} \alpha_1^s r}{c_s} - \frac{0.26465926210717\alpha_0^{c_s} \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.58335484068555\alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.082654189589007 \left(\alpha_0^s\right)^2 r^2 \log(c_s)}{c_s^2} \\
&- \frac{0.020833333333333\alpha_2^s r \log(c_s)^3}{c_s} \\
&- \frac{0.020833333333333 \left(\alpha_0^s\right)^3 r \log(c_s)^3}{c_s} \\
&- \frac{0.23087387125098\alpha_0^s \alpha_1^s r}{c_s} \\
&+ \frac{0.0625\alpha_0^s \alpha_1^s r \log(c_s)^3}{c_s} \\
&+ \frac{0.14189767784134\alpha_0^{c_s} \left(\alpha_0^s\right)^2 r \log(c_s)^2}{c_s} \\
&+ \frac{0.46478233218736\alpha_0^{c_s} \alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{0.33061675835603\alpha_0^s \alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{0.33325502603924\alpha_0^s \alpha_1^{c_s} r \log(c_s)}{c_s} \\
&- \frac{0.1875\alpha_0^s \alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{0.14189767784134\alpha_0^{c_s} \alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{0.20825502603924 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r \log(c_s)}{c_s} \\
&- \frac{0.46478233218736\alpha_0^{c_s} \left(\alpha_0^s\right)^2 r \log(c_s)}{c_s} \\
&- \frac{0.12235302850968\alpha_0^{c_s} \alpha_0^s r^2 \log(c_s)}{c_s^2}
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{0,5} = &- \frac{0.00093280100983415 r^5}{c_s^5} + \frac{0.125\alpha_3^s r}{c_s} \\
&+ \frac{0.077998779670661\alpha_1^{c_s} r^3}{c_s^3} + \frac{0.04315951309091\alpha_1^s r^3}{c_s^3} \\
&+ \frac{0.37743788705055\alpha_3^{c_s} r}{c_s} + \frac{0.22678710229221 \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&- \frac{0.038536574146577 \left(\alpha_0^s\right)^4 r}{c_s} - \frac{0.018206214478227\alpha_0^s r^4}{c_s^4} \\
&- \frac{0.17317013837934\alpha_2^{c_s} r^2}{c_s^2} - \frac{0.46393123483107 \left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.077020534433109\alpha_2^s r^2}{c_s^2} - \frac{0.044182861011306\left(\alpha_0^{c_s}\right)^4 r}{c_s} \\
&- \frac{0.14103704832205\left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.18695785481671\left(\alpha_0^s\right)^3 r^2}{c_s^2} \\
&- \frac{0.025790978878929\alpha_0^{c_s} r^4}{c_s^4} - \frac{0.21229467061039\left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
&- \frac{0.10777687087114\left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&+ \frac{0.0625\alpha_3^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.065244100954277\left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.77902151158975\alpha_0^{c_s}\alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.33322659757276\alpha_0^s\alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.2807614807417\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.073664680412863\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.0046068370119847\alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.0059705496172218\alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.91440223214728\left(\alpha_0^{c_s}\right)^2\alpha_1^s r}{c_s} \\
&+ \frac{0.25691700713838\left(\alpha_0^s\right)^2\alpha_1^s r}{c_s} \\
&+ \frac{0.585172074201116\alpha_0^{c_s}\alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.64688363806266\left(\alpha_0^s\right)^2\alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.0026041666666667\alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.0052083333333333\alpha_3^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.010416666666667\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.020833333333333\left(\alpha_0^s\right)^4 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.67279698790137\alpha_0^s\alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.20641298302274\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.23087387125098\left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$+\frac{0.060109233906372\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.015625\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.125\alpha_3^s r\log\left(c_s\right)}{c_s}-\frac{0.70604739146173\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$-\frac{0.24116427546487\alpha_0^s\alpha_2^s r}{c_s}-\frac{0.67016777237071\alpha_0^s\alpha_2^{c_s} r}{c_s}$$
$$-\frac{0.37631034727105\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r}{c_s}$$
$$-\frac{0.544109075112 08\alpha_0^{c_s}\alpha_2^s r}{c_s}-\frac{0.33187406198245\alpha_0^{c_s}\alpha_0^s r^3}{c_s^3}$$
$$-\frac{0.051404189589007\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{1.0075354959104\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2}{c_s^2}$$
$$-\frac{0.020663547397252\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.082654189589007\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.16530837917801\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.0052083333333333\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.020833333333333\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{1.3905396123749\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^2}{c_s^2}$$
$$-\frac{0.024554893470954\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$-\frac{0.025650784758331\alpha_0^{c_s}\alpha_2^{c_s} r}{c_s}$$
$$-\frac{0.41785905564102\left(\alpha_0^{c_s}\right)^3\alpha_0^s r}{c_s}$$
$$-\frac{0.8774300995702\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r}{c_s}$$
$$-\frac{0.013820511035954\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.03125\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.047299225947114\alpha_0^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.047299225947114\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.076954635474829\left(\alpha_0^{c_s}\right)^3\alpha_0^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.2543932705273\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r}{c_s}$$
$$+\frac{0.69932433312286\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$

$$+\frac{0.36074999115751\alpha_0^s\alpha_2^{c_s}r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.12398128438351\alpha_0^s\alpha_1^sr^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.2408463078633\alpha_0^{c_s}\alpha_0^s\alpha_1^sr}{c_s}$$
$$+\frac{0.10412751301962\left(\alpha_0^{c_s}\right)^2\alpha_1^sr\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.51223106590429\alpha_0^{c_s}\alpha_2^sr\log\left(c_s\right)}{c_s}$$
$$+\frac{0.58335484068555\alpha_1^{c_s}\alpha_1^sr\log\left(c_s\right)}{c_s}$$
$$+\frac{0.60410986672479\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.31952094794504\left(\alpha_0^s\right)^2\alpha_1^sr\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.16662751301962\left(\alpha_0^s\right)^2\alpha_1^{c_s}r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.083333333333333\alpha_0^s\alpha_2^sr\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.35587387125098\alpha_0^s\alpha_2^sr\log\left(c_s\right)}{c_s}$$
$$+\frac{0.42829339375497\left(\alpha_0^{c_s}\right)^2\alpha_0^sr^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.1087788078636\alpha_0^{c_s}\alpha_0^sr^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.46184803450948\alpha_0^{c_s}\left(\alpha_0^s\right)^3r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.061176514254839\alpha_0^{c_s}\alpha_1^sr^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.015625\alpha_0^s\alpha_1^sr^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.125\left(\alpha_0^s\right)^2\alpha_1^sr\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.34966216656143\alpha_0^{c_s}\alpha_1^sr^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.10412751301962\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.42660617411381\left(\alpha_0^s\right)^2\alpha_1^sr\log\left(c_s\right)}{c_s}$$
$$-\frac{0.23239116609368\alpha_0^{c_s}\alpha_2^sr\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.23239116609368\alpha_0^{c_s}\left(\alpha_0^s\right)^3r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.58335484068555\left(\alpha_0^s\right)^2\alpha_1^{c_s}r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.60410986672479\left(\alpha_0^{c_s}\right)^2\alpha_1^sr\log\left(c_s\right)}{c_s}$$
$$-\frac{0.32663145083548\alpha_0^s\alpha_1^sr^2\log\left(c_s\right)}{c_s^2}$$

$$- \frac{0.16662751301962\alpha_1^{c_s}\alpha_1^s r \log(c_s)^2}{c_s}$$

$$- \frac{0.020833333333333\alpha_0^s\alpha_2^s r \log(c_s)^4}{c_s}$$
$$- \frac{0.22780837917801\alpha_0^s\alpha_2^s r \log(c_s)^2}{c_s}$$
$$- \frac{0.23758419687749\alpha_0^s\alpha_1^{c_s} r^2 \log(c_s)}{c_s^2}$$
$$- \frac{0.12235302850968\alpha_0^{c_s}(\alpha_0^s)^2 r^2 \log(c_s)^2}{c_s^2}$$
$$+ \frac{0.69717349828103\alpha_0^{c_s}\alpha_0^s\alpha_1^s r \log(c_s)^2}{c_s}$$
$$- \frac{0.14189767784134\alpha_0^{c_s}\alpha_0^s\alpha_1^s r \log(c_s)^3}{c_s}$$
$$- \frac{0.26465926210717\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r \log(c_s)}{c_s}$$
$$- \frac{1.4359271349233\alpha_0^{c_s}\alpha_0^s\alpha_1^s r \log(c_s)}{c_s}$$

2.2 $n = 1$

For $p = 4$, the absolute cutoff is $W = 6$:

$$\alpha_1^t = \sum_{d=2}^{6} R_{1,d} + O(\mathrm{SR}^7).$$

Absolute order 2.

$$R_{1,2} = \frac{0.015625r^2}{c_s^2} + \frac{0.125\alpha_0^{c_s} r}{c_s} + \frac{0.125\alpha_0^s r}{c_s}$$

Absolute order 3.

$$R_{1,3} = -\frac{0.0029296875r^3}{c_s^3} + \frac{0.125\alpha_1^s r}{c_s} + \frac{0.28379535568268\alpha_1^{c_s} r}{c_s} - \frac{0.125(\alpha_0^s)^2 r}{c_s} - \frac{0.046875\alpha_0^s r^2}{c_s^2} - \frac{0.28379535568268(\alpha_0^{c_s})^2 r}{c_s} - \frac{0.086573838920671\alpha_0^{c_s} r^2}{c_s^2} + \frac{0.03125\alpha_0^s r^2 \log(c_s)}{c_s^2} + \frac{0.125(\alpha_0^s)^2 r \log(c_s)}{c_s} - \frac{0.125\alpha_1^s r \log(c_s)}{c_s} - \frac{0.53379535568268\alpha_0^{c_s}\alpha_0^s r}{c_s} + \frac{0.125\alpha_0^{c_s}\alpha_0^s r \log(c_s)}{c_s}$$

Absolute order 4.

$$R_{1,4} = \frac{0.0016576826498282r^4}{c_s^4} + \frac{0.125\alpha_2^s r}{c_s}$$

$$
\begin{aligned}
&+ \frac{0.12398128438351\left(\alpha_0^s\right)^2 r^2}{c_s^2} + \frac{0.10280837917801\left(\alpha_0^s\right)^3 r}{c_s} \\
&+ \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^3 r}{c_s} \\
&+ \frac{0.25495658892655\left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} + \frac{0.038806320518347\alpha_0^{c_s} r^3}{c_s^3} \\
&+ \frac{0.33325502603924\alpha_2^{c_s} r}{c_s} + \frac{0.027641022071908\alpha_0^s r^3}{c_s^3} \\
&- \frac{0.14449027076465\alpha_1^{c_s} r^2}{c_s^2} - \frac{0.072577094794504\alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.03125\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.0625\alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.046875\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.69259071136537\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r}{c_s} \\
&+ \frac{0.4077766400662\alpha_0^{c_s}\alpha_0^s r^2}{c_s^2} \\
&+ \frac{0.95683271390928\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} \\
&- \frac{0.015625\alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.125\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.125\left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.09375\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.0087890625\alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} - \frac{0.33061675835603\alpha_0^s\alpha_1^s r}{c_s} \\
&- \frac{0.71478233218736\alpha_0^{c_s}\alpha_1^s r}{c_s} - \frac{0.74976507811772\alpha_0^{c_s}\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.79803735822659\alpha_0^s\alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.375\alpha_0^s\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.28379535568268\alpha_0^s\alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.53379535568268\alpha_0^{c_s}\alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0625\alpha_0^{c_s}\alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.1875\alpha_0^s\alpha_1^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.17314767784134\alpha_0^{c_s} \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.53379535568268\alpha_0^{c_s} \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{1,5} = &- \frac{0.0010528680915447 r^5}{c_s^5} + \frac{0.125\alpha_3^s r}{c_s} \\
&+ \frac{0.088617304208522\alpha_1^{c_s} r^3}{c_s^3} + \frac{0.36074999115751\alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.050025543685844\alpha_1^s r^3}{c_s^3} - \frac{0.56573567099871\left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2} \\
&- \frac{0.076954635474829\left(\alpha_0^{c_s}\right)^4 r}{c_s} - \frac{0.091359233906372\alpha_2^s r^2}{c_s^2} \\
&- \frac{0.065244100954277\left(\alpha_0^s\right)^4 r}{c_s} - \frac{0.20897959622812\alpha_2^{c_s} r^2}{c_s^2} \\
&- \frac{0.24604962823258\left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
&- \frac{0.13027434666455\left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&- \frac{0.26465926210717\left(\alpha_1^{c_s}\right)^2 r}{c_s} - \frac{0.021075080402355\alpha_0^s r^4}{c_s^4} \\
&- \frac{0.23087387125098\left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.029498514193987\alpha_0^{c_s} r^4}{c_s^4} \\
&- \frac{0.24842610305581\left(\alpha_0^s\right)^3 r^2}{c_s^2} \\
&+ \frac{0.0625\alpha_3^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.00439453125\alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{1.5811232648164\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.24796256876702\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.082923066215725\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.71021433456542\alpha_0^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.10280837917801\left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.42660617411381\left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.045202875195\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.072577094794504\alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.33061675835603\left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.0052083333333333\alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.020833333333333\left(\alpha_0^s\right)^4 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.020833333333333\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.81639883012011\alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.41559991864823\alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.0066307305993129\alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.92119486345495\alpha_0^{c_s} \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.49552316853166\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.0625\left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125\alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0234375\alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.09375\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.1875\left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.01318359375\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{0.62990051464178\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{0.62540925326469\alpha_0^{c_s} \alpha_2^{c_s} r}{c_s} \\
&- \frac{0.39180031621678\alpha_0^{c_s} \alpha_0^s r^3}{c_s^3} - \frac{1.0955859065898\alpha_1^{c_s} \alpha_1^s r}{c_s} \\
&- \frac{1.5307402334216\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.7168681131348\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{0.88931952823886\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{1.2784021784808\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{0.020833333333333\alpha_3^s r \log\left(c_s\right)^3}{c_s} - \frac{0.35587387125098\alpha_0^s \alpha_2^s r}{c_s} \\
&- \frac{0.76223106590429\alpha_0^{c_s} \alpha_2^s r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.027641022071908\alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.94410483184306\alpha_0^s \alpha_2^{c_s} r}{c_s} \\
&+ \frac{0.375\left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.140625\alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.14189767784134\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.14189767784134\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.086573838920671\alpha_0^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.69259071136537\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.11641896155504\alpha_0^{c_s} \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{1.9974177645303\alpha_0^{c_s} \alpha_0^s \alpha_1^s r}{c_s} \\
&+ \frac{0.5099131778531\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{2.3894888622815\alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.33325502603924\alpha_0^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.020833333333333\alpha_0^{c_s} \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.020833333333333\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.083333333333333\alpha_0^s \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.71478233218736\alpha_0^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.79803735822659\alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.81555328013239\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.45561675835603\alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.95683271390928\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.03125\alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&- \frac{0.125\left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.25\alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.14189767784134\alpha_1^{c_s}\alpha_1^s r\log(c_s)^2}{c_s} \\
&- \frac{0.14189767784134\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r\log(c_s)^2}{c_s} \\
&- \frac{0.17314767784134\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log(c_s)^2}{c_s^2} \\
&- \frac{0.26689767784134\alpha_0^{c_s}\alpha_2^s r\log(c_s)^2}{c_s} \\
&- \frac{0.26689767784134\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log(c_s)^2}{c_s} \\
&- \frac{0.2889805415293\alpha_0^s\alpha_1^{c_s} r^2\log(c_s)}{c_s^2} \\
&- \frac{0.63904189589007\left(\alpha_0^s\right)^2\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{0.39311675835603\alpha_0^s\alpha_1^s r^2\log(c_s)}{c_s^2} \\
&- \frac{0.79803735822659\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log(c_s)}{c_s} \\
&- \frac{0.4077766400662\alpha_0^{c_s}\alpha_1^s r^2\log(c_s)}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.95683271390928\left(\alpha_0^{c_s}\right)^2\alpha_1^s r\log(c_s)}{c_s} \\
&+ \frac{0.80069303352403\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log(c_s)^2}{c_s} \\
&- \frac{0.0625\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log(c_s)^3}{c_s} \\
&- \frac{2.0999637549181\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{0.74976507811772\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r\log(c_s)}{c_s}
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{1,6} = &\frac{0.00079786539870909r^6}{c_s^6} + \frac{0.125\alpha_4^s r}{c_s} \\
&+ \frac{0.26720808608064\left(\alpha_0^{c_s}\right)^2 r^4}{c_s^4} \\
&+ \frac{0.038536574146577\left(\alpha_0^s\right)^5 r}{c_s} + \frac{0.41043044123784\left(\alpha_0^s\right)^4 r^2}{c_s^2} \\
&+ \frac{1.0259570770776\left(\alpha_0^{c_s}\right)^4 r^2}{c_s^2} \\
&+ \frac{0.14689479099556\left(\alpha_0^s\right)^2 r^4}{c_s^4} + \frac{0.018569586968286\alpha_0^s r^5}{c_s^5} \\
&+ \frac{1.033204312895\left(\alpha_0^{c_s}\right)^3 r^3}{c_s^3} + \frac{0.074971901003326\alpha_2^s r^3}{c_s^3} \\
&+ \frac{0.15139265948067\alpha_2^{c_s} r^3}{c_s^3} + \frac{0.044182861011306\left(\alpha_0^{c_s}\right)^5 r}{c_s} \\
&+ \frac{0.025257209491638\alpha_0^{c_s} r^5}{c_s^5} + \frac{0.37743788705055\alpha_4^{c_s} r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.38411085965327\left(\alpha_1^s\right)^2 r^2}{c_s^2} \\
&+ \frac{0.83338364839711\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2} \\
&+ \frac{0.44641771599844\left(\alpha_0^s\right)^3 r^3}{c_s^3} - \frac{0.043199996920584\alpha_1^s r^4}{c_s^4} \\
&- \frac{0.10827053443311\alpha_3^s r^2}{c_s^2} - \frac{0.26752961014198\alpha_3^{c_s} r^2}{c_s^2} \\
&- \frac{0.072596626026171\alpha_1^{c_s} r^4}{c_s^4} + \frac{0.0078125\alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.0078125\left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^4}{c_s^2} \\
&+ \frac{0.0625\alpha_4^s r \log\left(c_s\right)^2}{c_s} + \frac{1.0062984846724\alpha_0^{c_s}\alpha_2^s r^2}{c_s^2} \\
&+ \frac{1.0276061682351\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&+ \frac{0.091359233906372\alpha_3^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{1.3365328286442\alpha_0^{c_s}\alpha_2^{c_s} r^2}{c_s^2} \\
&+ \frac{2.0767524199373\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&+ \frac{0.4279234198261\alpha_1^{c_s}\alpha_2^{c_s} r}{c_s} \\
&+ \frac{0.48009102237191\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r}{c_s} \\
&+ \frac{0.24796256876702\left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{6.3868435353963\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&+ \frac{0.051404189589007\left(\alpha_0^s\right)^5 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.7155884813507\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r}{c_s} \\
&+ \frac{1.899399021936\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r}{c_s} \\
&+ \frac{0.12438459932359\left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.53899655212715\left(\alpha_0^{c_s}\right)^4 \alpha_0^s r}{c_s} \\
&+ \frac{0.5788087353052\alpha_0^s\alpha_2^s r^2}{c_s^2} \\
&+ \frac{4.7974121733276\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2}{c_s^2} \\
&+ \frac{0.30641226550003\left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r}{c_s} \\
&+ \frac{1.2768667055421\alpha_0^s\alpha_2^{c_s} r^2}{c_s^2}
\end{aligned}
$$

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\begin{aligned}
&+ \frac{0.65487106259881\alpha_0^s \left(\alpha_1^s\right)^2 r}{c_s} \\
&+ \frac{1.3170514104334 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r}{c_s} \\
&+ \frac{0.0052083333333333\alpha_4^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.0052083333333333 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.010416666666667 \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^4}{c_s^2} \\
&+ \frac{0.33473585919119\alpha_0^{c_s} \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&+ \frac{0.021075080402355\alpha_1^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{2.7660197079164 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3}{c_s^3} \\
&+ \frac{2.9375382620807\alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2}{c_s^2} \\
&+ \frac{1.5177946664778\alpha_1^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.19655837917801 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{1.6127572274363\alpha_0^{c_s} \left(\alpha_1^s\right)^2 r}{c_s} \\
&+ \frac{0.42065015124113\alpha_0^{c_s} \alpha_0^s r^4}{c_s^4} \\
&+ \frac{0.013261461198626 \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)^2}{c_s^4} \\
&+ \frac{0.013820511035954\alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{1.9707423731637 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r}{c_s} \\
&+ \frac{0.49808128260325 \left(\alpha_0^s\right)^2 \alpha_2^s r}{c_s} \\
&- \frac{0.0625 \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&- \frac{0.125\alpha_4^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.00146484375\alpha_2^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&- \frac{0.046875 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&- \frac{0.01318359375 \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^3}{c_s^3} \\
&- \frac{0.45749292478693 \left(\alpha_0^{c_s}\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.065244100954277 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

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\begin{aligned}
&- \frac{1.0476056594213 \alpha_0^s \alpha_3^{c_s} r}{c_s} \\
&- \frac{1.0231939853337 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.36616427546487 \alpha_0^s \alpha_3^s r}{c_s} - \frac{0.79146979525308 \alpha_0^{c_s} \alpha_1^s r^3}{c_s^3} \\
&- \frac{0.52323837210896 \alpha_1^s \alpha_2^s r}{c_s} - \frac{1.1192217785737 \alpha_0^{c_s} \alpha_1^{c_s} r^3}{c_s^3} \\
&- \frac{1.3762151638324 \alpha_1^s \alpha_2^{c_s} r}{c_s} - \frac{0.79410907511208 \alpha_0^{c_s} \alpha_3^s r}{c_s} \\
&- \frac{0.40308867180888 \alpha_0^{c_s} \alpha_3^{c_s} r}{c_s} \\
&- \frac{3.4819478369654 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{1.9363711545131 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{0.51344198472507 \alpha_0^s \alpha_1^s r^3}{c_s^3} \\
&- \frac{0.036288547397252 \alpha_3^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{1.2501564665738 \alpha_1^{c_s} \alpha_2^s r}{c_s} \\
&- \frac{0.0013020833333333 \alpha_3^s r^2 \log\left(c_s\right)^4}{c_s^2} \\
&- \frac{0.020833333333333 \alpha_4^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.020833333333333 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.0052643404577234 \alpha_0^s r^5 \log\left(c_s\right)}{c_s^5} \\
&- \frac{0.084300321609422 \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)}{c_s^4} \\
&- \frac{2.7292119095603 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{1.3749765739875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{0.39082303999365 \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.050025543685844 \alpha_2^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.41106330372469 \left(\alpha_0^s\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{3.309049304958 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{0.41559991864823 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.0033153652996565 \alpha_1^s r^4 \log\left(c_s\right)^2}{c_s^4}
\end{aligned}
$$

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\begin{aligned}
&- \frac{0.91770607360615\alpha_0^s \alpha_1^{c_s} r^3}{c_s^3} \\
&- \frac{0.49685220611163 \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.015625\alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.1875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.078125\alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.01318359375\alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.047299225947114\alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.047299225947114 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.34629535568268\alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.5764781464639 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{1.049981877459\alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.39180031621678\alpha_0^{c_s} \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.41062532438663\alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{1.0955859065898\alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{2.3819935296913\alpha_0^{c_s} \alpha_0^s \alpha_2^s r}{c_s} \\
&+ \frac{0.36074999115751\alpha_0^s \alpha_3^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.80435027506809\alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.26913547397252\alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{4.3725999669691\alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r}{c_s} \\
&+ \frac{0.54732932712305 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.17462844233256\alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.088965892613781\alpha_0^{c_s} \alpha_3^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.10412751301962 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.7168681131348 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

$$+\frac{0.88931952823886\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.5099131778531\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.045202875195\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.26585191262557\alpha_0^s\alpha_1^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{2.3109618188138\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r}{c_s}$$
$$+\frac{0.14449027076465\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.16662751301962\left(\alpha_0^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.0052083333333333\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.010416666666667\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.68758257793092\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.76223106590429\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.2406609754504\alpha_0^s\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{0.2038883200331\alpha_0^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.81555328013239\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.6327976602402\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.81762161375293\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.052083333333333\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.10416666666667\alpha_0^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.20833333333333\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.20833333333333\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.94410483184306\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.47841635695464\left(\alpha_0^{c_s}\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.47841635695464\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.48087387125098\alpha_0^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.03125\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^4}{c_s^2}$$

$$- \frac{0.0625\alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.3125\alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.047299225947114 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.047299225947114 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.26689767784134\alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.62990051464178\alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.1314713419974 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.076954635474829 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{4.8640065775119\alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2}{c_s^2}$$
$$- \frac{0.78248004536479 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.5811232648164 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.55842513753404\alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.1754009486504\alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{0.71021433456542\alpha_0^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.088965892613781\alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{3.0533005309287\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s}$$
$$- \frac{1.5307402334216 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.10412751301962 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.41795919245624\alpha_0^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{3.4337362262696 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.88931952823886 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.058209480777521\alpha_0^{c_s} \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{0.12438459932359\alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{1.9974177645303\alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$

$$- \frac{0.25495658892655 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.52513770630106 \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.26465926210717 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{4.2394919484159 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s}$$
$$- \frac{1.0840862194786 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.2889805415293 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.29030837917801 \alpha_0^s \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$

$$- \frac{0.59831838646097 \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{2.5568043569616 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.16662751301962 \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.0052083333333333 \alpha_0^{c_s} \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{5.3460358740743 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r}{c_s}$$
$$- \frac{0.35739116609368 \alpha_0^{c_s} \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.73814888469775 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{3.0533911802992 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r}{c_s}$$
$$- \frac{0.76506061150656 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.3990186791133 \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.3990186791133 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.81639883012011 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.026041666666667 \alpha_0^s \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.052083333333333 \alpha_1^s \alpha_2^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.052083333333333 \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.20833333333333 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{6.9429524392027 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2}$$

$$
\begin{aligned}
&- \frac{0.02885794630689\alpha_0^{c_s}\alpha_2^s r^2 \log(c_s)^3}{c_s^2} \\
&- \frac{0.11543178522756\alpha_0^{c_s}(\alpha_0^s)^3 r^2 \log(c_s)^3}{c_s^2} \\
&- \frac{0.11799405677595\alpha_0^{c_s}\alpha_0^s r^4 \log(c_s)}{c_s^4} \\
&- \frac{0.94410483184306(\alpha_0^s)^2\alpha_2^{c_s} r \log(c_s)}{c_s} \\
&+ \frac{0.14189767784134(\alpha_0^{c_s})^2\alpha_0^s\alpha_1^s r \log(c_s)^3}{c_s} \\
&+ \frac{0.17314767784134\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^2 \log(c_s)^3}{c_s^2} \\
&+ \frac{0.53379535568268\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^s r \log(c_s)^3}{c_s} \\
&+ \frac{1.1970560373399\alpha_0^s\alpha_1^{c_s}\alpha_1^s r \log(c_s)^2}{c_s} \\
&+ \frac{3.9772330260924\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^2 \log(c_s)}{c_s^2} \\
&+ \frac{4.6426037316597(\alpha_0^{c_s})^2\alpha_0^s\alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{2.3894888622815\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^{c_s} r \log(c_s)}{c_s} \\
&+ \frac{0.020833333333333\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log(c_s)^4}{c_s} \\
&+ \frac{1.4073730435527\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log(c_s)^2}{c_s} \\
&+ \frac{0.37488253905886\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r \log(c_s)^2}{c_s} \\
&+ \frac{1.8423897269099\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r^2 \log(c_s)}{c_s^2} \\
&+ \frac{3.8871193084556\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{0.49552316853166(\alpha_0^{c_s})^2\alpha_0^s\alpha_1^{c_s} r \log(c_s)}{c_s} \\
&- \frac{0.03125\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^s r \log(c_s)^4}{c_s} \\
&- \frac{0.14189767784134\alpha_0^s\alpha_1^{c_s}\alpha_1^s r \log(c_s)^3}{c_s} \\
&- \frac{0.62540925326469\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r \log(c_s)}{c_s} \\
&- \frac{2.0888679445071\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{2.7596488304346\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log(c_s)}{c_s} \\
&- \frac{0.35586357045512\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log(c_s)^3}{c_s} \\
&- \frac{2.3894888622815\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r \log(c_s)}{c_s} \\
&- \frac{1.2233299201986\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^2 \log(c_s)^2}{c_s^2}
\end{aligned}
$$

$$- \frac{1.4352490708639 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.37488253905886 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{3.1859916569799 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s}$$

2.3 $n = 2$

For $p = 4$, the absolute cutoff is $W = 7$:

$$\alpha_2^t = \sum_{d=3}^{7} R_{2,d} + O(\mathrm{SR}^8).$$

Absolute order 3.

$$\begin{aligned} R_{2,3} = &- \frac{0.00390625 r^3}{c_s^3} + \frac{0.125 \alpha_1^{c_s} r}{c_s} \\ &+ \frac{0.125 \alpha_1^s r}{c_s} - \frac{0.125 \left(\alpha_0^{c_s}\right)^2 r}{c_s} \\ &- \frac{0.125 \left(\alpha_0^s\right)^2 r}{c_s} - \frac{0.046875 \alpha_0^{c_s} r^2}{c_s^2} \\ &- \frac{0.046875 \alpha_0^s r^2}{c_s^2} - \frac{0.25 \alpha_0^{c_s} \alpha_0^s r}{c_s} \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{2,4} = &\frac{0.0013427734375 r^4}{c_s^4} + \frac{0.125 \alpha_2^s r}{c_s} \\ &+ \frac{0.125 \left(\alpha_0^s\right)^3 r}{c_s} + \frac{0.125 \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\ &+ \frac{0.025390625 \alpha_0^s r^3}{c_s^3} + \frac{0.28379535568268 \alpha_2^{c_s} r}{c_s} \\ &+ \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^3 r}{c_s} \\ &+ \frac{0.24409651676201 \left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} + \frac{0.040277689595252 \alpha_0^{c_s} r^3}{c_s^3} \\ &- \frac{0.0625 \alpha_1^s r^2}{c_s^2} - \frac{0.12204825838101 \alpha_1^{c_s} r^2}{c_s^2} \\ &+ \frac{0.046875 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\ &+ \frac{0.94259071136537 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} \\ &+ \frac{0.78379535568268 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r}{c_s} \\ &+ \frac{0.41597151676201 \alpha_0^{c_s} \alpha_0^s r^2}{c_s^2} - \frac{0.125 \alpha_2^s r \log\left(c_s\right)}{c_s} \\ &- \frac{0.125 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\ &- \frac{0.01171875 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\ &- \frac{0.09375 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} - \frac{0.375 \alpha_0^s \alpha_1^s r}{c_s} \end{aligned}$$

$$
\begin{aligned}
&- \frac{0.81759071136537\alpha_0^s \alpha_1^{c_s} r}{c_s} - \frac{0.78379535568268\alpha_0^{c_s} \alpha_1^s r}{c_s} \\
&- \frac{0.85138606704805\alpha_0^{c_s} \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.125\alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.25\alpha_0^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.375\alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.125\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.25\alpha_0^{c_s} \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.09375\alpha_0^{c_s} \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{2,5} = &- \frac{0.00099348164677838 r^5}{c_s^5} + \frac{0.125\alpha_3^s r}{c_s} \\
&+ \frac{0.050668108270534\alpha_1^s r^3}{c_s^3} + \frac{0.33325502603924\alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.087560258989506\alpha_1^{c_s} r^3}{c_s^3} - \frac{0.24566880066715\left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
&- \frac{0.13499234851023\left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&- \frac{0.28928966356152\left(\alpha_0^s\right)^3 r^2}{c_s^2} - \frac{0.020502543360439\alpha_0^s r^4}{c_s^4} \\
&- \frac{0.10280837917801\left(\alpha_0^s\right)^4 r}{c_s} - \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^4 r}{c_s} \\
&- \frac{0.027325781299929\alpha_0^{c_s} r^4}{c_s^4} - \frac{0.64251673826997\left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2} \\
&- \frac{0.33061675835603\left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.088202094794504\alpha_2^s r^2}{c_s^2} \\
&- \frac{0.18614714901955\alpha_2^{c_s} r^2}{c_s^2} - \frac{0.74976507811772\left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&+ \frac{0.0625\alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} + \frac{0.0625\alpha_3^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.125\left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.25\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.005859375\alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.375\left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.00537109375\alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.076171875\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{2.2054104017793 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.490628069592 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.0148110363091 \alpha_0^{c_s} \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.63904189589007 \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.3745301562354 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.74025304063896 \alpha_0^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.87364314908906 \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.46569385315053 \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&- \frac{0.0625 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125 \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0234375 \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.09375 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.1875 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.017578125 \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{0.025390625 \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.92039909054338 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{1.1312923842658 \alpha_0^s \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.4488830956312 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{2.1832187809573 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.083020104157 \alpha_0^{c_s} \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.4169873656614 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{1.512819690414 \alpha_1^{c_s} \alpha_1^s r}{c_s} - \frac{0.38983219984103 \alpha_0^{c_s} \alpha_0^s r^3}{c_s^3} \\
&- \frac{0.45561675835603 \alpha_0^s \alpha_2^s r}{c_s} \\
&- \frac{1.8893109571319 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{0.96478233218736 \alpha_0^{c_s} \alpha_2^s r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.0625 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.5 \alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.046875 \alpha_0^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.375 \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.140625 \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.28379535568268 \alpha_0^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.81759071136537 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.94259071136537 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.78379535568268 \alpha_0^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.78379535568268 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.48819303352403 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.83194303352403 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.12083306878576 \alpha_0^{c_s} \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{2.8055805132741 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r}{c_s} \\
&+ \frac{3.7452632198456 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.0625 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.0625 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125 \alpha_0^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.25 \alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.09375 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.375 \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.75 \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.81759071136537\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{0.94259071136537\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.24409651676201\alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.41597151676201\alpha_0^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.375\alpha_0^{c_s} \alpha_0^s \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.85138606704805\alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{2.3513860670481\alpha_0^{c_s} \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{2,6} = &\frac{0.0007834832445242 r^6}{c_s^6} + \frac{0.125\alpha_4^s r}{c_s} \\
&+ \frac{0.076954635474829\left(\alpha_0^{c_s}\right)^5 r}{c_s} + \frac{0.065244100954277\left(\alpha_0^s\right)^5 r}{c_s} \\
&+ \frac{0.51447193930681\left(\alpha_0^s\right)^3 r^3}{c_s^3} + \frac{0.018883632267472\alpha_0^s r^5}{c_s^5} \\
&+ \frac{0.36074999115751\alpha_4^{c_s} r}{c_s} + \frac{1.0488534998662\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2} \\
&+ \frac{0.15473919538016\alpha_2^{c_s} r^3}{c_s^3} + \frac{0.077748164662437\alpha_2^s r^3}{c_s^3} \\
&+ \frac{1.2689136192411\left(\alpha_0^{c_s}\right)^4 r^2}{c_s^2} + \frac{0.025194043126048\alpha_0^{c_s} r^5}{c_s^5} \\
&+ \frac{0.5388653261447\left(\alpha_0^s\right)^4 r^2}{c_s^2} \\
&+ \frac{0.27180459639332\left(\alpha_0^{c_s}\right)^2 r^4}{c_s^4} \\
&+ \frac{1.1085144407626\left(\alpha_0^{c_s}\right)^3 r^3}{c_s^3} \\
&+ \frac{0.15691226448912\left(\alpha_0^s\right)^2 r^4}{c_s^4} \\
&+ \frac{0.4600841525546\left(\alpha_1^s\right)^2 r^2}{c_s^2} - \frac{0.10698423390637\alpha_3^s r^2}{c_s^2} \\
&- \frac{0.044329625847117\alpha_1^s r^4}{c_s^4} - \frac{0.072500515380315\alpha_1^{c_s} r^4}{c_s^4} \\
&- \frac{0.25407334512281\alpha_3^{c_s} r^2}{c_s^2} + \frac{0.0078125\alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.0625\alpha_4^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^5 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.25\left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.1875\left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}
\end{aligned}
$$

$$+\frac{0.0107421875\left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{0.0126953125\alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.1142578125\left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{1.4391217537063\alpha_0^s \alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{5.812063941528\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2}{c_s^2}$$
$$+\frac{7.7501463158245\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2}{c_s^2}$$
$$+\frac{0.78248004536479\left(\alpha_0^s\right)^2 \alpha_2^s r}{c_s}$$
$$+\frac{1.111888199871\alpha_0^{c_s} \alpha_2^s r^2}{c_s^2}$$
$$+\frac{1.1745291897529\left(\alpha_0^{c_s}\right)^4 \alpha_0^s r}{c_s}$$
$$+\frac{1.2557055991705\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{1.7406927007245\alpha_1^{c_s} \alpha_1^s r^2}{c_s^2}$$
$$+\frac{1.1209324217963\left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r}{c_s}$$
$$+\frac{0.020502543360439\alpha_1^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$+\frac{0.79795299477407\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r}{c_s}$$
$$+\frac{3.6749700194140\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2}{c_s^2}$$
$$+\frac{1.0840862194786\alpha_0^s\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{2.3085962344727\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3}{c_s^3}$$
$$+\frac{2.5589083941373\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{2.6541481243887\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{0.67405262036735\alpha_0^s \alpha_2^s r^2}{c_s^2}$$
$$+\frac{0.088202094794504\alpha_3^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$+\frac{2.8532314594288\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r}{c_s}$$
$$+\frac{2.995589067438\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3}{c_s^3}$$
$$+\frac{3.0581366629081\left(\alpha_0^{c_s}\right)^2 \alpha_2^s r}{c_s}$$

$$+\frac{1.55700019011 18\alpha_0^{c_s}\alpha_2^{c_s}r^2}{c_s^2}$$
$$+\frac{3.3768924755698\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r}{c_s}$$
$$+\frac{0.43745143869001\alpha_0^{c_s}\alpha_0^s r^4}{c_s^4}$$
$$+\frac{1.9893077070381\left(\alpha_0^s\right)^2\alpha_2^{c_s}r}{c_s}$$
$$-\frac{0.001953125\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.03125\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.0625\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.125\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.046875\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.017578125\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.002685546875\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$-\frac{0.98015911825867\alpha_0^s\alpha_1^{c_s}r^3}{c_s^3}$$
$$-\frac{4.0010774285941\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}r^2}{c_s^2}$$
$$-\frac{4.071100917586\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2}{c_s^2}$$
$$-\frac{0.0049674082338919\alpha_0^s r^5\log\left(c_s\right)}{c_s^5}$$
$$-\frac{3.2853576417753\left(\alpha_0^s\right)^2\alpha_1^{c_s}r^2}{c_s^2}$$
$$-\frac{0.84666127388087\alpha_0^{c_s}\alpha_1^s r^3}{c_s^3}-\frac{2.0396907384329\alpha_1^s\alpha_2^{c_s}r}{c_s}$$
$$-\frac{0.56686562073618\alpha_0^s\alpha_1^s r^3}{c_s^3}$$
$$-\frac{0.57857932712305\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{1.1743023189517\alpha_0^{c_s}\alpha_1^{c_s}r^3}{c_s^3}$$
$$-\frac{0.082010173441755\left(\alpha_0^s\right)^2 r^4\log\left(c_s\right)}{c_s^4}$$
$$-\frac{0.40497704553068\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)}{c_s^3}$$
$$-\frac{0.050668108270534\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$-\frac{0.10280837917801\left(\alpha_0^s\right)^5 r\log\left(c_s\right)}{c_s}$$

$$
\begin{aligned}
&- \frac{3.4272755069646 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{1.0122310659043 \alpha_0^{c_s} \alpha_3^s r}{c_s} - \frac{1.154727777479 \alpha_1^{c_s} \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.3048548230006 \alpha_0^s \alpha_3^{c_s} r}{c_s} \\
&- \frac{0.020833333333333 \alpha_4^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.020833333333333 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.68758257793092 \left(\alpha_0^s\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{0.80334171043098 \left(\alpha_0^{c_s}\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.81762161375293 \alpha_1^s \alpha_2^s r}{c_s} \\
&- \frac{1.6751033898368 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{1.7284662943263 \left(\alpha_0^s\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{1.8578169724941 \alpha_1^{c_s} \alpha_2^s r}{c_s} \\
&- \frac{0.46569385315053 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.48087387125098 \alpha_0^s \alpha_3^s r}{c_s} - \frac{0.9861592444222 \alpha_0^{c_s} \alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.25 \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.1875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.625 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.017578125 \alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.9375 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.14189767784134 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.14189767784134 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.47129535568268 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.47129535568268 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.39189767784134 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.12204825838101 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}
\end{aligned}
$$

$$+\frac{0.48819303352403\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.20798575838101\alpha_0^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.83194303352403\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.1312923842658\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.175693033524\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.4488830956312\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.490628069592\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.26268077696852\alpha_0^s\alpha_1^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.18124960317863\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{1.1168502750681\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.42198902183206\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{1.0502754126021\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{4.2922579734285\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r}{c_s}$$
$$+\frac{0.58061675835603\alpha_0^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{5.1834094215102\alpha_0^s\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{0.33325502603924\alpha_0^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.020833333333333\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.041666666666667\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.512819690414\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.38983219984103\alpha_0^{c_s}\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.10416666666667\alpha_0^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.20833333333333\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.20833333333333\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.7472862981781\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$

$$+ \frac{3.5711394600416\alpha_0^{c_s}\alpha_0^s\alpha_2^s r}{c_s}$$
$$+ \frac{1.7992566969856\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+ \frac{7.4453586567308\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+ \frac{1.8893109571319\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+ \frac{0.96478233218736\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$- \frac{0.015625\alpha_0^{c_s}\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.0625\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.125\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.75\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.3125\alpha_0^s\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.3125\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.625\alpha_1^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.625\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.1142578125\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{2.2054104017793\left(\alpha_0^{c_s}\right)^2\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$- \frac{0.14189767784134\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.14189767784134\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.40879535568268\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.40879535568268\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.39189767784134\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.24409651676201\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.24409651676201\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.92039909054338\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r\log\left(c_s\right)}{c_s}$$

$$- \frac{1.1312923842658 \left(\alpha_{0}^{s}\right)^{2} \alpha_{2}^{c_{s}} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{1.4488830956312 \left(\alpha_{0}^{c_{s}}\right)^{3} \left(\alpha_{0}^{s}\right)^{2} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{0.060416534392878 \alpha_{0}^{c_{s}} \alpha_{1}^{s} r^{3} \log\left(c_{s}\right)^{2}}{c_{s}^{3}}$$
$$- \frac{2.1832187809573 \left(\alpha_{0}^{c_{s}}\right)^{2} \left(\alpha_{0}^{s}\right)^{3} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{2.8055805132741 \alpha_{0}^{c_{s}} \left(\alpha_{1}^{s}\right)^{2} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{4.9527673784595 \alpha_{0}^{c_{s}} \left(\alpha_{0}^{s}\right)^{2} \alpha_{1}^{s} r}{c_{s}}$$
$$- \frac{1.6087005501362 \alpha_{0}^{s} \left(\alpha_{1}^{s}\right)^{2} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{8.2325850087098 \alpha_{0}^{c_{s}} \alpha_{0}^{s} \alpha_{1}^{c_{s}} r^{2}}{c_{s}^{2}}$$
$$- \frac{1.0946586542461 \left(\alpha_{0}^{s}\right)^{2} \alpha_{2}^{s} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{0.64209804273954 \alpha_{0}^{s} \alpha_{2}^{s} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{0.73700640200144 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{0}^{s} r^{3} \log\left(c_{s}\right)}{c_{s}^{3}}$$
$$- \frac{6.3027356936475 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{0}^{s} \alpha_{1}^{c_{s}} r}{c_{s}}$$
$$- \frac{0.20825502603924 \left(\alpha_{0}^{c_{s}}\right)^{4} \alpha_{0}^{s} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{0.10930312519972 \alpha_{0}^{c_{s}} \alpha_{0}^{s} r^{4} \log\left(c_{s}\right)}{c_{s}^{4}}$$
$$- \frac{8.7399852511081 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{0}^{s} \alpha_{1}^{s} r}{c_{s}}$$
$$- \frac{1.1694965995231 \alpha_{0}^{c_{s}} \left(\alpha_{0}^{s}\right)^{2} r^{3} \log\left(c_{s}\right)}{c_{s}^{3}}$$
$$- \frac{1.2850334765399 \left(\alpha_{0}^{c_{s}}\right)^{3} \alpha_{0}^{s} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{0.020833333333333 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{2}^{s} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{0.020833333333333 \left(\alpha_{0}^{c_{s}}\right)^{2} \left(\alpha_{0}^{s}\right)^{3} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{0.041666666666667 \alpha_{0}^{c_{s}} \left(\alpha_{0}^{s}\right)^{4} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{2.8339747313229 \alpha_{0}^{c_{s}} \left(\alpha_{0}^{s}\right)^{3} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{5.8521124624951 \alpha_{0}^{c_{s}} \alpha_{0}^{s} \alpha_{1}^{s} r^{2}}{c_{s}^{2}}$$

$$- \frac{0.74025304063896 \alpha_{0}^{c_{s}} \alpha_{2}^{s} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{0.37229429803911 \alpha_{0}^{s} \alpha_{2}^{c_{s}} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$

$$
\begin{aligned}
&- \frac{0.74976507811772\alpha_0^s \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.20833333333333 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.87364314908906\alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{7.2942095625464\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&- \frac{3.7786219142638 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.0625 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.25\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.09375\alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{1.5675907113654\alpha_0^{c_s} \alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.42569303352403\alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.2263860670481\alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{5.5667777849043\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{6.571847963694 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{3.7452632198456\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{4.3144808126008\alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{2.0296220726182\alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{1.3745301562354 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0625\alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{3.7703628454615\alpha_0^{c_s} \alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.42569303352403\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&- \frac{1.4138860670481 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{2.3513860670481\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{4.4940758295979\alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{1.247914550286\alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{3.7452632198456\alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$- \frac{1.083020104157\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.16666666666667\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log\left(c_s\right)^3}{c_s}$$

Absolute order 7.

$$\begin{aligned}
R_{2,7} = &- \frac{0.00070216008727898 r^7}{c_s^7} + \frac{0.125\alpha_5^s r}{c_s} \\
&+ \frac{0.4279234198261\left(\alpha_2^{c_s}\right)^2 r}{c_s} + \frac{0.1069223471116\alpha_3^s r^3}{c_s^3} \\
&+ \frac{0.070634875645736\alpha_1^{c_s} r^5}{c_s^5} + \frac{0.65487106259881\left(\alpha_1^s\right)^3 r}{c_s} \\
&+ \frac{0.33473585919119\left(\alpha_1^{c_s}\right)^3 r}{c_s} + \frac{0.044680319835584\alpha_1^s r^5}{c_s^5} \\
&+ \frac{0.37743788705055\alpha_5^{c_s} r}{c_s} + \frac{0.23288835529241\alpha_3^{c_s} r^3}{c_s^3} \\
&- \frac{0.3276541902157\left(\alpha_0^{c_s}\right)^2 r^5}{c_s^5} \\
&- \frac{0.038536574146577\left(\alpha_0^s\right)^6 r}{c_s} - \frac{0.025251668868352\alpha_0^{c_s} r^6}{c_s^6} \\
&- \frac{0.076631375424485\alpha_2^s r^4}{c_s^4} - \frac{0.3147093460233\alpha_4^{c_s} r^2}{c_s^2} \\
&- \frac{0.85737561408011\left(\alpha_0^s\right)^5 r^2}{c_s^2} \\
&- \frac{1.8197663035556\left(\alpha_0^{c_s}\right)^3 r^4}{c_s^4} \\
&- \frac{0.52323837210896\left(\alpha_2^s\right)^2 r}{c_s} - \frac{0.019279820845041\alpha_0^s r^6}{c_s^6} \\
&- \frac{0.044182861011306\left(\alpha_0^{c_s}\right)^6 r}{c_s} \\
&- \frac{1.5778098507287\left(\alpha_0^s\right)^4 r^3}{c_s^3} - \frac{0.12389553443311\alpha_4^s r^2}{c_s^2} \\
&- \frac{2.1500087615706\left(\alpha_0^{c_s}\right)^5 r^2}{c_s^2} - \frac{0.14223990162492\alpha_2^{c_s} r^4}{c_s^4} \\
&- \frac{0.64133873745452\left(\alpha_1^s\right)^2 r^3}{c_s^3} \\
&- \frac{1.4525651079324\left(\alpha_1^{c_s}\right)^2 r^3}{c_s^3} \\
&- \frac{0.193795881651\left(\alpha_0^s\right)^2 r^5}{c_s^5} \\
&- \frac{0.85944179815707\left(\alpha_0^s\right)^3 r^4}{c_s^4} \\
&- \frac{3.9302124998924\left(\alpha_0^{c_s}\right)^4 r^3}{c_s^3} \\
&+ \frac{0.00048828125\alpha_3^s r^3 \log\left(c_s\right)^4}{c_s^3} \\
&+ \frac{0.0625\alpha_5^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}$$

$$
\begin{aligned}
&+ \frac{0.078125\left(\alpha_1^s\right)^3 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.0380859375\left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.1142578125\left(\alpha_0^s\right)^4 r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.10698423390637\alpha_4^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.065244100954277\left(\alpha_0^s\right)^6 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.002483704116946\alpha_1^s r^5 \log\left(c_s\right)^2}{c_s^5} \\
&+ \frac{0.67840722984345\left(\alpha_0^{c_s}\right)^4 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.044329625847117\alpha_2^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{1.4436852511254\alpha_1^s \alpha_2^s r^2}{c_s^2} + \frac{0.024834748017219\alpha_1^{c_s} \alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.80435027506809\left(\alpha_1^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{7.1901163394516\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^3}{c_s^3} \\
&+ \frac{2.0715370408386\alpha_0^{c_s} \alpha_3^{c_s} r^2}{c_s^2} \\
&+ \frac{8.33371742252\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r^3}{c_s^3} \\
&+ \frac{0.56686562073618\left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{4.6123541905271\left(\alpha_0^s\right)^3 \alpha_1^s r^2}{c_s^2} \\
&+ \frac{1.1971995800375\alpha_0^s \alpha_1^{c_s} r^4}{c_s^4} \\
&+ \frac{0.0047008994671452\alpha_0^s r^6 \log\left(c_s\right)}{c_s^6} \\
&+ \frac{5.4002061352042\left(\alpha_1^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{2.7574940071872\alpha_1^{c_s} \alpha_2^s r^2}{c_s^2} \\
&+ \frac{0.70950093730891\left(\alpha_0^{c_s}\right)^2 \alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.72830526168963\alpha_0^s \alpha_1^s r^4}{c_s^4} \\
&+ \frac{1.4604100336711\alpha_0^{c_s} \alpha_1^{c_s} r^4}{c_s^4} \\
&+ \frac{6.0729067208546\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^3}{c_s^3} \\
&+ \frac{1.5434158179204\left(\alpha_0^s\right)^4 r^3 \log\left(c_s\right)}{c_s^3}
\end{aligned}
$$

$$+\frac{0.025334054135267\alpha_3^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.87237033860452\alpha_0^s \alpha_3^s r^2}{c_s^2}$$
$$+\frac{3.52708251418\left(\alpha_0^{c_s}\right)^2 \alpha_3^s r}{c_s}$$
$$+\frac{2.0351501180297\alpha_0^s \alpha_3^{c_s} r^2}{c_s^2}$$
$$+\frac{1.0337660189957\alpha_0^{c_s} \alpha_1^s r^4}{c_s^4}$$
$$+\frac{1.0777306522894\left(\alpha_0^s\right)^5 r^2 \log\left(c_s\right)}{c_s^2}$$
$$+\frac{8.9431138078501\left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2}{c_s^2}$$
$$+\frac{2.3646570698546\left(\alpha_0^s\right)^2 \alpha_3^{c_s} r}{c_s}$$
$$+\frac{0.60374617445758\left(\alpha_0^s\right)^4 \alpha_1^s r}{c_s}$$
$$+\frac{4.8534182028867\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{0.62764905795649\left(\alpha_0^s\right)^3 r^4 \log\left(c_s\right)}{c_s^4}$$
$$+\frac{0.0052083333333333\alpha_5^s r \log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.010416666666667\alpha_4^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.020833333333333\left(\alpha_0^s\right)^6 r \log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.16666666666667\left(\alpha_0^s\right)^5 r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{11.277328710343\left(\alpha_0^{c_s}\right)^3 \alpha_1^{c_s} r^2}{c_s^2}$$
$$+\frac{1.4645618412944\alpha_0^{c_s} \alpha_3^s r^2}{c_s^2}$$
$$+\frac{1.5032850077056\left(\alpha_0^s\right)^4 \alpha_1^{c_s} r}{c_s}$$
$$+\frac{0.094418161337362\left(\alpha_0^s\right)^2 r^5 \log\left(c_s\right)}{c_s^5}$$
$$+\frac{3.0534388946485\alpha_1^s \alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{3.1632477823936\alpha_1^{c_s} \alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{0.81762161375293\left(\alpha_2^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$+\frac{0.20833333333333\left(\alpha_2^s\right)^2 r \log\left(c_s\right)^3}{c_s}$$
$$+\frac{3.3646872348791\left(\alpha_0^{c_s}\right)^4 \alpha_1^s r}{c_s}$$

$$+\frac{0.86424555806812\left(\alpha_0^s\right)^2\alpha_3^s r}{c_s}$$
$$+\frac{3.481878776513\left(\alpha_0^s\right)^2\alpha_1^s r^3}{c_s^3}$$
$$+\frac{14.613115091474\left(\alpha_0^{c_s}\right)^3\alpha_1^s r^2}{c_s^2}$$
$$+\frac{0.00089518229166667\alpha_2^s r^4\log\left(c_s\right)^3}{c_s^4}$$
$$+\frac{0.014322916666667\left(\alpha_0^s\right)^3 r^4\log\left(c_s\right)^3}{c_s^4}$$
$$-\frac{0.001953125\alpha_4^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.03125\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.125\alpha_5^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.3125\left(\alpha_1^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.00439453125\left(\alpha_1^s\right)^2 r^3\log\left(c_s\right)^4}{c_s^3}$$
$$-\frac{0.01318359375\left(\alpha_0^s\right)^4 r^3\log\left(c_s\right)^4}{c_s^3}$$

$$-\frac{0.88940264757383\alpha_1^s\alpha_3^s r}{c_s}-\frac{2.6263716304063\alpha_2^{c_s}\alpha_2^s r}{c_s}$$
$$-\frac{1.4317167452955\alpha_0^{c_s}\alpha_2^s r^3}{c_s^3}$$
$$-\frac{0.78052655885942\alpha_0^{c_s}\alpha_4^{c_s} r}{c_s}$$
$$-\frac{5.7287159804501\left(\alpha_0^s\right)^2\alpha_2^{c_s} r^2}{c_s^2}$$
$$-\frac{8.9562108660409\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r}{c_s}$$
$$-\frac{0.018883632267472\alpha_1^s r^5\log\left(c_s\right)}{c_s^5}$$
$$-\frac{27.820691072328\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^2}{c_s^2}$$
$$-\frac{0.94272853191881\alpha_0^s\alpha_2^s r^3}{c_s^3}$$
$$-\frac{1.8880609737729\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r}{c_s}$$
$$-\frac{0.49116427546487\alpha_0^s\alpha_4^s r}{c_s}-\frac{2.1363321100741\alpha_0^{c_s}\alpha_2^{c_s} r^3}{c_s^3}$$
$$-\frac{8.6984406208029\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2}$$
$$-\frac{0.55842513753404\left(\alpha_2^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{2.2356514460006\alpha_1^{c_s}\alpha_1^s r^3}{c_s^3}$$

$$- \frac{0.57857932712305 \left(\alpha_0^s\right)^5 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.58387169747276 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r}{c_s}$$
$$- \frac{9.708715958061 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^3}{c_s^3}$$
$$- \frac{19.738732504714 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^3}{c_s^3}$$
$$- \frac{0.077748164662437 \alpha_3^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{0.010251271680219 \alpha_2^s r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{0.16402034688351 \left(\alpha_0^s\right)^3 r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{0.76390519028696 \left(\alpha_0^{c_s}\right)^3 \alpha_2^{c_s} r}{c_s}$$
$$- \frac{0.01241852058473 \left(\alpha_0^s\right)^2 r^5 \log\left(c_s\right)^2}{c_s^5}$$
$$- \frac{0.051404189589007 \left(\alpha_0^s\right)^6 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.21099451091603 \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{3.6713788131456 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^4}{c_s^4}$$
$$- \frac{7.6134560419899 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r^2}{c_s^2}$$
$$- \frac{2.0442655416859 \alpha_1^{c_s} \alpha_3^s r}{c_s} - \frac{1.0441090751121 \alpha_0^{c_s} \alpha_4^s r}{c_s}$$
$$- \frac{0.5254419502199 \alpha_0^{c_s} \alpha_0^s r^5}{c_s^5}$$
$$- \frac{0.0042317708333333 \alpha_3^s r^3 \log\left(c_s\right)^3}{c_s^3}$$
$$- \frac{1.0840862194786 \left(\alpha_1^s\right)^3 r \log\left(c_s\right)}{c_s}$$
$$- \frac{9.1744078821009 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2}$$
$$- \frac{2.3402453957671 \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r}{c_s}$$
$$- \frac{4.7283638398433 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^4}{c_s^4}$$
$$- \frac{2.4238208232537 \alpha_1^s \alpha_3^{c_s} r}{c_s}$$
$$- \frac{0.60746556829603 \left(\alpha_0^s\right)^4 r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{5.1457277367084 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^3 r}{c_s}$$
$$- \frac{5.2670981166509 \left(\alpha_0^s\right)^2 \left(\alpha_1^{c_s}\right)^2 r}{c_s}$$

$$
\begin{aligned}
&- \frac{0.66013404861329 \left(\alpha_0^{c_s}\right)^5 \alpha_0^s r}{c_s} \\
&- \frac{0.0052083333333333 \left(\alpha_0^s\right)^6 r \log\left(c_s\right)^4}{c_s} \\
&- \frac{0.020833333333333 \alpha_5^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{2.69796002066 \left(\alpha_0^s\right)^2 \alpha_2^s r^2}{c_s^2} \\
&- \frac{5.4882367924932 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r}{c_s} \\
&- \frac{0.044101047397252 \alpha_4^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{2.8255800183644 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r}{c_s} \\
&- \frac{22.729086187968 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r^2}{c_s^2} \\
&- \frac{1.4250435464718 \alpha_0^s \alpha_4^{c_s} r}{c_s} \\
&- \frac{6.3728922911847 \left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r^2}{c_s^2} \\
&- \frac{6.4859501071075 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r^2}{c_s^2} \\
&- \frac{3.327715102302 \left(\alpha_0^{c_s}\right)^4 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{0.052083333333333 \left(\alpha_2^s\right)^2 r \log\left(c_s\right)^4}{c_s} \\
&- \frac{6.7969395060861 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r^2}{c_s^2} \\
&- \frac{1.707214633552 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&- \frac{13.673923812628 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r^2}{c_s^2} \\
&- \frac{0.90914458632794 \left(\alpha_0^s\right)^3 \alpha_2^s r}{c_s} \\
&- \frac{1.8409032303964 \alpha_0^s \alpha_2^{c_s} r^3}{c_s^3} \\
&- \frac{3.7167089437539 \alpha_0^s \left(\alpha_1^s\right)^2 r^2}{c_s^2} \\
&- \frac{15.502242761859 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^3}{c_s^3} \\
&+ \frac{0.00390625 \alpha_0^{c_s} \alpha_3^s r^2 \log\left(c_s\right)^4}{c_s^2} \\
&+ \frac{0.015625 \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.125 \alpha_0^s \alpha_4^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.01953125 \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)^4}{c_s^2}
\end{aligned}
$$

$$+\frac{0.0390625\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.078125\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.078125\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.15625\left(\alpha_0^s\right)^3\alpha_1^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.3125\alpha_1^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.625\alpha_0^s\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.05078125\alpha_0^s\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{0.9375\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.0263671875\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$+\frac{0.047299225947114\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.047299225947114\left(\alpha_0^s\right)^3\alpha_2^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.047299225947114\left(\alpha_0^{c_s}\right)^3\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.047299225947114\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.15709845189423\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{2.8782435074127\left(\alpha_0^s\right)^2\alpha_2^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.1027052008897\left(\alpha_0^{c_s}\right)^2\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.47129535568268\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020138844797626\alpha_0^{c_s}\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{0.41597151676201\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2}$$

$$+\frac{36.08012389062\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^2}{c_s^2}$$
$$+\frac{0.60587387125098\alpha_0^s\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.076954635474829\left(\alpha_0^{c_s}\right)^5\alpha_0^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{11.624127883056\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.74531403479598\left(\alpha_0^s\right)^4\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$

$$+\frac{15.500292631649\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.98015911825867\alpha_1^{c_s}\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{4.071100917586\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{2.2470379147989\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.1468339532713\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{17.433221157511\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r}{c_s}$$
$$+\frac{4.3771008937617\alpha_0^s\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.1745291897529\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.36074999115751\alpha_0^s\alpha_4^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.18124960317863\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{0.39402116545278\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.84666127388087\alpha_0^{c_s}\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{6.925788703418\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.46421758614047\alpha_0^s\alpha_2^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{15.35180563443\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r^3}{c_s^3}$$
$$+\frac{3.9382507354578\alpha_0^s\alpha_1^{c_s}\alpha_2^{c_s} r}{c_s}$$
$$+\frac{0.12597021563024\alpha_0^{c_s}\alpha_0^s r^5\log\left(c_s\right)}{c_s^5}$$
$$+\frac{2.0396907384329\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.0977069997324\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{8.986767202314\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{2.5378272384823\left(\alpha_0^{c_s}\right)^4\alpha_0^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.081365505587338\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.32546202234935\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{10.65982708646\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r}{c_s}$$
$$+\frac{2.8291430625131\alpha_0^s\alpha_1^s\alpha_2^s r}{c_s}$$

$$+\frac{0.36850320100072\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{3.0765715350609\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.79795299477407\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.80011011472349\alpha_0^s\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.40925111616427\alpha_0^s\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{3.3255433222878\left(\alpha_0^{c_s}\right)^3\alpha_0^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.10412751301962\left(\alpha_0^{c_s}\right)^4\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.4272755069646\left(\alpha_0^{c_s}\right)^3\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{6.9483679704258\alpha_0^{c_s}\alpha_1^s\alpha_2^s r}{c_s}$$
$$+\frac{0.054651562599858\alpha_0^{c_s}\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{7.3499400388279\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{15.380569056805\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^{c_s} r}{c_s}$$
$$+\frac{4.0231407515192\alpha_0^{c_s}\alpha_0^s\alpha_3^s r}{c_s}$$
$$+\frac{1.0122310659043\alpha_0^{c_s}\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{4.1224061412015\alpha_0^{c_s}\alpha_0^s\alpha_3^{c_s} r}{c_s}$$
$$+\frac{0.13063255928045\alpha_0^{c_s}\alpha_4^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.13063255928045\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.0872183855733\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{0.13626511856089\alpha_1^{c_s}\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{8.7399852511081\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.55462868901602\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{9.0038817814584\alpha_0^{c_s}\alpha_1^s\alpha_2^{c_s} r}{c_s}$$
$$+\frac{36.268069337383\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2}{c_s^2}$$
$$+\frac{9.7390657716461\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s} r}{c_s}$$
$$+\frac{39.364077231656\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^{c_s} r^2}{c_s^2}$$

$$+\frac{1.2404491010441\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.64251673826997\left(\alpha_0^{c_s}\right)^3\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.2984954850039\alpha_1^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.3048548230006\alpha_1^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.6541481243887\left(\alpha_0^s\right)^2\left(\alpha_1^{c_s}\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{10.660458351696\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3}{c_s^3}$$
$$+\frac{0.16662751301962\left(\alpha_0^s\right)^2\alpha_3^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.0052083333333333\left(\alpha_0^s\right)^4\alpha_1^{c_s} r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.0052083333333333\left(\alpha_0^{c_s}\right)^2\alpha_3^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{2.8532314594288\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r\log\left(c_s\right)}{c_s}$$
$$+\frac{5.8521124624951\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.4700626232957\left(\alpha_0^s\right)^3\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.9564153512317\left(\alpha_0^s\right)^3\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.37012652031948\alpha_0^{c_s}\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.18614714901955\alpha_1^s\alpha_2^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.37488253905886\left(\alpha_1^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{6.0723105020719\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r}{c_s}$$
$$+\frac{24.365411855672\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2}{c_s^2}$$
$$+\frac{6.1623934821264\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^{c_s} r}{c_s}$$
$$+\frac{0.7821586542461\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.7867428745203\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.41666666666667\left(\alpha_0^s\right)^3\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.41666666666667\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.83763770630106\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$

$$+\frac{3.3768924755698\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r\log\left(c_s\right)}{c_s}$$

$$+\frac{0.85437282955636\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^{c_s}r}{c_s}$$
$$+\frac{6.8728109717154\alpha_0^s\alpha_1^{c_s}\alpha_2^s r}{c_s}$$
$$+\frac{0.43682157454453\alpha_1^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.7472862981781\left(\alpha_0^s\right)^3\alpha_1^{c_s}r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.7498057547601\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{7.2509789601495\alpha_0^s\alpha_1^s\alpha_2^{c_s}r}{c_s}$$
$$+\frac{1.8578169724941\alpha_1^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{15.723808804119\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r}{c_s}$$
$$+\frac{0.12301526016263\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{1.9893077070381\left(\alpha_0^s\right)^3\alpha_2^{c_s}r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.015625\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.03125\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.03125\alpha_0^s\alpha_4^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.005859375\alpha_0^s\alpha_2^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$-\frac{0.0234375\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.078125\alpha_1^s\alpha_3^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.078125\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.3125\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.3125\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0107421875\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)^3}{c_s^4}$$
$$-\frac{0.1171875\alpha_0^s\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.234375\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.228515625\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)^3}{c_s^3}$$

$$- \frac{0.15709845189423 \left(\alpha_0^{c_s}\right)^2 \alpha_3^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.4391217537063 \alpha_1^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.40879535568268 \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.46019954527169 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.56564619213292 \alpha_2^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.56564619213292 \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.28699667332183 \alpha_0^s \alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{0.7244415478156 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.7244415478156 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{5.812063941528 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{1.5942209254765 \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.13134038848426 \alpha_1^{c_s} \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{1.0916093904787 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{3.285923981847 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.111888199871 \alpha_0^{c_s} \alpha_3^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{9.3495442100534 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r^2}{c_s^2}$$
$$- \frac{1.1745291897529 \left(\alpha_0^{c_s}\right)^4 \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.2633539166158 \left(\alpha_0^s\right)^2 \alpha_3^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.940477354776 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{6.5707152835506 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{1.7406927007245 \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{7.7337836501318 \alpha_0^{c_s} \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$- \frac{0.49114303236671 \alpha_0^s \alpha_1^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{1.0138030827023 \left(\alpha_0^s\right)^4 \alpha_1^s r \log\left(c_s\right)}{c_s}$$

$$
- \frac{2.1572404063004\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}
- \frac{2.2649505501362\alpha_0^s\left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}
- \frac{0.29000206152126\alpha_0^s\alpha_1^{c_s} r^4 \log\left(c_s\right)}{c_s^4}
- \frac{9.3635434623802\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r}{c_s}
- \frac{0.040682752793669\alpha_1^{c_s}\alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{0.16273101117468\left(\alpha_0^s\right)^3\alpha_1^{c_s} r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{5.6123938932313\left(\alpha_0^s\right)^3\alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}
- \frac{3.2440126801289\left(\alpha_0^s\right)^2\alpha_1^s r^3 \log\left(c_s\right)}{c_s^3}
- \frac{0.10412751301962\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s}
- \frac{6.715855388588\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r}{c_s}
- \frac{13.657124688295\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r^2}{c_s^2}
- \frac{0.21860625039943\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)^2}{c_s^4}
- \frac{1.7542448992846\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3}
- \frac{0.50814669024562\alpha_0^s\alpha_3^{c_s} r^2 \log\left(c_s\right)}{c_s^2}
- \frac{1.0724670334241\left(\alpha_0^s\right)^3\alpha_2^s r \log\left(c_s\right)^2}{c_s}
- \frac{0.13626511856089\left(\alpha_0^s\right)^4\alpha_1^{c_s} r \log\left(c_s\right)^3}{c_s}
- \frac{8.8116100095061\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^{c_s} r}{c_s}
- \frac{1.1055096030022\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3}
- \frac{0.069328586127002\alpha_0^{c_s}\alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{9.4583155713711\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r}{c_s}
- \frac{2.3797633939713\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}
- \frac{19.214966032848\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}
- \frac{1.2850334765399\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}
- \frac{5.1834094215102\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}
$$

$$- \frac{1.3048548230006 \left(\alpha_0^s\right)^2 \alpha_3^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.6541481243887 \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.16662751301962 \alpha_1^s \alpha_3^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.0052083333333333 \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.0052083333333333 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.010416666666667 \alpha_0^{c_s} \alpha_4^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.010416666666667 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.35280837917801 \alpha_0^s \alpha_4^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{2.8339747313229 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^2}{c_s^2}$$

$$- \frac{0.37229429803911 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{2.995589067438 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{0.37488253905886 \left(\alpha_0^s\right)^2 \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.75640984520698 \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{3.0581366629081 \left(\alpha_0^{c_s}\right)^2 \alpha_3^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.5417263912324 \left(\alpha_0^s\right)^2 \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{12.458420607364 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r}{c_s}$$
$$- \frac{0.19491609992051 \alpha_0^{c_s} \alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{6.2961745178777 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^{c_s} r}{c_s}$$
$$- \frac{0.10416666666667 \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.10416666666667 \left(\alpha_0^s\right)^3 \alpha_2^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.20833333333333 \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.83333333333333 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{1.6751033898368 \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.84873351671206 \alpha_1^s \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$

$$
\begin{aligned}
&- \frac{0.43745143869001\alpha_0^{c_s}\alpha_1^s r^4 \log(c_s)}{c_s^4} \\
&- \frac{0.88802108818009\alpha_0^s\alpha_3^s r^2 \log(c_s)}{c_s^2} \\
&- \frac{14.336681167464\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2} \\
&- \frac{0.94465547856594\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^2 \log(c_s)^2}{c_s^2} \\
&- \frac{3.7786219142638\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^2 \log(c_s)^2}{c_s^2} \\
&- \frac{0.48239116609368\alpha_0^{c_s}\alpha_4^s r \log(c_s)^2}{c_s} \\
&- \frac{15.624593473089\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_1^s r}{c_s} \\
&+ \frac{0.03125\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r \log(c_s)^4}{c_s} \\
&+ \frac{0.09375\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2 \log(c_s)^4}{c_s^2} \\
&+ \frac{0.3125\alpha_0^s\alpha_1^s\alpha_2^s r \log(c_s)^4}{c_s} \\
&+ \frac{0.62839380757691\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^s r \log(c_s)^3}{c_s} \\
&+ \frac{0.81759071136537\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r \log(c_s)^3}{c_s} \\
&+ \frac{4.6241870735389\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r \log(c_s)^2}{c_s} \\
&+ \frac{0.24409651676201\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^2 \log(c_s)^3}{c_s^2} \\
&+ \frac{8.523906838501\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r \log(c_s)}{c_s} \\
&+ \frac{8.565442909585\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2 \log(c_s)^2}{c_s^2} \\
&+ \frac{1.9594883892067\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r \log(c_s)^3}{c_s} \\
&+ \frac{1.6969385763988\alpha_0^s\alpha_1^s\alpha_2^{c_s} r \log(c_s)^2}{c_s} \\
&+ \frac{1.8726316099228\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r \log(c_s)^2}{c_s} \\
&+ \frac{1.8726316099228\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s} r \log(c_s)^2}{c_s} \\
&+ \frac{8.0758888622372\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2 \log(c_s)}{c_s^2} \\
&+ \frac{2.1733246434468\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r \log(c_s)^2}{c_s} \\
&+ \frac{4.6907619360049\alpha_0^{c_s}\alpha_1^s\alpha_2^s r \log(c_s)^2}{c_s} \\
&+ \frac{2.3675725888244\alpha_0^{c_s}\alpha_0^s\alpha_3^s r \log(c_s)^2}{c_s}
\end{aligned}
$$

$$+\frac{1.2557055991705\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.2617843419163\alpha_0^s\alpha_1^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{8.2325850087098\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.1209324217963\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{6.3027356936475\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{6.560752153283\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.7542448992846\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{2.0296220726182\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{4.2922579734285\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.54151005207848\alpha_0^{c_s}\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.55462868901602\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{10.052100685\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{10.208719591021\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.3063255928045\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\alpha_0^s\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.68726507811772\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{5.6679328713957\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{11.798121914016\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.0034477600059\alpha_0^s\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.1140003802236\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{12.464443151056\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.052083333333333\alpha_0^{c_s}\alpha_0^s\alpha_3^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.10416666666667\alpha_0^{c_s}\alpha_1^s\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.10416666666667\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^4}{c_s}$$

$$- \frac{0.03125\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2 \log(c_s)^4}{c_s^2}$$
$$- \frac{0.03125(\alpha_0^s)^2\alpha_1^{c_s}\alpha_1^s r \log(c_s)^4}{c_s}$$
$$- \frac{0.15625\alpha_0^{c_s}\alpha_0^s(\alpha_1^s)^2 r \log(c_s)^4}{c_s}$$
$$- \frac{1.25\alpha_0^s\alpha_1^s\alpha_2^s r \log(c_s)^3}{c_s}$$
$$- \frac{0.14189767784134\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r \log(c_s)^3}{c_s}$$
$$- \frac{0.14189767784134\alpha_0^{c_s}(\alpha_0^s)^3\alpha_1^{c_s} r \log(c_s)^3}{c_s}$$
$$- \frac{0.14189767784134\alpha_0^s\alpha_1^s\alpha_2^{c_s} r \log(c_s)^3}{c_s}$$
$$- \frac{0.14189767784134(\alpha_0^{c_s})^3\alpha_0^s\alpha_1^s r \log(c_s)^3}{c_s}$$
$$- \frac{0.94259071136537(\alpha_0^{c_s})^2(\alpha_0^s)^2\alpha_1^s r \log(c_s)^3}{c_s}$$
$$- \frac{0.48819303352403(\alpha_0^{c_s})^2\alpha_0^s\alpha_1^s r^2 \log(c_s)^3}{c_s^2}$$
$$- \frac{4.5507541434397\alpha_0^s\alpha_1^s\alpha_2^s r \log(c_s)}{c_s}$$
$$- \frac{1.6638860670481\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^s r^2 \log(c_s)^3}{c_s^2}$$
$$- \frac{8.0021548571882(\alpha_0^{c_s})^2\alpha_0^s\alpha_1^{c_s} r^2 \log(c_s)}{c_s^2}$$
$$- \frac{4.6685703151829\alpha_0^{c_s}(\alpha_0^s)^2\alpha_2^s r \log(c_s)^2}{c_s}$$
$$- \frac{2.6209294472672\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^2 \log(c_s)^2}{c_s^2}$$
$$- \frac{4.3886291827367(\alpha_0^{c_s})^2\alpha_0^s\alpha_2^s r \log(c_s)^2}{c_s}$$
$$- \frac{22.729134983232\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^s r^2 \log(c_s)}{c_s^2}$$

$$- \frac{0.18124960317863\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3 \log(c_s)^3}{c_s^3}$$
$$- \frac{23.642494466821(\alpha_0^{c_s})^2\alpha_0^s\alpha_1^s r^2 \log(c_s)}{c_s^2}$$
$$- \frac{16.46517001742\alpha_0^{c_s}(\alpha_0^s)^2\alpha_1^{c_s} r^2 \log(c_s)}{c_s^2}$$
$$- \frac{4.4829800191869(\alpha_0^s)^2\alpha_1^{c_s}\alpha_1^s r \log(c_s)^2}{c_s}$$
$$- \frac{4.5833705259459\alpha_0^{c_s}\alpha_0^s\alpha_3^s r \log(c_s)}{c_s}$$
$$- \frac{10.181060458104(\alpha_0^{c_s})^3\alpha_0^s\alpha_1^s r \log(c_s)}{c_s}$$
$$- \frac{6.3027356936475(\alpha_0^{c_s})^2(\alpha_0^s)^2\alpha_1^{c_s} r \log(c_s)}{c_s}$$

$$- \frac{6.9695680415413 \alpha_0^{c_s} \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{29.97030824438 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r}{c_s}$$
$$- \frac{1.0148110363091 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{8.1445793575557 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{4.2922579734285 \alpha_0^{c_s} \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{17.299679629394 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.54151005207848 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{8.6889562483161 \alpha_0^{c_s} \alpha_1^s \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.54506047424358 \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.154727777479 \alpha_0^s \alpha_1^{c_s} \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.65316279640224 \alpha_0^{c_s} \alpha_0^s \alpha_3^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.3063255928045 \alpha_0^{c_s} \alpha_1^s \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.3063255928045 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.020833333333333 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.68726507811772 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{2.8974934469393 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{6.0183061525091 \alpha_0^s \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.80334171043098 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.10416666666667 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{7.0412263940043 \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{3.522906956855 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{7.1571762905879 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{7.2942095625464 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{7.4453586567308 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.9861592444222 \alpha_0^{c_s} \alpha_0^s \alpha_3^{c_s} r \log\left(c_s\right)}{c_s}$$

$$+\frac{0.42569303352403\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_1^s r\log(c_s)^3}{c_s}$$
$$+\frac{22.033777781824\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_1^s r\log(c_s)}{c_s}$$
$$-\frac{5.6178948297683\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_1^s r\log(c_s)^2}{c_s}$$

2.4 $n = 3$

For $p = 4$, the absolute cutoff is $W = 8$:

$$\alpha_3^t = \sum_{d=4}^{8} R_{3,d} + O(\mathrm{SR}^9).$$

Absolute order 4.

$$R_{3,4} = \frac{0.00146484375r^4}{c_s^4} + \frac{0.125\alpha_2^{c_s}r}{c_s}$$
$$+\frac{0.125\alpha_2^s r}{c_s} + \frac{0.125(\alpha_0^{c_s})^3 r}{c_s}$$
$$+\frac{0.125(\alpha_0^s)^3 r}{c_s} + \frac{0.0234375\alpha_0^{c_s}r^3}{c_s^3}$$
$$+\frac{0.0234375\alpha_0^s r^3}{c_s^3} + \frac{0.109375(\alpha_0^{c_s})^2 r^2}{c_s^2}$$
$$+\frac{0.109375(\alpha_0^s)^2 r^2}{c_s^2} - \frac{0.0625\alpha_1^{c_s}r^2}{c_s^2}$$
$$-\frac{0.0625\alpha_1^s r^2}{c_s^2} + \frac{0.375\alpha_0^{c_s}(\alpha_0^s)^2 r}{c_s}$$
$$+\frac{0.375(\alpha_0^{c_s})^2\alpha_0^s r}{c_s} + \frac{0.21875\alpha_0^{c_s}\alpha_0^s r^2}{c_s^2}$$
$$-\frac{0.375\alpha_0^{c_s}\alpha_1^{c_s}r}{c_s} - \frac{0.375\alpha_0^{c_s}\alpha_1^s r}{c_s}$$
$$-\frac{0.375\alpha_0^s\alpha_1^{c_s}r}{c_s} - \frac{0.375\alpha_0^s\alpha_1^s r}{c_s}$$

Absolute order 5.

$$R_{3,5} = -\frac{0.000762939453125r^5}{c_s^5} + \frac{0.125\alpha_3^s r}{c_s}$$
$$+\frac{0.0419921875\alpha_1^s r^3}{c_s^3} + \frac{0.28379535568268\alpha_3^{c_s}r}{c_s}$$
$$+\frac{0.071766316690503\alpha_1^{c_s}r^3}{c_s^3} - \frac{0.125(\alpha_0^s)^4 r}{c_s}$$
$$-\frac{0.375(\alpha_1^s)^2 r}{c_s} - \frac{0.078125\alpha_2^s r^2}{c_s^2}$$
$$-\frac{0.28125(\alpha_0^s)^3 r^2}{c_s^2} - \frac{0.01708984375\alpha_0^s r^4}{c_s^4}$$
$$-\frac{0.1201171875(\alpha_0^s)^2 r^3}{c_s^3} - \frac{0.28379535568268(\alpha_0^{c_s})^4 r}{c_s}$$
$$-\frac{0.15752267784134\alpha_2^{c_s}r^2}{c_s^2} - \frac{0.85138606704805(\alpha_1^{c_s})^2 r}{c_s}$$
$$-\frac{0.024533376047626\alpha_0^{c_s}r^4}{c_s^4} - \frac{0.5591418724447(\alpha_0^{c_s})^3 r^2}{c_s^2}$$

$$
\begin{aligned}
&- \frac{0.20943957507151 \left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
&+ \frac{0.0625 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.125 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.005859375 \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.375 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.75 \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.21875 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.4375 \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.0703125 \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.72129535568268 \alpha_0^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{1.7027721340961 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.6013860670481 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.9763860670481 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.7778918724447 \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.87418722812738 \alpha_0^{c_s} \alpha_1^{c_s} r^2}{c_s^2} - \frac{0.125 \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.5 \alpha_0^s \alpha_2^s r}{c_s} - \frac{0.0234375 \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{1.1351814227307 \alpha_0^{c_s} \alpha_2^{c_s} r}{c_s} - \frac{1.0337953556827 \alpha_0^{c_s} \alpha_2^s r}{c_s} \\
&- \frac{1.0337953556827 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{1.1013860670481 \alpha_0^s \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.3513860670481 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{1.6013860670481 \alpha_1^{c_s} \alpha_1^s r}{c_s} \\
&- \frac{1.9763860670481 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.6182837448894 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{1.3403918724447 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{0.35299426257151 \alpha_0^{c_s} \alpha_0^s r^3}{c_s^3}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.125\alpha_0^s\alpha_2^{c_s} r\log(c_s)}{c_s} \\
&+ \frac{0.125\left(\alpha_0^{c_s}\right)^3\alpha_0^s r\log(c_s)}{c_s} \\
&+ \frac{0.5\alpha_0^s\alpha_2^s r\log(c_s)}{c_s} \\
&+ \frac{0.375\alpha_0^{c_s}\alpha_2^s r\log(c_s)}{c_s} \\
&+ \frac{0.375\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log(c_s)}{c_s} \\
&+ \frac{0.375\alpha_1^{c_s}\alpha_1^s r\log(c_s)}{c_s} \\
&+ \frac{0.375\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r\log(c_s)}{c_s} \\
&+ \frac{0.21875\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^2\log(c_s)}{c_s^2} \\
&+ \frac{0.4375\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log(c_s)}{c_s^2} \\
&+ \frac{0.0703125\alpha_0^{c_s}\alpha_0^s r^3\log(c_s)}{c_s^3} \\
&+ \frac{3.1013860670481\alpha_0^{c_s}\alpha_0^s\alpha_1^s r}{c_s} \\
&+ \frac{3.6791582011442\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.125\alpha_0^s\alpha_1^{c_s} r^2\log(c_s)}{c_s^2} \\
&- \frac{0.375\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log(c_s)}{c_s} \\
&- \frac{0.375\left(\alpha_0^{c_s}\right)^2\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{0.75\left(\alpha_0^s\right)^2\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{0.21875\alpha_0^{c_s}\alpha_1^s r^2\log(c_s)}{c_s^2} \\
&- \frac{0.34375\alpha_0^s\alpha_1^s r^2\log(c_s)}{c_s^2} \\
&- \frac{0.375\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r\log(c_s)}{c_s} \\
&- \frac{1.125\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log(c_s)}{c_s}
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{3,6} = &\frac{0.00069887861331449r^6}{c_s^6} + \frac{0.125\alpha_4^s r}{c_s} \\
&+ \frac{1.065570626349\left(\alpha_0^{c_s}\right)^3 r^3}{c_s^3} \\
&+ \frac{1.4176370369568\left(\alpha_0^{c_s}\right)^4 r^2}{c_s^2} \\
&+ \frac{0.52264594794504\left(\alpha_1^s\right)^2 r^2}{c_s^2} + \frac{0.10280837917801\left(\alpha_0^s\right)^5 r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.20825502603924 \left(\alpha_0^{c_s}\right)^5 r}{c_s} \\
&+ \frac{0.24809448714071 \left(\alpha_0^{c_s}\right)^2 r^4}{c_s^4} \\
&+ \frac{0.52433019631832 \left(\alpha_0^s\right)^3 r^3}{c_s^3} + \frac{0.13631419746067 \alpha_2^{c_s} r^3}{c_s^3} \\
&+ \frac{0.017257388611671 \alpha_0^s r^5}{c_s^5} + \frac{1.1501885493287 \left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2} \\
&+ \frac{0.073695194469 16 \alpha_2^s r^3}{c_s^3} + \frac{0.1495700174229 \left(\alpha_0^s\right)^2 r^4}{c_s^4} \\
&+ \frac{0.61990642191755 \left(\alpha_0^s\right)^4 r^2}{c_s^2} + \frac{0.33325502603924 \alpha_4^{c_s} r}{c_s} \\
&+ \frac{0.022103438284604 \alpha_0^{c_s} r^5}{c_s^5} - \frac{0.042328393296221 \alpha_1^s r^4}{c_s^4} \\
&- \frac{0.066357117878927 \alpha_1^{c_s} r^4}{c_s^4} - \frac{0.1038270947945 \alpha_3^s r^2}{c_s^2} \\
&- \frac{0.22780402727446 \alpha_3^{c_s} r^2}{c_s^2} + \frac{0.0625 \alpha_4^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.01171875 \alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.01171875 \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)^2}{c_s^4} \\
&+ \frac{0.078125 \alpha_3^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.21875 \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.171875 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.10546875 \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.01708984375 \alpha_1^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{4.5746925879539 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r}{c_s} \\
&+ \frac{1.1352294412117 \alpha_0^{c_s} \alpha_2^s r^2}{c_s^2} \\
&+ \frac{2.6219204538578 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r}{c_s} \\
&+ \frac{3.8874132271834 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r}{c_s} \\
&+ \frac{2.2571479479676 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&+ \frac{1.9409334773531 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r}{c_s} \\
&+ \frac{3.5111972716302 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r}{c_s}
\end{aligned}
$$

$$+\frac{8.2127818560984\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2}{c_s^2}$$
$$+\frac{4.4950282979633\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{1.6087005501362\alpha_0^s\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{1.0946586542461\left(\alpha_0^s\right)^2\alpha_2^s r}{c_s}$$
$$+\frac{0.71467513753404\alpha_0^s\alpha_2^s r^2}{c_s^2}$$
$$+\frac{1.6030226512749\alpha_0^{c_s}\alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{0.40191656771049\alpha_0^{c_s}\alpha_0^s r^4}{c_s^4}$$
$$+\frac{1.1482074697214\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r}{c_s}$$
$$+\frac{2.4575502603924\left(\alpha_0^{c_s}\right)^2\alpha_2^{c_s} r}{c_s}$$
$$+\frac{1.4228234146217\alpha_0^s\alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{2.8877107655697\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^3}{c_s^3}$$
$$+\frac{6.1460976449703\left(\alpha_0^{c_s}\right)^3\alpha_0^s r^2}{c_s^2}$$
$$+\frac{3.4988253905886\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{1.8602805292847\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}$$
$$+\frac{3.9539880896494\left(\alpha_0^{c_s}\right)^2\alpha_2^s r}{c_s}$$
$$+\frac{3.9948526700024\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2}{c_s^2}$$
$$-\frac{0.03125\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}-\frac{0.125\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.125\left(\alpha_0^s\right)^5 r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.0029296875\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$-\frac{0.4375\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.5625\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.068359375\left(\alpha_0^s\right)^2 r^4\log\left(c_s\right)}{c_s^4}$$
$$-\frac{0.0419921875\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$-\frac{0.003814697265625\alpha_0^s r^5\log\left(c_s\right)}{c_s^5}$$

$$
\begin{aligned}
&- \frac{0.3603515625\left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.57385180946901\alpha_0^s\alpha_1^s r^3}{c_s^3} \\
&- \frac{2.4110271601353\left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{1.2147823321874\alpha_0^{c_s}\alpha_3^s r}{c_s} - \frac{1.4645474103051\alpha_0^s\alpha_3^{c_s} r}{c_s} \\
&- \frac{2.6441120746798\alpha_1^s\alpha_2^{c_s} r}{c_s} \\
&- \frac{4.5968842087759\left(\alpha_0^{c_s}\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{1.9544879813691\left(\alpha_0^s\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{4.4382967251885\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{1.1168502750681\alpha_1^s\alpha_2^s r}{c_s} - \frac{1.1225357717973\alpha_0^{c_s}\alpha_1^{c_s} r^3}{c_s^3} \\
&- \frac{0.81325404477083\alpha_0^{c_s}\alpha_1^s r^3}{c_s^3} \\
&- \frac{1.0502754126021\left(\alpha_0^s\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{4.3255592153975\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{2.2075502603924\left(\alpha_0^{c_s}\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.58061675835603\alpha_0^s\alpha_3^s r}{c_s} - \frac{2.4776020226013\alpha_1^{c_s}\alpha_2^s r}{c_s} \\
&- \frac{2.5825502603924\alpha_1^{c_s}\alpha_2^{c_s} r}{c_s} - \frac{1.4162751301962\alpha_0^{c_s}\alpha_3^{c_s} r}{c_s} \\
&- \frac{3.5073089371114\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{0.9442318016501\alpha_0^s\alpha_1^{c_s} r^3}{c_s^3} \\
&+ \frac{0.0625\alpha_1^{c_s}\alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.625\alpha_0^s\alpha_3^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.625\left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$+\frac{1.25\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.25\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.109375\alpha_0^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$

$$+\frac{0.21875\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.4375\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.5625\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.234375\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.9375\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.10546875\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{1.71875\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.3662109375\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.28379535568268\alpha_0^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.0337953556827\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.1013860670481\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.3513860670481\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.6013860670481\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.6013860670481\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.5557837448894\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{6.243371063951\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r}{c_s}$$
$$+\frac{1.6182837448894\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.21529895007151\alpha_0^s\alpha_1^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{4.7259796038175\alpha_0^{c_s}\alpha_0^s\alpha_2^s r}{c_s}$$
$$+\frac{0.35299426257151\alpha_0^{c_s}\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{7.299656342872\alpha_0^s\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{10.486494425184\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r}{c_s}$$

$$- \frac{0.0625 \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.0625 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.125 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.1875 \alpha_0^{c_s} \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.1875 \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.1875 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.3125 \alpha_0^s \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.625 \alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.625 \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.25 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.109375 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.03515625 \alpha_0^{c_s} \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{1.875 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.59375 \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.10546875 \alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{0.671875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.31504535568268 \alpha_0^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.72129535568268 \alpha_0^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.85138606704805 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.0337953556827 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.1013860670481 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.3513860670481 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.098133504190503 \alpha_0^{c_s} \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4}$$

$$
\begin{aligned}
&- \frac{1.9763860670481 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{1.9763860670481 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s} \\
&- \frac{1.1182837448894 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.1013860670481 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.7778918724447 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.2365674897788 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{2.6807837448894 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.62831872521453 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{10.317828730023 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&- \frac{10.419919562718 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&- \frac{6.2653115524259 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&- \frac{1.0589827877145 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{11.728814544016 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r}{c_s} \\
&- \frac{8.7476479574518 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{6.9558196807943 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.1875 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.75 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.5625 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.7027721340961 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{6.2027721340961 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.7483744562548 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{3.6791582011442 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{5.9291582011442 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{4.1233744562548 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.1875\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s} \\
&- \frac{0.5625\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r\log\left(c_s\right)^2}{c_s} \\
&- \frac{1.125\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s} \\
&- \frac{0.65625\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&- \frac{1.1351814227307\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s} \\
&- \frac{4.1351814227307\alpha_0^{c_s}\alpha_0^s\alpha_2^s r\log\left(c_s\right)}{c_s} \\
&- \frac{3.6791582011442\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s} \\
&- \frac{4.8041582011442\alpha_0^s\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}
\end{aligned}
$$

Absolute order 7.

$$
\begin{aligned}
R_{3,7} = &- \frac{0.00066040799045703 r^7}{c_s^7} + \frac{0.125\alpha_5^s r}{c_s} \\
&+ \frac{1.2557055991705\left(\alpha_1^{c_s}\right)^3 r}{c_s} + \frac{0.36074999115751\alpha_5^{c_s} r}{c_s} \\
&+ \frac{0.044182253542082\alpha_1^s r^5}{c_s^5} + \frac{0.10547078563903\alpha_3^s r^3}{c_s^3} \\
&+ \frac{0.067614788198182\alpha_1^{c_s} r^5}{c_s^5} + \frac{1.0840862194786\left(\alpha_1^s\right)^3 r}{c_s} \\
&+ \frac{0.22086108655179\alpha_3^{c_s} r^3}{c_s^3} - \frac{0.076954635474829\left(\alpha_0^{c_s}\right)^6 r}{c_s} \\
&- \frac{0.12260923390637\alpha_4^s r^2}{c_s^2} - \frac{0.065244100954277\left(\alpha_0^s\right)^6 r}{c_s} \\
&- \frac{0.31475559837547\left(\alpha_0^{c_s}\right)^2 r^5}{c_s^5} \\
&- \frac{1.8011431833565\left(\alpha_0^s\right)^4 r^3}{c_s^3} \\
&- \frac{1.8035217754453\left(\alpha_0^{c_s}\right)^3 r^4}{c_s^4} \\
&- \frac{1.1197437723225\left(\alpha_0^s\right)^5 r^2}{c_s^2} - \frac{0.2991670940175\alpha_4^{c_s} r^2}{c_s^2} \\
&- \frac{0.076712092161027\alpha_2^s r^4}{c_s^4} - \frac{0.70760825653778\left(\alpha_1^s\right)^2 r^3}{c_s^3} \\
&- \frac{0.19445546235151\left(\alpha_0^s\right)^2 r^5}{c_s^5} \\
&- \frac{4.1620983497444\left(\alpha_0^{c_s}\right)^4 r^3}{c_s^3} - \frac{0.023975376743708\alpha_0^{c_s} r^6}{c_s^6} \\
&- \frac{0.13853540110911\alpha_2^{c_s} r^4}{c_s^4} - \frac{1.154727777479\left(\alpha_2^{c_s}\right)^2 r}{c_s} \\
&- \frac{0.018648400200653\alpha_0^s r^6}{c_s^6} - \frac{2.675269515726\left(\alpha_0^{c_s}\right)^5 r^2}{c_s^2} \\
&- \frac{1.5385686369455\left(\alpha_1^{c_s}\right)^2 r^3}{c_s^3}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.81762161375293 \left(\alpha_2^s\right)^2 r}{c_s} - \frac{0.9077746502428 \left(\alpha_0^s\right)^3 r^4}{c_s^4} \\
&+ \frac{0.0009765625 \alpha_2^s r^4 \log\left(c_s\right)^3}{c_s^4} \\
&+ \frac{0.015625 \left(\alpha_0^s\right)^3 r^4 \log\left(c_s\right)^3}{c_s^4} \\
&+ \frac{0.0625 \alpha_5^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.03515625 \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.9375 \left(\alpha_1^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.10546875 \left(\alpha_0^s\right)^4 r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.02099609375 \alpha_3^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.0019073486328125 \alpha_1^s r^5 \log\left(c_s\right)^2}{c_s^5} \\
&+ \frac{3.9387181395716 \alpha_1^{c_s} \alpha_2^{c_s} r^2}{c_s^2} \\
&+ \frac{0.57385180946901 \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{10.09950678112 \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.4409653155906 \alpha_0^{c_s} \alpha_1^{c_s} r^4}{c_s^4} \\
&+ \frac{2.2829930145581 \alpha_0^{c_s} \alpha_3^{c_s} r^2}{c_s^2} \\
&+ \frac{6.0506877923487 \left(\alpha_0^{c_s}\right)^4 \alpha_1^s r}{c_s} \\
&+ \frac{0.75724388321833 \alpha_0^s \alpha_1^s r^4}{c_s^4} \\
&+ \frac{1.1881148878051 \left(\alpha_0^{c_s}\right)^4 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.2633539166158 \left(\alpha_0^s\right)^2 \alpha_3^s r}{c_s} \\
&+ \frac{10.84818865722 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.086286943058355 \left(\alpha_0^s\right)^2 r^5 \log\left(c_s\right)}{c_s^5} \\
&+ \frac{5.8694148411763 \left(\alpha_0^s\right)^3 \alpha_1^s r^2}{c_s^2} \\
&+ \frac{3.899202830862 \left(\alpha_0^s\right)^2 \alpha_1^s r^3}{c_s^3} \\
&+ \frac{1.0138030827023 \left(\alpha_0^s\right)^4 \alpha_1^s r}{c_s} \\
&+ \frac{1.0488038466557 \alpha_0^{c_s} \alpha_1^s r^4}{c_s^4}
\end{aligned}
$$

$$+\frac{2.1070916662185\left(\alpha_0^{c_s}\right)^2\alpha_3^{c_s}r}{c_s}$$
$$+\frac{0.004193271679887\alpha_0^s r^6\log\left(c_s\right)}{c_s^6}$$
$$+\frac{8.7674965667948\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}r^3}{c_s^3}$$
$$+\frac{1.1168502750681\left(\alpha_2^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{5.0351500609997\left(\alpha_0^{c_s}\right)^2\alpha_3^s r}{c_s}$$
$$+\frac{0.042328393296221\alpha_2^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{6.5251062314093\left(\alpha_0^s\right)^2\alpha_1^{c_s}r^3}{c_s^3}$$
$$+\frac{0.10280837917801\left(\alpha_0^s\right)^6 r\log\left(c_s\right)}{c_s}$$
$$+\frac{7.5978803659804\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^3}{c_s^3}$$
$$+\frac{17.170223355941\left(\alpha_0^{c_s}\right)^3\alpha_1^s r^2}{c_s^2}$$
$$+\frac{2.1520321750819\alpha_0^s\alpha_3^{c_s}r^2}{c_s^2}$$
$$+\frac{0.14583333333333\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.5982800696916\left(\alpha_0^s\right)^3 r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{1.200524716176\alpha_0^s\alpha_1^{c_s}r^4}{c_s^4}$$
$$+\frac{2.4730563846109\left(\alpha_0^s\right)^4\alpha_1^{c_s}r}{c_s}$$
$$+\frac{1.2398128438351\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.010416666666667\alpha_4^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.020833333333333\left(\alpha_0^s\right)^6 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{3.1420899004122\alpha_1^{c_s}\alpha_2^s r^2}{c_s^2}$$
$$+\frac{1.572990588955\left(\alpha_0^s\right)^4 r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{1.5760620651767\alpha_0^{c_s}\alpha_3^s r^2}{c_s^2}$$
$$+\frac{3.2941625300387\left(\alpha_0^s\right)^2\alpha_3^{c_s}r}{c_s}$$
$$+\frac{0.1038270947945\alpha_4^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.20833333333333\left(\alpha_2^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$

$$
\begin{aligned}
&+ \frac{1.7276736271957\alpha_1^s\alpha_2^s r^2}{c_s^2} \\
&+ \frac{13.88289571595\left(\alpha_0^{c_s}\right)^3\alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{3.51534396509\alpha_1^s\alpha_2^{c_s} r^2}{c_s^2} + \frac{0.96375532208646\alpha_0^s\alpha_3^s r^2}{c_s^2} \\
&+ \frac{7.7423178156474\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r}{c_s} \\
&- \frac{0.00390625\alpha_3^s r^3\log\left(c_s\right)^3}{c_s^3} \\
&- \frac{0.0625\left(\alpha_0^s\right)^6 r\log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125\alpha_5^s r\log\left(c_s\right)}{c_s} \\
&- \frac{0.0390625\alpha_4^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.3125\left(\alpha_1^s\right)^3 r\log\left(c_s\right)^3}{c_s} \\
&- \frac{0.625\left(\alpha_2^s\right)^2 r\log\left(c_s\right)^2}{c_s} \\
&- \frac{0.5625\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.008544921875\alpha_2^s r^4\log\left(c_s\right)^2}{c_s^4} \\
&- \frac{0.13671875\left(\alpha_0^s\right)^3 r^4\log\left(c_s\right)^2}{c_s^4} \\
&- \frac{0.18310546875\left(\alpha_1^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3} \\
&- \frac{0.0095367431640625\left(\alpha_0^s\right)^2 r^5\log\left(c_s\right)^2}{c_s^5} \\
&- \frac{0.54052734375\left(\alpha_0^s\right)^4 r^3\log\left(c_s\right)^2}{c_s^3}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{1.4855117920869\alpha_0^{c_s}\alpha_2^s r^3}{c_s^3} - \frac{1.0111179362879\alpha_0^s\alpha_2^s r^3}{c_s^3} \\
&- \frac{0.60587387125098\alpha_0^s\alpha_4^s r}{c_s} - \frac{2.3683440805453\alpha_1^{c_s}\alpha_1^s r^3}{c_s^3} \\
&- \frac{3.1468339532713\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r}{c_s} \\
&- \frac{3.3445455614335\alpha_1^s\alpha_3^{c_s} r}{c_s} \\
&- \frac{7.7328550642049\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^2}{c_s^2} \\
&- \frac{0.51503239853261\alpha_0^{c_s}\alpha_0^s r^5}{c_s^5} \\
&- \frac{10.786415997547\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2} \\
&- \frac{11.101166727647\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2}
\end{aligned}
$$

$$
- \frac{1.6087005501362\left(\alpha_1^s\right)^3 r \log\left(c_s\right)}{c_s}
- \frac{27.261332258597\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^2}{c_s^2}
- \frac{7.6667459952952\left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r^2}{c_s^2}
- \frac{3.897507710927\alpha_2^{c_s}\alpha_2^s r}{c_s}
- \frac{2.1408870219012\alpha_1^{c_s}\alpha_3^{c_s} r}{c_s}
- \frac{2.1930683622079\alpha_0^{c_s}\alpha_2^{c_s} r^3}{c_s^3}
- \frac{4.6296729998974\alpha_0^s\left(\alpha_1^s\right)^2 r^2}{c_s^2}
- \frac{9.4629411206417\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2}{c_s^2}
- \frac{4.7412780427983\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^4}{c_s^4}
- \frac{10.6792492855\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^3}{c_s^3}
- \frac{1.3469092355797\alpha_0^{c_s}\alpha_4^{c_s} r}{c_s}
- \frac{3.2598096829233\left(\alpha_0^s\right)^2 \alpha_2^s r^2}{c_s^2}
- \frac{1.6656048141581\alpha_0^s\alpha_4^{c_s} r}{c_s}
- \frac{3.7668956958108\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^4}{c_s^4}
- \frac{7.8136689732526\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2}{c_s^2}
- \frac{16.377283754452\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^3}{c_s^3}
- \frac{16.453977631147\left(\alpha_0^{c_s}\right)^4 \alpha_0^s r^2}{c_s^2}
- \frac{32.974734898143\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^2}{c_s^2}
- \frac{8.4133427159559\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r}{c_s}
- \frac{8.690822571652\left(\alpha_0^{c_s}\right)^3 \alpha_2^s r}{c_s}
- \frac{0.017257388611671\alpha_1^s r^5 \log\left(c_s\right)}{c_s^5}
- \frac{4.5715835447462\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r}{c_s}
- \frac{0.07369519446916\alpha_3^s r^3 \log\left(c_s\right)}{c_s^3}
- \frac{9.9483576869351\left(\alpha_0^s\right)^2\left(\alpha_1^{c_s}\right)^2 r}{c_s}
- \frac{1.2622310659043\alpha_0^{c_s}\alpha_4^s r}{c_s}
- \frac{1.2984954850039\alpha_1^s\alpha_3^s r}{c_s}
$$

$$
\begin{aligned}
&- \frac{21.116091248333 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&- \frac{0.020833333333333 \alpha_5^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{2.8700480383984 \alpha_1^{c_s} \alpha_3^s r}{c_s} \\
&- \frac{1.459738851267 \left(\alpha_0^{c_s}\right)^5 \alpha_0^s r}{c_s} \\
&- \frac{1.4700626232957 \left(\alpha_0^s\right)^3 \alpha_2^s r}{c_s} \\
&- \frac{6.0003047609539 \left(\alpha_0^{c_s}\right)^4 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{6.6279464500316 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r^2}{c_s^2} \\
&- \frac{14.104474158519 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^s\right)^2 r}{c_s} \\
&- \frac{3.6644110968749 \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r}{c_s} \\
&- \frac{3.6657307304634 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&- \frac{1.9050809010702 \alpha_0^s \alpha_2^{c_s} r^3}{c_s^3} \\
&- \frac{1.9242741322273 \left(\alpha_0^{c_s}\right)^3 \alpha_2^{c_s} r}{c_s} \\
&- \frac{0.96600547490636 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r}{c_s} \\
&+ \frac{0.0625 \alpha_0^{c_s} \alpha_4^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.0625 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.0625 \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.0625 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.125 \alpha_0^s \alpha_4^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.01171875 \alpha_0^{c_s} \alpha_2^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.046875 \alpha_0^s \alpha_2^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.1875 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.375 \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.3125 \alpha_1^s \alpha_3^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.21875 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^3}{c_s^2}
\end{aligned}
$$

$$+\frac{0.5625\alpha_0^s\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.9375\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.9375\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.9375\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.10546875\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{0.734375\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{2.84375\left(\alpha_0^s\right)^3\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.1025390625\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{1.08984375\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{4.5746925879539\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.14189767784134\left(\alpha_0^{c_s}\right)^4\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.14189767784134\left(\alpha_0^s\right)^2\alpha_3^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.15752267784134\alpha_1^s\alpha_2^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{10.623428137137\alpha_0^{c_s}\alpha_1^s\alpha_2^s r}{c_s}$$
$$+\frac{0.36064767784134\alpha_0^{c_s}\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.42569303352403\left(\alpha_1^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{4.5507541434397\alpha_0^s\alpha_1^s\alpha_2^s r}{c_s}$$
$$+\frac{2.6219204538578\left(\alpha_0^s\right)^3\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.049066752095252\alpha_0^{c_s}\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{0.80069303352403\left(\alpha_0^s\right)^4\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.8874132271834\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.98819303352403\left(\alpha_0^{c_s}\right)^2\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.5591418724447\left(\alpha_0^{c_s}\right)^3\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.2147823321874\alpha_0^{c_s}\alpha_4^s r\log\left(c_s\right)}{c_s}$$

$$
\begin{aligned}
&+ \frac{1.4645474103051 \alpha_1^s \alpha_3^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.38894593622235 \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{1.5557837448894 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{25.492893672906 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^s r}{c_s} \\
&+ \frac{1.9409334773531 \left(\alpha_0^{c_s}\right)^4 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{16.183034565821 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^3}{c_s^3} \\
&+ \frac{2.4020791005721 \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{2.6441120746798 \alpha_2^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{5.6448611555529 \alpha_0^{c_s} \alpha_0^s \alpha_3^s r}{c_s} \\
&+ \frac{6.7714438439027 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{14.908599752858 \alpha_0^{c_s} \alpha_1^s \alpha_2^{c_s} r}{c_s} \\
&+ \frac{16.425563712197 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{4.4950282979633 \left(\alpha_0^s\right)^2 \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{4.5968842087759 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{9.8824373480888 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r}{c_s} \\
&+ \frac{0.31415936260727 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.32294842510727 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{11.201715574019 \alpha_0^s \alpha_1^s \alpha_2^{c_s} r}{c_s} \\
&+ \frac{2.8352740739136 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{5.9080038194127 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^{c_s} r}{c_s} \\
&+ \frac{6.2653115524259 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{8.6631322967092 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{4.9542678586284 \alpha_0^s \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{10.612365854046 \alpha_0^s \alpha_1^{c_s} \alpha_2^s r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{42.618229821464 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&+ \frac{10.755281999685 \alpha_0^s \alpha_1^{c_s} \alpha_2^{c_s} r}{c_s} \\
&+ \frac{11.559920669238 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^3}{c_s^3} \\
&+ \frac{11.728814544016 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{3.196711879047 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{1.6076662708419 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.81325404477083 \alpha_0^{c_s} \alpha_2^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.20825502603924 \left(\alpha_0^{c_s}\right)^5 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{6.9165270668905 \alpha_0^{c_s} \alpha_0^s \alpha_3^{c_s} r}{c_s} \\
&+ \frac{27.819224792758 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.11051719142302 \alpha_0^{c_s} \alpha_0^s r^5 \log\left(c_s\right)}{c_s^5} \\
&+ \frac{29.668801756997 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.99237794856283 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{16.166403940833 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{16.932268645455 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^s r}{c_s} \\
&+ \frac{2.1449340668482 \left(\alpha_0^s\right)^3 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{4.3255592153975 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{1.1482074697214 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r \log\left(c_s\right)}{c_s} \\
&+ \frac{2.3003770986574 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.036458333333333 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.14583333333333 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.29166666666667 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{4.7595267879426 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{2.4776020226013 \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.33325502603924 \alpha_0^s \alpha_4^{c_s} r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$+\frac{0.020833333333333\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^s\right)^3\alpha_2^{c_s}r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^{c_s}\right)^3\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{43.051818854924\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s}r^2}{c_s^2}$$
$$+\frac{0.70561675835603\alpha_0^s\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.8456468292435\left(\alpha_0^s\right)^2\alpha_2^{c_s}r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{47.050382883379\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^{c_s}r^2}{c_s^2}$$
$$+\frac{0.38020833333333\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{12.292195289941\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.79493739287649\alpha_0^s\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.408942592382\alpha_0^s\alpha_2^{c_s}r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{13.178724319326\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^{c_s}r}{c_s}$$
$$+\frac{0.41666666666667\left(\alpha_0^s\right)^3\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{3.3838382564372\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.6974670334241\alpha_1^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{27.472885902749\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^{c_s}r}{c_s}$$
$$+\frac{0.9442318016501\alpha_1^{c_s}\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{7.9897053400048\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.0625\left(\alpha_0^s\right)^4\alpha_1^{c_s}r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\left(\alpha_0^{c_s}\right)^2\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.01171875\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)^3}{c_s^4}$$
$$-\frac{0.1875\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.375\alpha_0^s\alpha_4^s r\log\left(c_s\right)^2}{c_s}$$

$$- \frac{0.3125 \left(\alpha_0^s\right)^4 \alpha_1^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.3125 \left(\alpha_0^s\right)^2 \alpha_3^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.25 \left(\alpha_0^s\right)^3 \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.9375 \alpha_1^s \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.2109375 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 \log\left(c_s\right)^3}{c_s^3}$$
$$- \frac{2.8125 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.24609375 \alpha_0^s \alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3}$$

$$- \frac{2.15625 \alpha_0^s \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{1.453125 \left(\alpha_0^s\right)^2 \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.14189767784134 \alpha_1^s \alpha_3^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.14189767784134 \left(\alpha_0^{c_s}\right)^4 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.31504535568268 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.42569303352403 \left(\alpha_0^s\right)^2 \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.51689767784134 \alpha_0^{c_s} \alpha_4^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.51689767784134 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.55069303352403 \alpha_2^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.55069303352403 \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.1352294412117 \alpha_0^{c_s} \alpha_3^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.67569303352403 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.67569303352403 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.19626700838101 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{0.80069303352403 \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.98819303352403 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.1182837448894 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}$$

$$- \frac{2.4110271601353 \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.4645474103051 \left(\alpha_0^s\right)^2 \alpha_3^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.8091418724447 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{3.2365674897788 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.10764947503576 \alpha_1^{c_s} \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{1.9409334773531 \left(\alpha_0^{c_s}\right)^4 \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.6807837448894 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{10.851732081256 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r^2}{c_s^2}$$
$$- \frac{0.17649713128576 \alpha_0^{c_s} \alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{2.9645791005721 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{3.294546017362 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{27.775184091118 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} \alpha_1^s r}{c_s}$$
$$- \frac{2.0616872281274 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{4.4950282979633 \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{22.554554719686 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r^2}{c_s^2}$$
$$- \frac{7.299656342872 \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{15.812179043725 \alpha_0^{c_s} \alpha_0^s \alpha_2^{c_s} r^2}{c_s^2}$$
$$- \frac{12.798946193029 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^{c_s} r}{c_s}$$
$$- \frac{0.40191656771049 \alpha_0^{c_s} \alpha_1^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{0.92232932712305 \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.46845360803068 \alpha_0^s \alpha_1^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{0.9424780878218 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{15.157229920671 \alpha_0^{c_s} \alpha_0^s \left(\alpha_1^s\right)^2 r}{c_s}$$
$$- \frac{0.26542847151571 \alpha_0^s \alpha_1^{c_s} r^4 \log\left(c_s\right)}{c_s^4}$$

$$- \frac{17.070189849163 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2}{c_s^2}$$
$$- \frac{17.265090188972 \alpha_0^{c_s} \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$- \frac{0.036458333333333 \alpha_0^{c_s} \alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{19.139624219519 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r}{c_s}$$
$$- \frac{10.401045538112 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r}{c_s}$$
$$- \frac{0.020833333333333 \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.083333333333333 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{2.8326954049503 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{1.4228234146217 \alpha_1^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{2.8877107655697 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{0.73958333333333 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{6.1460976449703 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.19270833333333 \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{1.5643173084922 \left(\alpha_0^s\right)^4 \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.098958333333333 \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{1.5884741815718 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{6.3886016504085 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{1.6752754126021 \left(\alpha_0^s\right)^2 \alpha_3^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{7.0146178742227 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{1.7599670334241 \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.45560805454892 \alpha_0^s \alpha_3^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{14.707067627279 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r}{c_s}$$
$$- \frac{1.8602805292847 \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{15.16148696948 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r}{c_s}$$

$$
\begin{aligned}
&- \frac{3.9539880896494 \left(\alpha_0^{c_s}\right)^2 \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.125 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.25 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.375 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.625 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.9375 \alpha_0^{c_s} \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{3.75 \alpha_0^s \alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.56759071136537 \alpha_0^{c_s} \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.85138606704805 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{3.2027721340961 \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{6.243371063951 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.6520791005721 \alpha_0^s \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.7483744562548 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{1.8395791005721 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.8395791005721 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{2.0270791005721 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{2.5844883892067 \alpha_0^{c_s} \alpha_0^s \alpha_3^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{5.1689767784134 \alpha_0^{c_s} \alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{5.1689767784134 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{5.9291582011442 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{4.8548512346682 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{10.317828730023 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{10.735178932792 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{15.682802633666 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$+\frac{8.5357704348493\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{11.681799284612\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.2060453025497\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{14.532737823278\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{8.1445500735889\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{17.422836633219\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{8.7476479574518\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.29166666666667\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{2.4575502603924\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.5884741815718\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{3.4988253905886\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.0625\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\alpha_0^s\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.25\alpha_0^s\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.375\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.3125\alpha_0^{c_s}\alpha_0^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.625\alpha_0^{c_s}\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.625\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{1.25\alpha_0^s\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.21875\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.875\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.10546875\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^3}{c_s^3}$$

$$- \frac{0.56759071136537\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{11.748374147078\alpha_0^{c_s} \alpha_1^s \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.85138606704805 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{3.9527721340961 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{6.243371063951\alpha_0^{c_s} \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.87418722812738\alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{4.8041582011442 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{5.1689767784134\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{2.3336756173341\alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{9.7772583654733\alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{10.317828730023 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{10.419919562718\alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.7829825838101\alpha_0^{c_s} \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{5.9407619360049\alpha_0^{c_s} \alpha_0^s \alpha_3^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{49.191233535148\alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r}{c_s}$$
$$- \frac{6.5235686838326\alpha_0^s \alpha_1^s \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{7.7534651676201\alpha_0^{c_s} \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{7.8879529823954\alpha_0^s \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{8.876593450377 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{10.486494425184\alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{25.076682142992 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{13.746269384684 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{17.495295914904\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{2.2075502603924 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.5825502603924\alpha_0^s \alpha_1^{c_s} \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$

$$
\begin{aligned}
&- \frac{1.4162751301962\alpha_0^{c_s}\alpha_0^s\alpha_3^{c_s} r\log(c_s)}{c_s} \\
&- \frac{11.54864955968\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{23.391054225566\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{24.515181114859\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2\log(c_s)}{c_s^2} \\
&- \frac{3.3676073153919\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r^3\log(c_s)}{c_s^3} \\
&- \frac{6.9540580302476\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3\log(c_s)}{c_s^3} \\
&+ \frac{0.1875\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_1^s r\log(c_s)^3}{c_s} \\
&+ \frac{31.326333550619\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{5.5187373017162\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_1^s r\log(c_s)^2}{c_s}
\end{aligned}
$$

Absolute order 8.

$$
\begin{aligned}
R_{3,8} = &\frac{0.00068532023746048 r^8}{c_s^8} + \frac{0.125\alpha_6^s r}{c_s} \\
&+ \frac{4.3619461768873\left(\alpha_0^s\right)^4 r^4}{c_s^4} + \frac{0.038536574146577\left(\alpha_0^s\right)^7 r}{c_s} \\
&+ \frac{3.0274572868282\left(\alpha_0^{c_s}\right)^3 r^5}{c_s^5} + \frac{0.31438405110415\alpha_4^{c_s} r^3}{c_s^3} \\
&+ \frac{5.2029254536162\left(\alpha_0^s\right)^5 r^3}{c_s^3} + \frac{0.021703788578011\alpha_0^s r^7}{c_s^7} \\
&+ \frac{1.545267118531\left(\alpha_0^s\right)^3 r^5}{c_s^5} \\
&+ \frac{4.3981121305567\left(\alpha_0^{c_s}\right)^6 r^2}{c_s^2} \\
&+ \frac{10.143916233326\left(\alpha_0^{c_s}\right)^4 r^4}{c_s^4} \\
&+ \frac{0.044182861011306\left(\alpha_0^{c_s}\right)^7 r}{c_s} \\
&+ \frac{3.2121365269684\left(\alpha_2^{c_s}\right)^2 r^2}{c_s^2} \\
&+ \frac{1.7512659597646\left(\alpha_0^s\right)^6 r^2}{c_s^2} + \frac{0.027576492995468\alpha_0^{c_s} r^7}{c_s^7} \\
&+ \frac{1.010573413351\left(\alpha_1^s\right)^2 r^4}{c_s^4} \\
&+ \frac{0.26453516473845\left(\alpha_0^s\right)^2 r^6}{c_s^6} \\
&+ \frac{2.1474415948096\left(\alpha_1^{c_s}\right)^2 r^4}{c_s^4} + \frac{0.13887279321988\alpha_4^s r^3}{c_s^3} \\
&+ \frac{0.1509154889293\alpha_2^{c_s} r^5}{c_s^5} + \frac{0.087120557839057\alpha_2^s r^5}{c_s^5} \\
&+ \frac{0.37743788705055\alpha_6^{c_s} r}{c_s} + \frac{1.524715047639\left(\alpha_2^s\right)^2 r^2}{c_s^2}
\end{aligned}
$$

$$+\frac{13.489795581172\left(\alpha_0^{c_s}\right)^5 r^3}{c_s^3}$$
$$+\frac{0.42180491085016\left(\alpha_0^{c_s}\right)^2 r^6}{c_s^6}$$
$$-\frac{3.8575245524135\left(\alpha_1^s\right)^3 r^2}{c_s^2}-\frac{0.23898953217492\alpha_3^{c_s} r^4}{c_s^4}$$
$$-\frac{0.076159312110837\alpha_1^{c_s} r^6}{c_s^6}-\frac{0.05080989468861\alpha_1^s r^6}{c_s^6}$$
$$-\frac{0.13952053443311\alpha_5^s r^2}{c_s^2}-\frac{8.9617971960467\left(\alpha_1^{c_s}\right)^3 r^2}{c_s^2}$$
$$-\frac{0.36188908190462\alpha_5^{c_s} r^2}{c_s^2}-\frac{0.12024763108045\alpha_3^s r^4}{c_s^4}$$
$$+\frac{0.0009765625\alpha_4^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$+\frac{0.015625\left(\alpha_0^s\right)^4 r^4\log\left(c_s\right)^4}{c_s^4}$$
$$+\frac{0.0625\alpha_6^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.0029296875\left(\alpha_1^s\right)^2 r^4\log\left(c_s\right)^4}{c_s^4}$$
$$+\frac{0.71875\left(\alpha_1^s\right)^3 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.0791015625\left(\alpha_0^s\right)^5 r^3\log\left(c_s\right)^4}{c_s^3}$$
$$+\frac{98.264458246806\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^2 r^2}{c_s^2}$$
$$+\frac{10.824539214502\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^4 r}{c_s}$$
$$+\frac{3.8435304034727\left(\alpha_0^s\right)^4\alpha_2^{c_s} r}{c_s}$$
$$+\frac{0.12260923390637\alpha_5^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.1971819419038\alpha_0^s\alpha_4^s r^2}{c_s^2}+\frac{6.2197975239915\alpha_2^{c_s}\alpha_2^s r^2}{c_s^2}$$
$$+\frac{33.420004921268\left(\alpha_0^{c_s}\right)^3\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{4.4208435529258\left(\alpha_0^{c_s}\right)^3\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{18.5185882453 08\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r^2}{c_s^2}$$
$$+\frac{37.766986363082\left(\alpha_0^{c_s}\right)^3\alpha_0^s r^4}{c_s^4}$$
$$+\frac{0.076712092161027\alpha_3^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{39.647700620735\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^3}{c_s^3}$$
$$+\frac{10.320494578314\left(\alpha_0^s\right)^3\alpha_2^s r^2}{c_s^2}$$

$$+\frac{1.4593688359096\alpha_0^s\alpha_2^s r^4}{c_s^4}$$
$$+\frac{24.768976538392\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2 r^3}{c_s^3}$$
$$+\frac{6.2414676313022\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^5}{c_s^5}$$
$$+\frac{1.8585804071299\left(\alpha_1^{c_s}\right)^2\alpha_2^{c_s} r}{c_s}$$
$$+\frac{3.7897006163264\left(\alpha_0^s\right)^2\alpha_4^{c_s} r}{c_s}$$
$$+\frac{4.2074047106112\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^5 r}{c_s}$$
$$+\frac{34.146037755265\left(\alpha_0^{c_s}\right)^3\alpha_2^s r^2}{c_s^2}$$
$$+\frac{17.633161008062\left(\alpha_0^{c_s}\right)^2\alpha_2^{c_s} r^3}{c_s^3}$$
$$+\frac{35.316943241358\left(\alpha_0^{c_s}\right)^5\alpha_0^s r^2}{c_s^2}$$
$$+\frac{4.6296729998974\left(\alpha_1^s\right)^3 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{4.7937562503095\left(\alpha_1^s\right)^2\alpha_2^s r}{c_s}$$
$$+\frac{10.176804101055\alpha_0^s\left(\alpha_1^s\right)^2 r^3}{c_s^3}$$
$$+\frac{2.614407465451\alpha_0^s\alpha_2^{c_s} r^4}{c_s^4}$$
$$+\frac{21.205426936281\left(\alpha_0^s\right)^3\alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{0.68765237257361\alpha_0^{c_s}\left(\alpha_0^s\right)^6 r}{c_s}$$
$$+\frac{5.5834226551964\left(\alpha_0^{c_s}\right)^2\alpha_4^s r}{c_s}$$
$$+\frac{23.201685922758\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r^3}{c_s^3}$$
$$+\frac{12.280199534337\left(\alpha_0^{c_s}\right)^4\alpha_2^s r}{c_s}$$
$$+\frac{3.0830831008158\alpha_0^{c_s}\alpha_2^{c_s} r^4}{c_s^4}$$
$$+\frac{0.012579815039661\left(\alpha_0^s\right)^2 r^6\log\left(c_s\right)^2}{c_s^6}$$
$$+\frac{0.051404189589007\left(\alpha_0^s\right)^7 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.3523814346221\alpha_0^s\left(\alpha_2^s\right)^2 r}{c_s}$$
$$+\frac{108.50062532693\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^3}{c_s^3}$$
$$+\frac{0.42644940973027\alpha_0^{c_s}\left(\alpha_2^{c_s}\right)^2 r}{c_s}$$

$$+\frac{3.5103273156317\alpha_0^s\left(\alpha_2^{c_s}\right)^2 r}{c_s}$$
$$+\frac{56.710339876453\left(\alpha_0^s\right)^2\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2}$$
$$+\frac{7.272533332723\left(\alpha_0^s\right)^2\alpha_2^s r^3}{c_s^3}$$
$$+\frac{0.23422680401534\left(\alpha_1^s\right)^2 r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{1.9853251979163\alpha_0^{c_s}\alpha_4^s r^2}{c_s^2}$$
$$+\frac{16.060033221664\left(\alpha_1^{c_s}\right)^2\alpha_2^s r}{c_s}$$
$$+\frac{2.0520702772579\alpha_0^{c_s}\alpha_2^s r^4}{c_s^4}$$
$$+\frac{132.14217429804\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^3}{c_s^3}$$
$$+\frac{8.2892279562877\alpha_0^{c_s}\left(\alpha_2^s\right)^2 r}{c_s}$$
$$+\frac{4.3030456716032\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{70.352560051611\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r^2}{c_s^2}$$
$$+\frac{0.0086286943058355\alpha_2^s r^5\log\left(c_s\right)^2}{c_s^5}$$
$$+\frac{71.14123601326\left(\alpha_0^{c_s}\right)^4\alpha_0^s r^3}{c_s^3}$$
$$+\frac{4.5348709194472\alpha_1^{c_s}\alpha_3^s r^2}{c_s^2}$$
$$+\frac{0.072916666666667\left(\alpha_0^s\right)^6 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.03684759723458\alpha_4^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{2.3594858834324\left(\alpha_0^s\right)^5 r^3\log\left(c_s\right)^2}{c_s^3}$$

$$+\frac{0.018648400200653\alpha_1^s r^6\log\left(c_s\right)}{c_s^6}$$
$$+\frac{1.1965601393832\left(\alpha_0^s\right)^4 r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{2.4584809206766\alpha_1^s\alpha_3^s r^2}{c_s^2}$$
$$+\frac{1.2398128438351\left(\alpha_0^s\right)^6 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{5.1623783406682\left(\alpha_0^{c_s}\right)^5\left(\alpha_0^s\right)^2 r}{c_s}$$
$$+\frac{0.0052083333333333\alpha_6^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.0052083333333333\left(\alpha_0^s\right)^7 r\log\left(c_s\right)^4}{c_s}$$

$$+\frac{5.3899191279918\alpha_1^{c_s}\alpha_3^{c_s}r^2}{c_s^2}$$
$$+\frac{1.355409833533\left(\alpha_0^s\right)^2\alpha_4^s r}{c_s}$$
$$+\frac{21.892191226939\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^2}{c_s^2}$$
$$+\frac{5.4804032297212\alpha_1^s\alpha_3^{c_s}r^2}{c_s^2}$$
$$+\frac{0.68903390319704\alpha_0^{c_s}\alpha_0^s r^6}{c_s^6}$$
$$+\frac{1.4423124201304\left(\alpha_0^{c_s}\right)^4\alpha_2^{c_s}r}{c_s}$$
$$+\frac{2.88779300228\alpha_0^s\alpha_4^{c_s}r^2}{c_s^2}$$
$$+\frac{2.9009007247957\alpha_0^{c_s}\alpha_4^{c_s}r^2}{c_s^2}$$
$$+\frac{0.0028483072916667\alpha_3^s r^4\log\left(c_s\right)^3}{c_s^4}$$
$$+\frac{11.85033531458\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^3 r}{c_s}$$
$$+\frac{1.4900274961683\left(\alpha_0^{c_s}\right)^2\alpha_4^{c_s}r}{c_s}$$
$$+\frac{47.906945044441\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^4}{c_s^4}$$
$$+\frac{1.5128907607855\left(\alpha_0^s\right)^4\alpha_2^s r}{c_s}$$
$$+\frac{12.104397163036\left(\alpha_1^s\right)^2\alpha_2^{c_s}r}{c_s}$$
$$+\frac{24.641603124434\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^4}{c_s^4}$$
$$+\frac{0.048177083333333\left(\alpha_2^s\right)^2 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{24.789686125124\left(\alpha_0^{c_s}\right)^3\alpha_2^{c_s}r^2}{c_s^2}$$
$$+\frac{0.78127154509942\left(\alpha_0^{c_s}\right)^6\alpha_0^s r}{c_s}$$
$$+\frac{3.2376109726285\alpha_1^{c_s}\alpha_1^s r^4}{c_s^4}$$
$$+\frac{52.644918053431\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2}$$
$$+\frac{0.013020833333333\alpha_5^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{13.692096829721\left(\alpha_0^s\right)^2\alpha_2^{c_s}r^3}{c_s^3}$$
$$+\frac{0.21571735764589\left(\alpha_0^s\right)^3 r^5\log\left(c_s\right)^2}{c_s^5}$$
$$+\frac{0.87998351671206\left(\alpha_2^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$

$$+\frac{0.88068158766941\alpha_2^{c_s}\alpha_3^{c_s}r}{c_s}$$
$$+\frac{57.627812385852\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2r^2}{c_s^2}$$
$$+\frac{14.930341170833\left(\alpha_0^{c_s}\right)^2\alpha_2^sr^3}{c_s^3}$$
$$+\frac{15.006163888297\left(\alpha_0^s\right)^3\left(\alpha_1^{c_s}\right)^2r}{c_s}$$
$$+\frac{121.08586126937\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3r^2}{c_s^2}$$
$$+\frac{15.456758869991\alpha_0^{c_s}\left(\alpha_1^s\right)^2r^3}{c_s^3}$$
$$+\frac{7.8212238121398\left(\alpha_0^{c_s}\right)^2\alpha_0^sr^5}{c_s^5}$$
$$-\frac{0.000244140625\alpha_3^sr^4\log\left(c_s\right)^4}{c_s^4}$$
$$-\frac{0.125\alpha_6^sr\log\left(c_s\right)}{c_s}$$
$$-\frac{0.375\left(\alpha_0^s\right)^6r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.140625\left(\alpha_1^s\right)^3r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.0341796875\left(\alpha_1^s\right)^2r^4\log\left(c_s\right)^3}{c_s^4}$$
$$-\frac{0.54052734375\left(\alpha_0^s\right)^5r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{1.6168598031015\alpha_0^s\alpha_1^{c_s}r^5}{c_s^5}$$
$$-\frac{13.246681635025\left(\alpha_0^{c_s}\right)^2\alpha_1^sr^4}{c_s^4}$$
$$-\frac{0.065244100954277\left(\alpha_0^s\right)^7r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.75724388321833\left(\alpha_1^s\right)^2r^4\log\left(c_s\right)}{c_s^4}$$
$$-\frac{3.7491651262951\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^3r}{c_s}$$
$$-\frac{2.2394875446449\left(\alpha_0^s\right)^6r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{4.4850321146896\alpha_1^{c_s}\alpha_2^sr^3}{c_s^3}$$
$$-\frac{4.7049024656217\left(\alpha_0^s\right)^3\alpha_3^{c_s}r}{c_s}$$
$$-\frac{10.339146348078\left(\alpha_0^s\right)^2\alpha_3^{c_s}r^2}{c_s^2}$$
$$-\frac{11.001103173563\left(\alpha_0^{c_s}\right)^2\alpha_3^{c_s}r^2}{c_s^2}$$
$$-\frac{11.197272853449\alpha_0^{c_s}\left(\alpha_1^s\right)^3r}{c_s}$$

$$- \frac{1.4121292935308 \alpha_0^{c_s} \alpha_1^s r^5}{c_s^5}$$
$$- \frac{11.607760056455 \left(\alpha_0^{c_s}\right)^2 \alpha_3^s r^2}{c_s^2}$$
$$- \frac{1.4734061275959 \left(\alpha_0^{c_s}\right)^3 \alpha_3^{c_s} r}{c_s}$$
$$- \frac{12.073455969581 \left(\alpha_0^{c_s}\right)^3 \alpha_3^s r}{c_s}$$
$$- \frac{26.296432524778 \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r^2}{c_s^2}$$
$$- \frac{1.7733901443961 \left(\alpha_0^s\right)^3 \alpha_3^s r}{c_s}$$
$$- \frac{0.11189040120392 \left(\alpha_0^s\right)^2 r^6 \log\left(c_s\right)}{c_s^6}$$
$$- \frac{31.86080314228 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^3}{c_s^3}$$
$$- \frac{8.068519509323 \alpha_0^s \left(\alpha_1^{c_s}\right)^3 r}{c_s}$$
$$- \frac{2.0871567051784 \left(\alpha_0^s\right)^5 \alpha_1^{c_s} r}{c_s}$$
$$- \frac{1.0496651482817 \alpha_0^s \alpha_1^s r^5}{c_s^5}$$
$$- \frac{0.0020966358399435 \alpha_1^s r^6 \log\left(c_s\right)^2}{c_s^6}$$
$$- \frac{2.2843695385284 \alpha_0^{c_s} \alpha_3^s r^3}{c_s^3}$$
$$- \frac{4.6393166739312 \left(\alpha_0^s\right)^2 \alpha_3^s r^2}{c_s^2}$$
$$- \frac{0.61616427546487 \alpha_0^s \alpha_5^s r}{c_s}$$
$$- \frac{2.4771339293142 \left(\alpha_1^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{5.4034295500695 \left(\alpha_0^s\right)^5 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{0.021164196648111 \alpha_3^s r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{0.044182253542082 \alpha_2^s r^5 \log\left(c_s\right)}{c_s^5}$$
$$- \frac{47.324281612349 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r^3}{c_s^3}$$
$$- \frac{6.1087812576758 \alpha_1^{c_s} \alpha_2^{c_s} r^3}{c_s^3}$$
$$- \frac{0.10547078563903 \alpha_4^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{13.841809923869 \left(\alpha_0^s\right)^4 \alpha_1^s r^2}{c_s^2}$$
$$- \frac{1.8024814335223 \alpha_0^s \alpha_5^{c_s} r}{c_s}$$

$$- \frac{0.94350439591129 \left(\alpha_0^{c_s}\right)^5 \alpha_1^{c_s} r}{c_s}$$

$$- \frac{16.202813491373 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r^4}{c_s^4}$$

$$- \frac{0.015894571940104 \left(\alpha_0^{s}\right)^3 r^5 \log\left(c_s\right)^3}{c_s^5}$$

$$- \frac{34.154386842901 \left(\alpha_0^{c_s}\right)^4 \alpha_1^{c_s} r^2}{c_s^2}$$

$$- \frac{4.4309930101446 \alpha_0^{s} \left(\alpha_1^{s}\right)^3 r}{c_s}$$

$$- \frac{1.15796444591 \alpha_0^{c_s} \alpha_5^{c_s} r}{c_s} - \frac{4.6706371720921 \alpha_2^{c_s} \alpha_3^{s} r}{c_s}$$

$$- \frac{18.743138308883 \left(\alpha_0^{s}\right)^3 \alpha_1^{s} r^3}{c_s^3}$$

$$- \frac{0.0046228559331992 \alpha_0^{s} r^7 \log\left(c_s\right)}{c_s^7}$$

$$- \frac{4.9861850088163 \alpha_1^{s} \alpha_2^{c_s} r^3}{c_s^3} - \frac{5.0501924536599 \alpha_2^{s} \alpha_3^{c_s} r}{c_s}$$

$$- \frac{1.2941090751121 \alpha_0^{c_s} \alpha_5^{s} r}{c_s}$$

$$- \frac{5.1993504732453 \left(\alpha_0^{c_s}\right)^5 \alpha_1^{s} r}{c_s}$$

$$- \frac{0.00063578287760417 \alpha_2^{s} r^5 \log\left(c_s\right)^3}{c_s^5}$$

$$- \frac{2.6475317332077 \alpha_1^{s} \alpha_2^{s} r^3}{c_s^3}$$

$$- \frac{0.0026041666666667 \alpha_5^{s} r^2 \log\left(c_s\right)^4}{c_s^2}$$

$$- \frac{0.020833333333333 \alpha_6^{s} r \log\left(c_s\right)^3}{c_s}$$

$$- \frac{0.020833333333333 \left(\alpha_0^{s}\right)^7 r \log\left(c_s\right)^3}{c_s}$$

$$- \frac{21.834476262622 \alpha_1^{c_s} \left(\alpha_1^{s}\right)^2 r^2}{c_s^2}$$

$$- \frac{1.3805669230387 \alpha_1^{s} \alpha_4^{s} r}{c_s}$$

$$- \frac{11.608444073165 \left(\alpha_0^{s}\right)^2 \alpha_1^{c_s} r^4}{c_s^4}$$

$$- \frac{0.18229166666667 \left(\alpha_0^{s}\right)^4 r^4 \log\left(c_s\right)^3}{c_s^4}$$

$$- \frac{47.29931057018 \left(\alpha_0^{c_s}\right)^3 \alpha_1^{c_s} r^3}{c_s^3}$$

$$- \frac{0.7556918108422 \alpha_1^{c_s} \alpha_4^{c_s} r}{c_s} - \frac{1.5289756803498 \alpha_0^{s} \alpha_3^{s} r^3}{c_s^3}$$

$$- \frac{3.088374616798 \alpha_1^{c_s} \alpha_4^{s} r}{c_s} - \frac{3.1712256043293 \alpha_0^{s} \alpha_3^{c_s} r^3}{c_s^3}$$

$$- \frac{50.916277011958 \left(\alpha_0^{c_s}\right)^4 \alpha_1^{s} r^2}{c_s^2}$$

$$
\begin{aligned}
&- \frac{0.051913547397252\alpha_5^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.83496561933704\left(\alpha_0^s\right)^5 \alpha_1^s r}{c_s} \\
&- \frac{1.7276736271957\left(\alpha_2^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.5605676587173\alpha_0^{c_s} \alpha_3^{c_s} r^3}{c_s^3} \\
&- \frac{0.0069869679166667\alpha_4^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&- \frac{7.2513766025917\left(\alpha_0^s\right)^2 \alpha_1^s r^4}{c_s^4} \\
&- \frac{3.6310986009712\left(\alpha_0^s\right)^4 r^4 \log\left(c_s\right)}{c_s^4} \\
&- \frac{29.722182618082\left(\alpha_1^{c_s}\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{3.8488643697255\alpha_1^s \alpha_4^{c_s} r}{c_s} - \frac{1.9358793917918\alpha_2^s \alpha_3^s r}{c_s} \\
&- \frac{0.97227731175753\left(\alpha_0^s\right)^3 r^5 \log\left(c_s\right)}{c_s^5} \\
&- \frac{1.9528034530067\alpha_0^{c_s} \alpha_1^{c_s} r^5}{c_s^5} \\
&- \frac{0.24479166666667\left(\alpha_2^s\right)^2 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.00390625\alpha_0^s \alpha_2^s r^4 \log\left(c_s\right)^4}{c_s^4} \\
&+ \frac{0.015625\alpha_0^{c_s}\left(\alpha_0^s\right)^6 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.015625\left(\alpha_1^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.015625\left(\alpha_0^{c_s}\right)^2 \alpha_4^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.015625\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.015625\left(\alpha_0^{c_s}\right)^3 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.375\left(\alpha_0^s\right)^3 \alpha_2^s r^2 \log\left(c_s\right)^4}{c_s^2} \\
&+ \frac{0.15625\alpha_0^{c_s}\left(\alpha_2^s\right)^2 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.0546875\left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^4}{c_s^2} \\
&+ \frac{0.109375\left(\alpha_0^s\right)^2 \alpha_4^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.4375\alpha_1^s \alpha_4^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.4375\left(\alpha_0^s\right)^5 \alpha_1^s r \log\left(c_s\right)^3}{c_s}
\end{aligned}
$$

$$+\frac{0.0263671875\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^3\log\left(c_s\right)^4}{c_s^3}$$
$$+\frac{0.2734375\left(\alpha_0^s\right)^2\alpha_1^s r^4\log\left(c_s\right)^3}{c_s^4}$$
$$+\frac{0.546875\left(\alpha_1^s\right)^2\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.546875\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{2.1875\alpha_0^s\left(\alpha_1^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.087890625\left(\alpha_0^s\right)^2\alpha_2^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$+\frac{0.734375\left(\alpha_0^s\right)^2\alpha_3^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.0791015625\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^3\log\left(c_s\right)^4}{c_s^3}$$
$$+\frac{0.8359375\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.1318359375\alpha_0^s\left(\alpha_1^s\right)^2 r^3\log\left(c_s\right)^4}{c_s^3}$$
$$+\frac{0.2041015625\alpha_1^s\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{0.10302734375\alpha_0^s\alpha_3^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{0.0095367431640625\alpha_0^s\alpha_1^s r^5\log\left(c_s\right)^3}{c_s^5}$$
$$+\frac{1.810546875\left(\alpha_0^s\right)^3\alpha_1^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{2.287346293977\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.047299225947114\alpha_2^s\alpha_3^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.047299225947114\left(\alpha_0^s\right)^3\alpha_3^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{32.710142004631\alpha_0^{c_s}\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{0.17229922594711\alpha_0^{c_s}\alpha_5^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.26689767784134\alpha_1^{c_s}\alpha_4^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.26689767784134\left(\alpha_0^s\right)^5\alpha_1^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{81.342255008749\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{21.448993995275\left(\alpha_0^s\right)^3\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{45.131403991963\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_2^s r}{c_s}$$

$$+\frac{5.8741870735389\alpha_0^{c_s}\left(\alpha_2^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.4855117920869\alpha_0^{c_s}\alpha_3^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{13.159406854523\alpha_0^{c_s}\alpha_1^s\alpha_3^s r}{c_s}$$
$$+\frac{1.0371891632596\left(\alpha_0^s\right)^4\alpha_1^{c_s} r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.56761472060586\alpha_0^{c_s}\alpha_4^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.035883158345252\alpha_1^{c_s}\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{1.3109602269289\left(\alpha_0^s\right)^4\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{53.639890865421\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r^2}{c_s^2}$$
$$+\frac{7.3777541260213\alpha_0^s\left(\alpha_1^s\right)^3 r\log\left(c_s\right)}{c_s}$$

$$+\frac{0.058832377095252\alpha_0^{c_s}\alpha_3^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{1.9437066135917\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^5 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{33.196509743454\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r^3}{c_s^3}$$
$$+\frac{0.73087387125098\alpha_0^s\alpha_5^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.7778918724447\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.97046673867656\left(\alpha_0^{c_s}\right)^4\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.97046673867656\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.3683440805453\alpha_1^{c_s}\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{2.5844883892067\alpha_0^{c_s}\left(\alpha_1^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{12.940217264921\alpha_0^s\alpha_1^{c_s}\alpha_3^s r}{c_s}$$
$$+\frac{3.3445455614335\alpha_2^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{7.7328550642049\left(\alpha_0^{c_s}\right)^2\alpha_3^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{16.425563712197\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{65.74541190349\left(\alpha_0^{c_s}\right)^2\alpha_0^s\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{0.51503239853261\alpha_0^{c_s}\alpha_1^s r^5\log\left(c_s\right)}{c_s^5}$$

$$+\frac{2.2475141489816\left(\alpha_1^{c_s}\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.2475141489816\left(\alpha_0^s\right)^3\left(\alpha_1^{c_s}\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.60982219241342\alpha_0^s\alpha_1^s r^5\log\left(c_s\right)}{c_s^5}$$
$$+\frac{0.613413888573\left(\alpha_0^s\right)^2\alpha_2^{c_s} r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{1.2557055991705\alpha_0^s\left(\alpha_1^{c_s}\right)^3 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.32294842510727\left(\alpha_0^s\right)^3\alpha_1^{c_s} r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{1.327530293205\alpha_0^s\alpha_3^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{21.696377314439\left(\alpha_0^s\right)^4\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{11.101166727647\left(\alpha_1^{c_s}\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{44.647816264883\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r^3}{c_s^3}$$
$$+\frac{2.8352740739136\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{5.7522994836017\left(\alpha_0^s\right)^2\alpha_1^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{0.36074999115751\alpha_0^s\alpha_5^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{6.6887262376902\alpha_0^{c_s}\alpha_0^s\alpha_4^{c_s} r}{c_s}$$
$$+\frac{3.8202094794504\alpha_0^s\left(\alpha_2^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.897507710927\alpha_2^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.9439764911977\left(\alpha_1^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{4.1553237315655\alpha_0^s\left(\alpha_1^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.27629297855756\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^5\log\left(c_s\right)^2}{c_s^5}$$
$$+\frac{4.7412780427983\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{1.1904460854791\left(\alpha_0^s\right)^2\alpha_4^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.4263344493635\alpha_1^s\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{19.575318694228\left(\alpha_0^s\right)^3\alpha_1^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{10.157165765854\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^3\log\left(c_s\right)^2}{c_s^3}$$

$$+\frac{41.548471588379\left(\alpha_0^{c_s}\right)^3\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{1.3411481802736\alpha_1^s\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{86.024005584334\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^{c_s}r^2}{c_s^2}$$
$$+\frac{5.4470368262232\alpha_0^s\alpha_1^s\alpha_3^s r}{c_s}$$
$$+\frac{22.000057340758\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^3}{c_s^3}$$
$$+\frac{5.5083980109557\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{46.416168004781\alpha_0^{c_s}\left(\alpha_0^s\right)^2\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{23.742544787325\alpha_1^{c_s}\alpha_1^s\alpha_2^{c_s}r}{c_s}$$
$$+\frac{12.538341071496\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.20095828385524\alpha_0^{c_s}\alpha_2^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{3.2153325416839\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{1.6656048141581\alpha_1^s\alpha_4^{c_s}r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.10412751301962\left(\alpha_0^{c_s}\right)^5\left(\alpha_0^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{6.8731346923418\left(\alpha_0^{c_s}\right)^3\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{14.404113923112\alpha_0^s\alpha_1^s\alpha_3^{c_s}r}{c_s}$$
$$+\frac{0.22523101117468\left(\alpha_0^{c_s}\right)^3\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{15.157229920671\alpha_0^{c_s}\left(\alpha_1^s\right)^3 r\log\left(c_s\right)}{c_s}$$
$$+\frac{15.167373100284\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_2^s r}{c_s}$$
$$+\frac{1.9847558971257\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{8.0358221286421\alpha_0^s\alpha_1^{c_s}\alpha_3^{c_s}r}{c_s}$$
$$+\frac{16.377283754452\left(\alpha_0^{c_s}\right)^3\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{16.453977631147\left(\alpha_0^{c_s}\right)^4\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{16.750161562271\alpha_0^s\alpha_2^{c_s}\alpha_2^s r}{c_s}$$
$$+\frac{16.854963568914\alpha_0^{c_s}\alpha_1^s\alpha_3^{c_s}r}{c_s}$$
$$+\frac{0.13271423575785\alpha_1^{c_s}\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$

$$+\frac{136.03500112836\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}$$
$$+\frac{17.070189849163\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{4.3040643501639\left(\alpha_0^s\right)^2\alpha_3^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.26971395748157\left(\alpha_0^{c_s}\right)^2\alpha_3^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{34.579297391127\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^3}{c_s^3}$$
$$+\frac{17.337548543965\left(\alpha_0^s\right)^4\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{8.690822571652\left(\alpha_0^{c_s}\right)^3\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{34.780611557617\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_2^{c_s} r}{c_s}$$
$$+\frac{2.248539956157\alpha_0^{c_s}\alpha_1^{c_s}\alpha_3^{c_s} r}{c_s}$$
$$+\frac{0.56499175835603\alpha_0^s\alpha_4^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.5741037348607\alpha_0^{c_s}\left(\alpha_0^s\right)^6 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.3003770986574\left(\alpha_0^s\right)^2\left(\alpha_1^{c_s}\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.0091145833333333\alpha_0^{c_s}\alpha_4^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.072916666666667\alpha_1^s\alpha_3^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.072916666666667\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.14583333333333\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)^4}{c_s^2}$$

$$+\frac{0.14583333333333\alpha_0^s\alpha_5^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{18.902181226012\left(\alpha_0^s\right)^3\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{76.390826297246\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r^2}{c_s^2}$$
$$+\frac{4.7950678185705\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{4.8020988647039\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{2.4730563846109\left(\alpha_0^s\right)^5\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{80.30483127493\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^s r^2}{c_s^2}$$
$$+\frac{1.2622310659043\alpha_0^{c_s}\alpha_5^s r\log\left(c_s\right)}{c_s}$$

$$+\frac{0.31888359060778\alpha_0^s\alpha_2^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{82.850057099823\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}$$
$$+\frac{5.1873203270962\left(\alpha_0^s\right)^2\alpha_3^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.66258325965537\alpha_0^s\alpha_3^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.16662751301962\left(\alpha_0^s\right)^2\alpha_4^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.0052083333333333\alpha_1^{c_s}\alpha_3^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{0.0052083333333333\left(\alpha_0^s\right)^4\alpha_2^{c_s} r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.0052083333333333\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{21.616074629\alpha_0^{c_s}\alpha_1^{c_s}\alpha_3^s r}{c_s}$$
$$+\frac{0.33807394099091\alpha_0^s\alpha_1^{c_s} r^5\log\left(c_s\right)}{c_s^5}$$
$$+\frac{2.7334165399115\left(\alpha_0^s\right)^3\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.405267688428\left(\alpha_0^s\right)^5\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.8333333333333\left(\alpha_0^s\right)^4\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{5.6748351671206\left(\alpha_1^s\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.8396790979957\left(\alpha_0^s\right)^2\alpha_2^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{45.468588117992\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_2^s r}{c_s}$$
$$+\frac{0.71141170731087\alpha_2^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{2.8456468292435\left(\alpha_0^s\right)^3\alpha_2^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{22.880228883957\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_2^{c_s} r}{c_s}$$
$$+\frac{2.8700480383984\alpha_1^{c_s}\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.4438553827849\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.18229166666667\left(\alpha_0^s\right)^4\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.36458333333333\alpha_0^s\left(\alpha_2^s\right)^2 r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.72916666666667\alpha_2^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$

$$+\frac{0.72916666666667\left(\alpha_0^s\right)^3\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.459738851267\left(\alpha_0^{c_s}\right)^5\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.9337387125098\alpha_2^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.18356434450801\alpha_2^{c_s}\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{23.528015347915\alpha_1^{c_s}\alpha_1^s\alpha_2^s r}{c_s}$$
$$+\frac{24.329771236758\alpha_0^{c_s}\alpha_2^{c_s}\alpha_2^s r}{c_s}$$
$$+\frac{3.0730488224852\left(\alpha_0^{c_s}\right)^3\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{12.292195289941\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{6.1642879733471\alpha_0^{c_s}\alpha_0^s\alpha_4^s r}{c_s}$$
$$+\frac{12.994698445064\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{6.5781390816415\left(\alpha_0^s\right)^3\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{3.2941625300387\left(\alpha_0^s\right)^3\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.4770290151238\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{14.343405612087\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_2^{c_s} r}{c_s}$$
$$+\frac{14.537170334241\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.22780402727446\alpha_1^s\alpha_3^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{29.183884329233\left(\alpha_0^{c_s}\right)^2\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{1.8546256876702\left(\alpha_0^s\right)^4\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.93014026464233\alpha_1^{c_s}\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{120.85823041059\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2}$$
$$+\frac{1.9043693562549\alpha_1^s\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.9050809010702\alpha_1^s\alpha_2^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.029947916666667\alpha_0^s\alpha_4^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{1.9769940448247\left(\alpha_0^{c_s}\right)^2\alpha_4^s r\log\left(c_s\right)^2}{c_s}$$

$$+\frac{7.9897053400048\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.015625\alpha_0^{c_s}\alpha_5^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.015625\alpha_1^{c_s}\alpha_4^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.015625\left(\alpha_0^s\right)^5\alpha_1^{c_s} r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.03125\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.0029296875\alpha_0^{c_s}\alpha_3^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$-\frac{0.0234375\left(\alpha_0^s\right)^2\alpha_1^s r^4\log\left(c_s\right)^4}{c_s^4}$$
$$-\frac{0.109375\alpha_1^s\alpha_4^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.109375\left(\alpha_0^s\right)^5\alpha_1^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.4375\left(\alpha_0^s\right)^2\alpha_4^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0146484375\alpha_0^s\alpha_3^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$-\frac{0.029296875\alpha_1^s\alpha_2^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$-\frac{0.234375\alpha_0^{c_s}\left(\alpha_1^s\right)^3 r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.546875\alpha_0^s\left(\alpha_1^s\right)^3 r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{2.1875\left(\alpha_1^s\right)^2\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{2.1875\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.14453125\left(\alpha_0^s\right)^2\alpha_3^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$-\frac{0.609375\left(\alpha_0^s\right)^2\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.263671875\left(\alpha_0^s\right)^3\alpha_1^s r^3\log\left(c_s\right)^4}{c_s^3}$$
$$-\frac{4.28125\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.90966796875\alpha_0^s\left(\alpha_1^s\right)^2 r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.047299225947114\left(\alpha_0^{c_s}\right)^4\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.047299225947114\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.14189767784134\left(\alpha_1^{c_s}\right)^2\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$

$$- \frac{0.14189767784134 \left(\alpha_{0}^{s}\right)^{3} \left(\alpha_{1}^{c_{s}}\right)^{2} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{0.17229922594711 \alpha_{0}^{c_{s}} \left(\alpha_{0}^{s}\right)^{6} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{0.32939767784134 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{4}^{s} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{0.32939767784134 \left(\alpha_{0}^{c_{s}}\right)^{2} \left(\alpha_{0}^{s}\right)^{5} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{10.623428137137 \alpha_{0}^{c_{s}} \left(\alpha_{2}^{s}\right)^{2} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{50.194489593759 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{1}^{c_{s}} \alpha_{2}^{s} r}{c_{s}}$$
$$- \frac{0.12964864540745 \alpha_{1}^{c_{s}} \alpha_{3}^{s} r^{2} \log\left(c_{s}\right)^{3}}{c_{s}^{2}}$$
$$- \frac{0.55069303352403 \left(\alpha_{1}^{s}\right)^{2} \alpha_{2}^{c_{s}} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{10.09950678112 \left(\alpha_{1}^{c_{s}}\right)^{2} \alpha_{2}^{s} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{0.67569303352403 \left(\alpha_{0}^{c_{s}}\right)^{3} \left(\alpha_{1}^{s}\right)^{2} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{46.008166558187 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{1}^{s} \alpha_{2}^{s} r}{c_{s}}$$
$$- \frac{24.592014399009 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{0}^{s} \alpha_{3}^{s} r}{c_{s}}$$
$$- \frac{1.7229922594711 \alpha_{0}^{c_{s}} \left(\alpha_{2}^{s}\right)^{2} r \log\left(c_{s}\right)^{3}}{c_{s}}$$
$$- \frac{1.7871891632596 \alpha_{0}^{c_{s}} \left(\alpha_{0}^{s}\right)^{5} r^{2} \log\left(c_{s}\right)^{3}}{c_{s}^{2}}$$
$$- \frac{147.72941224971 \alpha_{0}^{c_{s}} \left(\alpha_{0}^{s}\right)^{3} \alpha_{1}^{c_{s}} r^{2}}{c_{s}^{2}}$$
$$- \frac{1.2055135800677 \left(\alpha_{0}^{s}\right)^{5} \alpha_{1}^{c_{s}} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{0.60739116609368 \alpha_{0}^{c_{s}} \alpha_{5}^{s} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{0.076954635474829 \left(\alpha_{0}^{c_{s}}\right)^{6} \alpha_{0}^{s} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{5.5638770630106 \alpha_{0}^{s} \left(\alpha_{1}^{s}\right)^{3} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{0.73227370515254 \alpha_{2}^{s} \alpha_{3}^{c_{s}} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{0.73227370515254 \left(\alpha_{0}^{s}\right)^{3} \alpha_{3}^{c_{s}} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{25.492893672906 \left(\alpha_{0}^{c_{s}}\right)^{3} \left(\alpha_{1}^{s}\right)^{2} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{1.6182837448894 \left(\alpha_{0}^{c_{s}}\right)^{2} \left(\alpha_{1}^{s}\right)^{2} r^{2} \log\left(c_{s}\right)^{3}}{c_{s}^{2}}$$
$$- \frac{0.10471978753576 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{2}^{s} r^{3} \log\left(c_{s}\right)^{3}}{c_{s}^{3}}$$

$$- \frac{2.483865705998 \left(\alpha_{0}^{s}\right)^{4} \alpha_{2}^{s} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{1.3220560373399 \alpha_{2}^{c_{s}} \alpha_{3}^{s} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{6.0506877923487 \left(\alpha_{0}^{c_{s}}\right)^{4} \alpha_{2}^{s} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{7.2140871017812 \left(\alpha_{0}^{c_{s}}\right)^{3} \alpha_{0}^{s} r^{4} \log\left(c_{s}\right)}{c_{s}^{4}}$$
$$- \frac{261.07571894615 \left(\alpha_{0}^{c_{s}}\right)^{2} \left(\alpha_{0}^{s}\right)^{2} \alpha_{1}^{c_{s}} r^{2}}{c_{s}^{2}}$$
$$- \frac{2.2984421043879 \left(\alpha_{0}^{c_{s}}\right)^{3} \alpha_{3}^{s} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{0.598334188035 \alpha_{0}^{s} \alpha_{4}^{c_{s}} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{2.6142484553417 \left(\alpha_{0}^{s}\right)^{2} \alpha_{3}^{s} r^{2} \log\left(c_{s}\right)^{2}}{c_{s}^{2}}$$
$$- \frac{5.367589466396 \alpha_{1}^{c_{s}} \left(\alpha_{1}^{s}\right)^{2} r^{2} \log\left(c_{s}\right)^{2}}{c_{s}^{2}}$$
$$- \frac{2.6914289492821 \alpha_{1}^{s} \alpha_{3}^{s} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{22.202333455293 \left(\alpha_{0}^{s}\right)^{2} \left(\alpha_{1}^{c_{s}}\right)^{2} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{11.201715574019 \left(\alpha_{1}^{s}\right)^{2} \alpha_{2}^{c_{s}} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{1.4176370369568 \left(\alpha_{0}^{c_{s}}\right)^{4} \alpha_{1}^{s} r^{2} \log\left(c_{s}\right)^{2}}{c_{s}^{2}}$$
$$- \frac{23.284056978266 \left(\alpha_{0}^{s}\right)^{2} \alpha_{1}^{c_{s}} \alpha_{2}^{c_{s}} r}{c_{s}}$$
$$- \frac{47.037324855903 \alpha_{0}^{c_{s}} \left(\alpha_{1}^{c_{s}}\right)^{2} \alpha_{1}^{s} r}{c_{s}}$$
$$- \frac{188.82167015916 \left(\alpha_{0}^{c_{s}}\right)^{3} \alpha_{0}^{s} \alpha_{1}^{s} r^{2}}{c_{s}^{2}}$$
$$- \frac{54.522664517193 \left(\alpha_{0}^{c_{s}}\right)^{2} \left(\alpha_{0}^{s}\right)^{4} r^{2} \log\left(c_{s}\right)}{c_{s}^{2}}$$
$$- \frac{220.19533786365 \left(\alpha_{0}^{c_{s}}\right)^{2} \left(\alpha_{0}^{s}\right)^{2} \alpha_{1}^{s} r^{2}}{c_{s}^{2}}$$
$$- \frac{28.254233793598 \alpha_{0}^{c_{s}} \alpha_{1}^{s} \alpha_{2}^{s} r^{2}}{c_{s}^{2}}$$
$$- \frac{28.788861243767 \alpha_{0}^{c_{s}} \alpha_{0}^{s} \alpha_{1}^{c_{s}} r^{4}}{c_{s}^{4}}$$
$$- \frac{116.82046036669 \left(\alpha_{0}^{c_{s}}\right)^{2} \alpha_{0}^{s} \alpha_{1}^{s} r^{3}}{c_{s}^{3}}$$
$$- \frac{1.8692277878667 \left(\alpha_{0}^{s}\right)^{2} \alpha_{4}^{s} r \log\left(c_{s}\right)}{c_{s}}$$
$$- \frac{1.9101047397252 \left(\alpha_{0}^{s}\right)^{3} \alpha_{3}^{s} r \log\left(c_{s}\right)^{2}}{c_{s}}$$
$$- \frac{0.97132050590725 \alpha_{1}^{s} \alpha_{2}^{s} r^{3} \log\left(c_{s}\right)^{2}}{c_{s}^{3}}$$

$$- \frac{65.893892801149 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s}$$
$$- \frac{1.0488038466557 \alpha_0^{c_s} \alpha_2^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{0.52949139385727 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)^3}{c_s^3}$$
$$- \frac{4.2490431074254 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{4.6157059108364 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{37.650039199474 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^{c_s} r^2}{c_s^2}$$
$$- \frac{18.925882241283 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{18.965112171193 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{1.2015418958901 \alpha_1^s \alpha_4^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.2089737898992 \alpha_0^s \alpha_4^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{5.0351500609997 \left(\alpha_0^{c_s}\right)^2 \alpha_4^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{41.156558641787 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s}$$
$$- \frac{10.609218246596 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 \alpha_1^s r}{c_s}$$
$$- \frac{169.81322958978 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2}$$
$$- \frac{42.618229821464 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{85.525499102562 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r^3}{c_s^3}$$
$$- \frac{5.3683757571926 \alpha_0^s \left(\alpha_2^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{11.559920669238 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{23.636778628067 \alpha_0^{c_s} \alpha_0^s \alpha_3^{c_s} r^2}{c_s^2}$$
$$- \frac{0.18638062414823 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.74552249659293 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{47.851027601122 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s}$$
$$- \frac{12.11424084081 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_3^{c_s} r}{c_s}$$
$$- \frac{97.031211706403 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r^2}{c_s^2}$$

$$- \frac{1.5737779918774 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^5 \log\left(c_s\right)}{c_s^5}$$
$$- \frac{1.5983559395235 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{0.40662702238542 \alpha_0^{c_s} \alpha_3^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{1.6656048141581 \left(\alpha_0^s\right)^2 \alpha_4^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.10412751301962 \left(\alpha_0^{c_s}\right)^5 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{26.867590523324 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.052507559280447 \alpha_2^{c_s} \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{0.21003023712179 \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r^2 \log\left(c_s\right)^3}{c_s^2}$$

$$- \frac{6.9325566397257 \left(\alpha_0^s\right)^2 \alpha_2^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{0.055258595711511 \alpha_0^{c_s} \alpha_1^s r^5 \log\left(c_s\right)^2}{c_s^5}$$
$$- \frac{7.2020462840805 \left(\alpha_0^s\right)^3 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.22523101117468 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{15.067582783243 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{0.9424780878218 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^3}{c_s^3}$$
$$- \frac{7.5978803659804 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{0.12021589261378 \alpha_0^{c_s} \alpha_4^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{15.766668698063 \alpha_0^{c_s} \alpha_0^s \alpha_3^s r^2}{c_s^2}$$
$$- \frac{0.49618897428141 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{32.037747856499 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{8.0878002039529 \left(\alpha_0^s\right)^3 \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{0.50801148814198 \alpha_0^s \alpha_3^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{16.335385878746 \alpha_0^s \alpha_1^s \alpha_2^s r^2}{c_s^2}$$
$$- \frac{32.907955262293 \left(\alpha_0^{c_s}\right)^4 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{65.949469796285 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2}$$

$$- \frac{33.038820699018 \alpha_0^s \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r}{c_s}$$
$$- \frac{0.12943041458753 \alpha_0^s \alpha_1^s r^5 \log\left(c_s\right)^2}{c_s^5}$$
$$- \frac{16.603528043563 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_3^{c_s} r}{c_s}$$
$$- \frac{33.488309328657 \alpha_0^s \alpha_1^s \alpha_2^{c_s} r^2}{c_s^2}$$
$$- \frac{0.016355584031751 \alpha_0^{c_s} \alpha_2^s r^4 \log\left(c_s\right)^3}{c_s^4}$$
$$- \frac{0.26168934450801 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^4 \log\left(c_s\right)^3}{c_s^4}$$
$$- \frac{8.4133427159559 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.53085694303141 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{1.0640922518624 \alpha_0^s \alpha_2^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{17.170223355941 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{2.1520321750819 \alpha_1^s \alpha_3^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{2.1577116598525 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{2.1627796076988 \left(\alpha_0^{c_s}\right)^2 \alpha_3^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{8.7114183166094 \alpha_0^{c_s} \left(\alpha_1^s\right)^3 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.0905837917801 \left(\alpha_0^s\right)^5 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{35.216907745854 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s \alpha_2^{c_s} r}{c_s}$$
$$- \frac{0.55208333333333 \left(\alpha_0^s\right)^4 \alpha_1^s r^2 \log\left(c_s\right)^4}{c_s^2}$$
$$- \frac{141.3356375469 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r^3}{c_s^3}$$
$$- \frac{0.55414160443643 \alpha_0^s \alpha_2^{c_s} r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{35.924986701706 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s \alpha_1^s r}{c_s}$$
$$- \frac{4.5715835447462 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.1501885493287 \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.14385226046225 \alpha_0^{c_s} \alpha_0^s r^6 \log\left(c_s\right)}{c_s^6}$$
$$- \frac{1.154727777479 \alpha_0^s \left(\alpha_2^{c_s}\right)^2 r \log\left(c_s\right)}{c_s}$$

$$- \frac{18.568614205917 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r}{c_s}$$
$$- \frac{9.3326987690427 \left(\alpha_0^s\right)^2 \alpha_1^s \alpha_2^s r}{c_s}$$
$$- \frac{0.0091145833333333 \left(\alpha_0^{c_s}\right)^2 \alpha_3^s r^2 \log\left(c_s\right)^4}{c_s^2}$$
$$- \frac{0.036458333333333 \alpha_0^s \alpha_5^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{9.4881337599963 \left(\alpha_0^s\right)^4 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{1.200524716176 \alpha_1^{c_s} \alpha_1^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{0.15104166666667 \alpha_0^s \alpha_4^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{1.2388010113007 \alpha_1^{c_s} \alpha_4^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{9.9212304313373 \alpha_0^s \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{9.9483576869351 \left(\alpha_0^s\right)^3 \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{39.890337412321 \alpha_0^s \alpha_1^{c_s} \alpha_2^{c_s} r^2}{c_s^2}$$
$$- \frac{20.461421595607 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^4}{c_s^4}$$
$$- \frac{2.5751619926631 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^5 \log\left(c_s\right)}{c_s^5}$$
$$- \frac{42.453281836198 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^s r^2}{c_s^2}$$
$$- \frac{0.16662751301962 \alpha_1^s \alpha_4^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.0052083333333333 \alpha_2^{c_s} \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.0052083333333333 \left(\alpha_0^{c_s}\right)^3 \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.041666666666667 \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r^2 \log\left(c_s\right)^4}{c_s^2}$$
$$- \frac{42.689137071712 \alpha_0^{c_s} \alpha_1^s \alpha_2^{c_s} r^2}{c_s^2}$$
$$- \frac{5.350539031452 \left(\alpha_0^{c_s}\right)^5 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{42.952431090596 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 \alpha_1^s r}{c_s}$$
$$- \frac{10.944812665467 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s \alpha_1^{c_s} r}{c_s}$$
$$- \frac{5.7152427032107 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{22.952209822684 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_2^s r}{c_s}$$

$$- \frac{45.904710612886 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r}{c_s}$$
$$- \frac{0.045572916666667 \alpha_0^s \alpha_2^s r^4 \log\left(c_s\right)^3}{c_s^4}$$
$$- \frac{0.18229166666667 \alpha_2^s \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.18229166666667 \left(\alpha_0^s\right)^3 \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.72916666666667 \left(\alpha_0^s\right)^4 \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.4583333333333 \alpha_0^s \left(\alpha_2^s\right)^2 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.459738851267 \left(\alpha_0^{c_s}\right)^5 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.18356434450801 \left(\alpha_0^s\right)^4 \alpha_2^{c_s} r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{23.635258609831 \left(\alpha_0^s\right)^2 \alpha_1^s \alpha_2^{c_s} r}{c_s}$$
$$- \frac{0.36979166666667 \alpha_1^s \alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{6.0003047609539 \left(\alpha_0^{c_s}\right)^4 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s}$$
$$- \frac{49.131851263357 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{12.389034207023 \left(\alpha_0^s\right)^3 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{12.486295049233 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{3.1420899004122 \alpha_1^{c_s} \alpha_3^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{1.5760620651767 \alpha_0^{c_s} \alpha_4^s r^2 \log\left(c_s\right)}{c_s^2}$$

$$- \frac{1.5884741815718 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r^3 \log\left(c_s\right)^3}{c_s^3}$$
$$- \frac{12.866595614791 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_3^s r}{c_s}$$
$$- \frac{0.204471296191 \alpha_1^s \alpha_2^{c_s} r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{6.5605176675212 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} \alpha_2^{c_s} r}{c_s}$$
$$- \frac{13.255892900063 \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.41530837917801 \alpha_0^s \alpha_5^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{7.0146178742227 \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{3.51534396509 \alpha_2^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$

$$
\begin{aligned}
&- \frac{57.9195576008 58 \alpha_0^{c_s} \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r}{c_s} \\
&- \frac{0.45560805454892 \left(\alpha_0^s\right)^2 \alpha_3^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{3.6644110968749 \left(\alpha_0^s\right)^4 \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{1.8343273205988 \left(\alpha_0^s\right)^2 \alpha_1^s r^4 \log\left(c_s\right)^2}{c_s^4} \\
&- \frac{0.47211590082505 \alpha_1^{c_s} \alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{30.649642188027 \alpha_0^s \alpha_1^{c_s} \alpha_2^s r^2}{c_s^2} \\
&- \frac{1.9166666666667 \left(\alpha_0^s\right)^3 \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&- \frac{0.96600547490636 \alpha_0^{c_s} \left(\alpha_0^s\right)^6 r \log\left(c_s\right)}{c_s} \\
&- \frac{123.65098576076 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^3}{c_s^3} \\
&- \frac{7.8030128018755 \left(\alpha_1^s\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{1.9655837917801 \alpha_2^s \alpha_3^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{63.348273744999 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.015625 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.125 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^4}{c_s^2} \\
&+ \frac{0.09375 \alpha_0^{c_s} \alpha_0^s \alpha_4^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.078125 \alpha_0^s \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.15625 \alpha_1^{c_s} \alpha_1^s \alpha_2^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.15625 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.15625 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.3125 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_2^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.03515625 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r^3 \log\left(c_s\right)^4}{c_s^3} \\
&+ \frac{0.234375 \alpha_0^{c_s} \alpha_1^s \alpha_3^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.234375 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.546875 \alpha_0^{c_s} \alpha_0^s \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^4}{c_s^2}
\end{aligned}
$$

$$+\frac{0.546875\alpha_0^s\alpha_1^s\alpha_3^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{4.375\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.703125\alpha_0^{c_s}\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.14189767784134\left(\alpha_0^{c_s}\right)^4\alpha_0^s\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.28379535568268\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.28379535568268\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.31504535568268\alpha_0^s\alpha_1^s\alpha_2^{c_s} r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.42569303352403\alpha_0^s\left(\alpha_1^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{7.8774362791432\alpha_0^s\alpha_1^{c_s}\alpha_2^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.0371891632596\alpha_0^s\alpha_1^{c_s}\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{1.1013860670481\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.29139574270913\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{1.1655829708365\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s} r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.61319303352403\alpha_0^{c_s}\left(\alpha_0^s\right)^4\alpha_1^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.3513860670481\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.19626700838101\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)^3}{c_s^4}$$
$$+\frac{3.5777721340961\alpha_0^{c_s}\alpha_0^s\alpha_4^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.1182837448894\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{2.6689767784134\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{5.7638612623624\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{3.1216855319755\alpha_0^{c_s}\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.1216855319755\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{6.4731349795576\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{1.6469883892067\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$

$$+\frac{3.2939767784134\left(\alpha_0^{c_s}\right)^2\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{3.2939767784134\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.8395791005721\alpha_0^{c_s}\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{4.5659860291161\alpha_0^{c_s}\alpha_0^s\alpha_3^{c_s}r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.1680926464976\alpha_0^{c_s}\alpha_0^s\alpha_3^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{19.536928429616\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.5844883892067\alpha_0^{c_s}\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{2.5844883892067\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{46.105558971743\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{12.188285525609\alpha_1^{c_s}\alpha_1^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{50.732921820774\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{27.775184091118\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{231.76628399015\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}$$
$$+\frac{4.0034651676201\alpha_0^s\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{16.863900286893\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{2.1968211154576\alpha_0^s\alpha_1^s\alpha_3^{c_s}r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.1881148878051\left(\alpha_0^{c_s}\right)^4\alpha_0^s\alpha_1^{c_s}r\log\left(c_s\right)}{c_s}$$
$$+\frac{5.2660325285376\alpha_0^s\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{22.554554719686\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{11.615224422146\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{11.729006778244\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{48.549103697462\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s}r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{49.191233535148\alpha_0^{c_s}\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{15.812179043725\alpha_0^{c_s}\alpha_1^s\alpha_2^{c_s}r^2\log\left(c_s\right)}{c_s^2}$$

$$+\frac{16.045906693665\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{65.025823594607\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{2.1070916662185\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{4.2490431074254\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{4.4382967251885\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{18.051297314256\left(\alpha_0^{c_s}\right)^4\alpha_0^s\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{9.1715767967298\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{20.793360992279\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{5.2432472125918\alpha_0^{c_s}\alpha_1^{c_s}\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{10.755281999685\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{97.189367996561\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{100.28991650817\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{12.798946193029\left(\alpha_0^{c_s}\right)^2\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.2060453025497\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{25.773852823526\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{6.9165270668905\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{29.255461497574\left(\alpha_0^{c_s}\right)^2\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.9424780878218\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{30.780492667285\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{16.166403940833\alpha_0^{c_s}\left(\alpha_0^s\right)^4\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{66.717509864214\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{17.265090188972\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.1037751301962\left(\alpha_0^{c_s}\right)^3\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{8.8445680415413\alpha_0^{c_s}\alpha_1^s\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$

$$+\frac{9.0040783755784\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{145.56124402526\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{2.3021190133549\alpha_0^{c_s}\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.072916666666667\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{4.69691008228\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{38.065291754608\alpha_0^s\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{19.234312620033\alpha_0^s\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.2287751301962\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{20.286580830243\alpha_0^s\alpha_1^s\alpha_2^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.2912751301962\alpha_1^{c_s}\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{167.02045641872\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{21.257105744126\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.020833333333333\alpha_0^s\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$+\frac{86.103637709847\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{22.194948864644\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.7081375650981\alpha_0^{c_s}\alpha_1^s\alpha_3^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{5.7857932712305\alpha_0^s\alpha_1^s\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.8958333333333\alpha_0^s\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.36458333333333\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)^4}{c_s^2}$$
$$+\frac{23.354369649987\alpha_0^s\alpha_1^{c_s}\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{94.100765766759\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{47.348572536558\left(\alpha_0^{c_s}\right)^2\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{24.174774280519\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$

$$+\frac{12.08842323191\left(\alpha_0^s\right)^3\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{6.1274301940373\alpha_0^s\alpha_1^{c_s}\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.1769483631436\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{26.034392656129\alpha_0^{c_s}\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{26.302489700385\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{13.178724319326\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{26.479070027761\alpha_0^{c_s}\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.683803657696\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{27.472885902749\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{27.7657914319\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.7494126952943\alpha_0^{c_s}\left(\alpha_0^s\right)^2\left(\alpha_1^{c_s}\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{14.003856211609\alpha_0^{c_s}\alpha_0^s\alpha_3^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{113.62621092774\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{14.712464722621\alpha_0^{c_s}\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{29.99622215499\alpha_0^s\left(\alpha_1^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{15.254609919869\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.015625\alpha_0^{c_s}\left(\alpha_0^s\right)^4\alpha_1^{c_s} r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.03125\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.03125\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.046875\alpha_0^{c_s}\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.078125\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_3^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.15625\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.15625\left(\alpha_0^{c_s}\right)^2\alpha_1^s\alpha_2^s r\log\left(c_s\right)^4}{c_s}$$
$$-\frac{0.15625\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^4}{c_s}$$

$$- \frac{0.21875 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^4}{c_s^2}$$
$$- \frac{0.234375 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 \alpha_1^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.234375 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_3^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.234375 \alpha_0^s \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{1.09375 \left(\alpha_0^s\right)^2 \alpha_1^s \alpha_2^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{2.1875 \alpha_0^s \alpha_1^s \alpha_3^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.158203125 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r^3 \log\left(c_s\right)^4}{c_s^3}$$
$$- \frac{0.18919690378846 \alpha_0^{c_s} \alpha_2^{c_s} \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$

$$- \frac{0.18919690378846 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.14189767784134 \alpha_0^s \alpha_1^s \alpha_3^{c_s} r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{1.0337953556827 \alpha_0^{c_s} \alpha_0^s \alpha_4^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.61319303352403 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{3.4459845189423 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{66.684945669984 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{34.183716244558 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{8.8112806103084 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.3344883892067 \alpha_0^s \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{2.6689767784134 \alpha_1^{c_s} \alpha_1^s \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{2.6689767784134 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{3.1115674897788 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{3.2939767784134 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{16.183034565821 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{20.588252811225 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{2.5844883892067 \alpha_0^{c_s} \alpha_1^s \alpha_3^s r \log\left(c_s\right)^3}{c_s}$$

$$
\begin{aligned}
&- \frac{0.70598852514302\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&- \frac{6.8041582011442\alpha_0^{c_s}\alpha_0^s \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&- \frac{7.2278699956807\alpha_0^s\alpha_1^{c_s}\alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{14.908599752858\alpha_0^{c_s}\alpha_2^{c_s}\alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{34.007267678512\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}\alpha_2^{c_s} r}{c_s} \\
&- \frac{74.9649414605\alpha_0^{c_s}\alpha_0^s \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{4.9409651676201 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s} \\
&- \frac{20.155244704455\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{5.1589143650113 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s}\alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{5.1589143650113 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.32294842510727\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&- \frac{5.2099597813588\alpha_0^{c_s} \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&- \frac{21.572831995093\alpha_0^{c_s}\alpha_0^s \left(\alpha_1^{c_s}\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{2.9114002160297 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s\alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{58.083939949874 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s\alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{15.33349199059 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s\alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{7.7534651676201\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s} \\
&- \frac{31.624358087449\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{16.26828929269\alpha_0^{c_s}\alpha_1^s\alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{2.1408870219012\alpha_0^s\alpha_1^{c_s}\alpha_3^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{69.866695157509\alpha_0^{c_s}\alpha_0^s\alpha_1^s\alpha_2^{c_s} r}{c_s} \\
&- \frac{8.876593450377 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{4.5701717470697\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&- \frac{19.570215860849 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s\alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{9.9328706215108\alpha_0^s\alpha_1^s\alpha_3^{c_s} r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$- \frac{10.755281999685 \alpha_1^{c_s} \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.3469092355797 \alpha_0^{c_s} \alpha_0^s \alpha_4^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{5.4031146586364 \alpha_0^{c_s} \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{11.238712230131 \left(\alpha_0^s\right)^2 \alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{49.715027711287 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{12.798946193029 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.6030226512749 \alpha_0^{c_s} \alpha_1^s \alpha_2^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{26.097001485341 \alpha_1^{c_s} \alpha_1^s \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{6.7425424469449 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{6.9070922214571 \alpha_0^{c_s} \alpha_0^s \alpha_4^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{6.9165270668905 \alpha_0^{c_s} \alpha_1^s \alpha_3^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.9009240446987 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{7.8768571719844 \left(\alpha_0^s\right)^2 \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{16.016456045499 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{8.3759574616751 \alpha_0^s \alpha_1^s \alpha_3^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{16.932268645455 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{4.2684702438652 \alpha_0^s \alpha_1^s \alpha_2^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{2.1577116598525 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{17.265090188972 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{8.6448434541435 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{69.742923744036 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{4.3738239787259 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{17.495295914904 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{1.1037751301962 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.56770833333333 \alpha_0^s \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)^4}{c_s^2}$$

$$- \frac{18.182565994714 \alpha_0^s \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{18.438292934911 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{2.4114994062629 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{19.47035356715 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{9.8183953616574 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.2287751301962 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{79.461013911287 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_2^s r}{c_s}$$
$$- \frac{10.09807314849 \alpha_0^s \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{5.0514109730879 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{40.494591794869 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{20.283482886201 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.2912751301962 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{10.533196769851 \alpha_0^{c_s} \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.041666666666667 \alpha_0^s \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)^4}{c_s^2}$$

$$- \frac{44.804541964937 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.7081375650981 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_3^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.091145833333333 \alpha_0^{c_s} \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)^4}{c_s^2}$$
$$- \frac{0.18229166666667 \alpha_0^{c_s} \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)^4}{c_s^2}$$
$$- \frac{0.72916666666667 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \log\left(c_s\right)^4}{c_s^2}$$
$$- \frac{46.676607096297 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^s r}{c_s}$$
$$- \frac{46.958849012277 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.73425737803204 \alpha_0^s \alpha_2^{c_s} \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{47.050382883379 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{12.154998094376 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$

$$
\begin{aligned}
&- \frac{25.053712507924 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{51.372265919509 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{12.994698445064 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{6.5792050866236 \alpha_0^{c_s} \alpha_0^s \alpha_2^{c_s} r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{13.178724319326 \left(\alpha_0^{c_s}\right)^3 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{13.482413892444 \alpha_0^s \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{13.735173574273 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{1.7494126952943 \alpha_0^{c_s} \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{3.6657307304634 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{14.707067627279 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{15.099223284946 \alpha_0^s \alpha_2^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{1.9242741322273 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{15.663166775309 \alpha_0^{c_s} \alpha_1^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.09375 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^4}{c_s} \\
&+ \frac{0.56759071136537 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{2.4527721340961 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{10.337953556827 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{5.9080038194127 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{26.242943872355 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{15.476743095034 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{66.123502180603 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{44.322735007416 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{61.739979054703 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{31.293046119386 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.0625 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^4}{c_s}
\end{aligned}
$$

$$- \frac{0.9375 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^s r \log\left(c_s\right)^4}{c_s}$$
$$- \frac{0.85138606704805 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{82.720955896615 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.7483744562548 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2}$$
$$- \frac{3.6791582011442 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{9.3650565959265 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{131.21274714922 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{34.97882299137 \alpha_0^{c_s} \alpha_0^s \alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{97.690445357646 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{20.906413987901 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$